

Contracting with Unknown Complementarity: Ratchets and Replacement Risk*

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Abstract

How do a firm and a worker contract while they are learning how productive their relationship is? I study a setting in which a principal and an agent contract over two periods under moral hazard. Neither knows how much complementarity there is between the agent and his assignment, which both learn from output. The agent earns limited liability rents, and the principal can only commit to the current period. In each period, the cheapest contract implementing strictly positive effort pays a bonus on an upper tail region of output when effort is always productive, and pays on both tails when effort is sometimes counterproductive. The agent's dynamic incentive to influence the principal's belief decomposes into an *extensive* retention force and an *intensive* wage force, measured against a manipulation-free benchmark. When the two forces oppose, a pessimistic prior induces over-experimentation and an optimistic one under-experimentation, with a single threshold between them. A pure-strategy equilibrium exists and is unique for a sufficiently convex cost of effort.

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1 Introduction

When a worker is newly hired, assigned a new task, or given a new technology to work with, neither he nor his employer may know at the outset just how productive the pairing will be. That is, they may not know how much *complementarity* there is between the worker and his assignment. They learn it from what he produces. And if effort is not directly observable, this is also the sole basis for his compensation.

A first question arises: how should a firm design contracts when the productivity of the worker's effort is commonly unknown? If there is a chance that effort backfires, then mediocre output is evidence that the worker did not try, while disaster is evidence that he tried and failed. Rewarding very low output, in addition to very high output, is then optimal.

At the same time, if the firm cannot commit today to the terms it will offer tomorrow, it uses what it learns to decide both whether to continue the relationship, and conditional on doing so, the contract it proposes. So a forward-looking worker has the incentive to influence what the firm learns. How much they learn depends on how hard he works, since effort is what makes output informative about the complementarity. A second question then arises: how does this incentive affect how much is learned? If the worker fears replacement, the incentive pushes him to exert more effort to avoid being fired, whereas if replacement risk is low, it pushes him to exert less so as to face more lenient terms in the future and escape the ratchet.

I model the relationship as follows. A principal and an agent, both risk-neutral, interact over two periods. They share a common prior over the degree of complementarity between the agent's effort and a task, the marginal product of effort. In each period, a noisy output signal is realized. Importantly, three key frictions are present: limited commitment power, moral hazard, and limited liability. The principal has an outside option to the agent, and commitment limited to the current period means that following the first output realization, she updates her belief and reoptimizes both her hiring decision and the contract she proposes the agent. Together with privately chosen effort, this creates a belief manipulation channel for the agent: by distorting effort away from the principal's conjecture today, the principal may be more likely to update her beliefs in a way that is more favorable to the agent in the next period. Limited liability rents imply that the agent has a stake to protect, which generates a retention incentive for the agent.

The setting differs from the classical ratchet problem in what the agent knows *ex ante*. In the standard formulation (cf. [Freixas et al., 1985](#); [Laffont and Tirole, 1987, 1988](#)), the agent's production technology is privately known, constituting *exogenous* asymmetric information that the agent is tempted to conceal from the principal. Here, the complementarity is the object of learning, and the agent's output is the signal from which both parties learn. When effort is privately

chosen, this gives rise to *endogenous* asymmetric information, in that the agent can influence not only what the principal learns about the production technology but also how much is learned by both. Moreover, the agent is at risk of replacement after the principal learns, which may generate a concern in addition to the ratchet.

In each period, the optimal contract is a bonus contract. Depending on the support of the complementarity, to implement strictly positive effort via the wage, the principal pays on either *one* or *two* tail regions of output. When the effort is productive at every possible complementarity, we recover a standard bonus contract, in which the agent is paid whenever realized output exceeds a single threshold. The novel two-tail shape arises exactly when effort is *counterproductive* with strictly positive probability, in which case the standard assumption of the monotone likelihood ratio property of the output distribution in effort is violated. In this case, the bonus is paid whenever the output is sufficiently high *or* sufficiently low, since a very low output is itself informative that effort was exerted but backfired.

To isolate how the agent's dynamic incentive to manipulate beliefs affects learning, I introduce a benchmark in which the agent does not account for the effect of first-period effort on his continuation value, and I compare equilibrium first-period effort to the benchmark's. I say that the agent over-experiments when equilibrium effort exceeds the benchmark's, since more effort makes output more informative about the complementarity, and that he under-experiments when it falls short. The continuation incentive of the agent decomposes into an *extensive* retention force and an *intensive* wage force. The retention force always pushes towards more effort today, since the agent strictly prefers being employed to being fired, and increasing effort pushes strictly positive output mass past the firing threshold. The sign of the wage force is ambiguous: a higher output realization today raises expected productivity, and hence surplus, tomorrow, and whether the agent captures some of this increased surplus depends on how the principal adjusts the contract to her belief.

To make the comparison, I introduce a one-parameter family of priors on the complementarity for a fixed support, ordered from pessimistic to optimistic in the likelihood ratio order. For a given conjectured effort, Gaussian noise allows the posteriors to be indexed by a single parameter as well, affine in the prior parameter and in realized output. When the prior is sufficiently concentrated, meaning that both parties have some idea of how much complementarity there is at the beginning of the relationship, the sign of the wage force along the posterior index tracks its sign in the known-complementarity case, which restores tractability: it changes at most once, and if so, from positive to negative.

The result is that the two forces balance at a single threshold in the prior: below it the agent over-experiments, above it he under-experiments. When the prior is just optimistic enough for the

principal to engage with the agent in the first place, the retention force necessarily dominates: at initiation the agent is fired in expectation, so the agent places more weight on keeping the job at all. As the prior becomes more optimistic, retention becomes more and more likely, and the wage force dominates. Which case obtains far from the margin depends on the curvature of the cost of effort. When curvature is not too high, the agent's rent eventually falls as the belief improves, and the ratchet appears. When it is sufficiently high, it never does: implementing higher effort from a better-matched agent then requires loosening the contract, and good news benefits the agent at every prior in the family.

Answering these economic questions involves first answering a more technical one: does a pure-strategy equilibrium exist, and if so, is it unique? The non-existence of pure-strategy equilibrium recurs in environments with limited commitment and private action choice, in particular in the literature on agency with renegotiation ([Fudenberg and Tirole, 1990](#)), the hold-up problem ([Gul, 2001](#)), and the more recent literature on the ratchet effect ([Bhaskar and Roketskiy, 2023](#)). In the closest of these, [Bhaskar and Roketskiy \(2023\)](#), full rent extraction in the second period leaves the agent's objective with a convex kink exactly at the conjectured effort, so that no interior effort can be implemented in pure strategies. Here, limited liability and a zero outside option for the agent ensure that the agent's continuation rent is twice continuously differentiable in his first-period effort, so the kink is absent. Sufficient curvature of the effort cost then delivers three things: the validity of the first-order approach, a contraction mapping each conjecture to the effort it induces, and strict concavity of the principal's objective in the implemented effort. For sufficient curvature, a pure-strategy equilibrium exists and is unique.

Two further results complete the analysis. The first concerns the scope of the main result: the interval of cost curvatures for which a unique equilibrium exists and the two forces oppose is nonempty once the noise is large enough and the prior is concentrated enough, so the two-sided case is non-vacuous. The second applies the result to technology adoption: when the assignment is working with a new technology, the principal's ex ante decision is whether to invest in it at all. Adoption is again governed by a threshold in the prior: it is weakly higher than in the benchmark at pessimistic priors, where the technology is worth more to the principal because the agent's manipulation incentive pays part of the wage bill, and weakly lower at optimistic ones.

Related literature. In the classical ratchet problem the agent knows his productivity and the principal does not. [Freixas et al. \(1985\)](#) and [Laffont and Tirole \(1987, 1988\)](#) establish the tension: revealing a good type invites tougher terms, so separation is hard to sustain. [Gerardi and Maestri \(2020\)](#) show that under limited commitment, as the parties become patient, the principal either

learns nothing or dismisses the low type at once. [Ekmekci et al. \(2022\)](#) are closest in mechanism: below a firing cutoff their agent mimics the good type to save himself, and the incentive to save the relationship destroys learning. The agent near the firing margin here also raises effort to save the relationship, but since effort makes output more informative rather than imitating a type, the same incentive increases what the principal learns.

When uncertainty is symmetric, both parties learn the parameter from output and effort moves the principal's belief without revealing a type, as in the signal-jamming of [Fudenberg and Tirole \(1986\)](#) and the career-concerns literature ([Holmström, 1999](#); [Dewatripont et al., 1999](#)). [Cisternas \(2018\)](#) derives a learning-driven ratchet principle in a continuous-time game against a market. [Prat and Jovanovic \(2014\)](#) and [DeMarzo and Sannikov \(2017\)](#) characterize the optimal long-term contract under full commitment, with symmetric learning and an additive hidden action. In the latter, shirking lowers the principal's estimate of profitability and earns the agent information rents, which the contract controls by distorting the termination decision. [Gao and Xu \(2020\)](#) make the same point in a discrete setting, with a cutoff termination rule written into the contract. Three things differ here: the principal cannot commit, so retention and the second-period contract are re-optimized after she learns. Effort is multiplicative, so the agent controls both *how much* is learned and *what* is learned. Limited liability leaves a rent that the retention decision puts at risk, which makes the manipulation incentive two-sided.

[Kim \(2026\)](#) studies replacement risk in a career-concerns setting with *observable* effort and obtains the opposite sign on the same margin: since the principal sees effort, a failure after high effort is attributed to the agent's ability, so the agent shirks to make failure uninformative and avoid replacement. Here effort is hidden, so the agent cannot shift the blame and instead works to stay.

The closest model is [Bhaskar and Roketskiy \(2023\)](#): a principal who cannot commit, symmetric learning about the difficulty of the task with finitely many states and outputs, and a continuum of efforts that determine the mixture of experiments output is drawn from, so that effort moves both where beliefs land and how fast they resolve, as here. The difference is limited liability, which removes the kink discussed above and creates the extensive margin. The same non-existence appears in hold-up with unobservable investment ([Gul, 2001](#)) and in moral hazard with interim renegotiation ([Fudenberg and Tirole, 1990](#)), where the only equilibria are in mixed strategies. In a similar setting with finite actions, [Bhaskar and Mailath \(2019\)](#) show that effort becomes costlier to implement as the horizon lengthens.

A separate literature studies contracts for experimentation on a project whose quality is revealed over time: [Bergemann and Hege \(2005\)](#), [Manso \(2011\)](#) and [Halac et al. \(2016\)](#) under full commitment, and [Hörner and Samuelson \(2013\)](#) without it, with limited liability and termination, as here.

One main difference lies in the information structure: these papers rely on good-news bandits, in which a success resolves the uncertainty and effort determines whether the experiment is run, whereas here output is a continuous signal informative in both directions, whose informativeness scales with effort. One feature carries over: the lower tail of the contract here resembles the tolerance for early failure in [Manso \(2011\)](#), in that both pay for outcomes the principal would rather not see but are evidence of the action she wants, exploration there and effort here.

The optimal contract is the limited-liability bonus of [Innes \(1990\)](#). The second threshold is new, and arises because, when the complementarity can be negative, a very low output is itself evidence of effort. Its derivation involves replacing the agent's incentive constraint with his first-order condition, and the classical conditions under which that condition is sufficient ([Mirrlees, 1999](#); [Rogerson, 1985](#); [Jewitt, 1988](#)) fail here: Gaussian noise violates the convexity condition and counterproductive effort breaks the likelihood ratio order. The static moral hazard problem with bounded payments has been solved without the approach ([Chi and Olsen, 2025](#)), and [Azevedo and Wolff \(2025\)](#) restore it through a high reservation utility, a route unavailable when the agent's utility is the endogenous limited-liability rent. Here validity comes from curvature of the effort cost, established directly.

The remainder of the paper proceeds as follows. Section 2 introduces the model and the agent's and the principal's problems in each period. Section 3 validates the first-order approach, characterizes the optimal contract and retention decision, and establishes existence and uniqueness of equilibrium. Section 4 characterizes the belief-manipulation channel and compares equilibrium effort against that of a manipulation-free benchmark. Section 5 applies it to the adoption of a new technology. Section 6 concludes. All proofs are in Appendix A.

2 Model

2.1 Primitives

Players and output. A principal and an agent interact over two periods, $t = 1, 2$. There is no discounting and both parties are risk-neutral. Output in each period is

$$y_t = \theta e_t + \varepsilon_t,$$

where $\theta \in \Theta := [\underline{\theta}, \bar{\theta}]$ is a commonly unknown complementarity parameter, $e_t \in \mathbb{R}_+$ is the agent's private effort choice, and $\varepsilon_t \in \mathbb{R}$ is an unobserved noise term. θ has prior density g_θ and ε_t is i.i.d. across periods and independent of θ , with density g_ε . Whether effort is productive and by how

much depends on the degree of complementarity θ , which may be positive or negative.

Assumption 1 (Distributional regularity).

(i) (Noise.) The noise density g_ε is $C^2(\mathbb{R})$, has full support on $\mathbb{R}$ and mean zero, and satisfies $g_\varepsilon'' \in L^1(\mathbb{R})$. Its location kernel $(y, x) \mapsto g_\varepsilon(y - x)$ is strictly totally positive of order three, so in particular g_ε is log-concave. Its log-curvature is bounded: there exist $0 < m \leq M < \infty$ with $m \leq -(\log g_\varepsilon)'' \leq M$.¹

(ii) (Complementarity.) $g_\theta \in C^2(\Theta)$ is log-concave with compact support $\Theta = [\underline{\theta}, \bar{\theta}]$, where $\bar{\theta} > 0$.

Outside options. The principal has an outside option worth $\pi_A > 0$ per period, commonly known and independent of θ . Her outside option is therefore $2\pi_A$ at the start of the first period and π_A at the start of the second period. The agent's outside option is zero. At the beginning of each period, the principal may decide to fire the agent and take the outside option, and similarly, upon observing the contract, the agent has the option to exit the relationship. Denote by $\rho_t \in \{0, 1\}$ the principal's retention decision, with $\rho_t = 1$ when the agent is retained. The first-period decision is whether to form the relationship at all and the second whether to fire the agent after observing his output, and since only one such decision is taken in each period the subscript is dropped where no ambiguity arises.

Wages. The wage schedule $w_t : \mathbb{R} \rightarrow \mathbb{R}$ is subject to limited liability and a uniform cap:

$$0 \leq w_t(y_t) \leq \bar{w} \quad \forall y_t \in \mathbb{R}.^2$$

Limited liability generates positive rents for the agent. The cap is a bound on the reward the principal can promise. It is also what makes the optimal contract well defined, since without it the

¹Strict total positivity of order three means $\det[g_\varepsilon(y_i - x_j)]_{i,j \leq n} > 0$ for $y_1 < \dots < y_n$, $x_1 < \dots < x_n$ and $n \leq 3$. It implies strict log-concavity, hence unimodality and $g_\varepsilon' \in L^1(\mathbb{R})$ with $\int |g_\varepsilon'| = 2 \max g_\varepsilon$; see [Bagnoli and Bergstrom \(2005\)](#) for more properties of log-concave densities. Order three itself is used only for the two-tail contract of Theorem 1(ii). Densities of order three are characterized by [Weinberger \(1983\)](#): writing $g_\varepsilon = e^{-\psi}$, and given the other requirements of the assumption, the kernel is strictly totally positive of order three if and only if $\log \psi'' - 2\psi$ is strictly concave. The Gaussian satisfies all of these requirements, with $-(\log g_\varepsilon)'' = 1/\sigma^2$. The Laplace, exponential and gamma densities are totally positive of every order but fail these requirements: the first two are not C^2 , and all three have exponential tails, violating the floor m .

²The cap is uniform rather than tied to realized output as in [Innes \(1990\)](#). There the setting is a financial relationship and output is a nonnegative profit: the entrepreneur's payment comes out of it, so he cannot pay more than there is, and the optimum is a debt contract. Here, output can be read as a performance signal rather than a cash flow the agent hands over, and since the signal can be negative, such a cap would read $w_t \leq \max\{0, y_t\}$. Such a cap would rule out payments at low outputs, which Theorem 1 shows are part of the optimal contract when effort can be counterproductive.

principal could buy any effort with a vanishing payment far in the tail.³

An *admissible* contract in period t is a wage schedule w_t satisfying these bounds. Beyond limited liability and the wage cap the principal is unrestricted. The retention decision is separate, taken at the start of the period before any contract is proposed. For every public history (w_1, y_1) , we get a retention decision $\rho_2(w_1, y_1)$ and wage schedule $w_2(\cdot; w_1, y_1)$.

Limited commitment. The principal's commitment power is limited: although she can commit to the current wage schedule upon retention, she cannot commit to future terms, nor can she commit to retain the agent in the future. Therefore, the principal uses what she observes in the first period to reoptimize the second-period retention choice and proposed wage schedule based on her updated belief about θ .

Costs. The agent's effort cost $c : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is $C^2(\mathbb{R}_+)$, strictly convex, with $c(0) = 0$, $c'(0) = 0$, and c'' nondecreasing.

Timing. In each period, the principal first decides whether to retain the agent. If she retains the agent, she proposes a wage schedule. Then, the agent accepts or rejects the contract. If he accepts, he privately chooses effort. Finally, output is realized and both parties update their beliefs. If either party walks away at any point, each gets their outside option and they do not have the chance to interact again in the future.⁴

2.2 Equilibrium

The solution concept I use is pure-strategy perfect Bayesian equilibrium. The restriction to pure strategies is without loss here rather than a selection: under the cost curvature maintained for the first-order approach introduced in Section 3, the agent's objective is strictly concave in effort, uniformly over the space of contracts and retention rules. So his best response is a singleton at every history, and randomization is strictly unprofitable. The principal's belief over the agent's first-period effort is accordingly characterized by a single conjecture $\hat{e}_1 \in \mathbb{R}_+$ rather than a distribution over effort levels.

³This is the reward counterpart of the problem in [Mirrlees \(1999\)](#), where unbounded penalties let the principal approach the first best by punishing severely on a vanishing set of outputs, so that no optimum exists.

⁴This may be interpreted as a property of the outside options, rather than additional commitment power by the principal.

Beliefs. Updating beliefs about the complementarity depends on the effort one conditions on. For any effort level $e \geq 0$, Bayes' rule gives the posterior density

$$p(\theta \mid y_1, e) = \frac{g_\varepsilon(y_1 - e \theta) g_\theta(\theta)}{\int_{\Theta} g_\varepsilon(y_1 - e \theta') g_\theta(\theta') d\theta'}. \quad (1)$$

Both players update by this formula and differ only in the effort they substitute: the agent knows the effort he exerted and substitutes the true e_1 , while the principal, who cannot observe effort directly, substitutes her conjecture $\hat{e}_1$. On the equilibrium path, $e_1 = \hat{e}_1$, and the two coincide.

Stage 2: the agent's problem. Fix a public history (w_1, y_1) . Conditional on retention and acceptance, an agent who exerted e_1 observes the contract and solves

$$\max_{e_2 \geq 0} \int w_2(y_2; y_1) f_2(y_2 \mid e_2; y_1, e_1) dy_2 - c(e_2), \quad (2)$$

where the second-period output density is

$$f_2(y_2 \mid e_2; y_1, e_1) = \int_{\Theta} g_\varepsilon(y_2 - e_2 \theta) p(\theta \mid y_1, e_1) d\theta.$$

Let $e_2^*(w_2; y_1, e_1)$ denote a solution to (2). Write the agent's indirect utility from (2) as

$$U_2(w_2; y_1, e_1) = \int w_2(y_2; y_1) f_2(y_2 \mid e_2^*; y_1, e_1) dy_2 - c(e_2^*). \quad (3)$$

Stage 2: the principal's problem. The principal's second-period problem at history (w_1, y_1) and conjecture $\hat{e}_1$ is

$$\max_{0 \leq w_2(\cdot; y_1) \leq \bar{w}, e_2 \geq 0} \int [y_2 - w_2(y_2; y_1)] f_2(y_2 \mid e_2; y_1, \hat{e}_1) dy_2 \quad (4)$$

$$\text{s.t. } e_2 \in \arg \max_{\tilde{e}_2 \geq 0} \int w_2(y_2; y_1) f_2(y_2 \mid \tilde{e}_2; y_1, \hat{e}_1) dy_2 - c(\tilde{e}_2), \quad (\text{IC}_2)$$

$$\int w_2(y_2; y_1) f_2(y_2 \mid e_2; y_1, \hat{e}_1) dy_2 - c(e_2) \geq 0. \quad (\text{IR}_2)$$

By the observation above, (IR₂) is implied by the wage bounds together with (IC₂): limited liability makes the payoff from zero effort nonnegative, and incentive compatibility makes the agent's equilibrium payoff at least as high. Note the choice of the principal depends only on $y_1, \hat{e}_1$ via the posterior belief. Thus denote by $w_2^*(\cdot; y_1, \hat{e}_1)$ the solution and $\Pi_2^*(y_1, \hat{e}_1)$ the resulting value for the principal. The indirect utility of the agent is then $U_2^*(y_1, e_1, \hat{e}_1) = U_2(w_2^*(\cdot; y_1, \hat{e}_1); y_1, e_1)$.

Retention. At the beginning of the second period, unable to commit to future terms, the principal optimally retains if and only if her continuation value is at least as high as that of replacement:

$$\Pi_2^*(y_1, \hat{e}_1) \geq \pi_A. \quad (5)$$

Then define the optimal retention rule as

$$\rho^*(y_1, \hat{e}_1) := \mathbf{1} \{ \Pi_2^*(y_1, \hat{e}_1) \geq \pi_A \}.$$

Stage 1: the agent's problem. Fixing the contract w_1 and the continuation, which runs through the conjecture $\hat{e}_1$, the agent solves

$$\max_{e_1 \geq 0} \int [w_1(y_1) + \rho^*(y_1, \hat{e}_1) U_2^*(y_1, e_1, \hat{e}_1)] f_1(y_1 | e_1) dy_1 - c(e_1), \quad (6)$$

with

$$f_1(y_1 | e_1) = \int_{\Theta} g_{\varepsilon}(y_1 - e_1 \theta) g_{\theta}(\theta) d\theta.$$

Let $e_1^*(w_1, \hat{e}_1)$ denote a solution. Note that the agent's deviation affects both the output density $f_1(\cdot | e_1)$ and his own posterior inside U_2 , but does not affect the retention rule nor the stage-2 contract, both chosen by the principal under the conjecture $\hat{e}_1$.

Stage 1: the principal's problem. The principal's full first-period program is

$$\max_{0 \leq w_1(\cdot) \leq \bar{w}, e_1 \geq 0} \int [y_1 - w_1(y_1) + \rho^*(y_1, e_1) \Pi_2^*(y_1, e_1) + (1 - \rho^*(y_1, e_1)) \pi_A] f_1(y_1 | e_1) dy_1 \quad (7)$$

$$\text{s.t. } e_1 \in \arg \max_{\tilde{e}_1 \geq 0} \int [w_1(y_1) + \rho^*(y_1, e_1) U_2^*(y_1, \tilde{e}_1, e_1)] f_1(y_1 | \tilde{e}_1) dy_1 - c(\tilde{e}_1), \quad (\text{IC}_1)$$

$$\int [w_1(y_1) + \rho^*(y_1, e_1) U_2^*(y_1, e_1, e_1)] f_1(y_1 | e_1) dy_1 - c(e_1) \geq 0. \quad (\text{IR}_1)$$

Again (IR_1) is implied by the wage bounds together with (IC_1) : zero effort yields nonnegative wages and a nonnegative continuation rent at zero cost, and the agent's optimal choice does at least as well. Constraint (IC_1) embeds a *fixed-point problem* within the principal's program. In equilibrium, a target effort is implemented under w_1 if and only if it is a fixed point of the agent's best-response correspondence: a conjecture $\hat{e}_1$ is consistent with Bayes' rule if and only if the agent would indeed choose $\hat{e}_1$ given w_1 , anticipating the optimal retention and contract in the second period. Denote by Π_1^* the value of the principal's first-period problem.

3 The optimal contract and existence of equilibrium

This section characterizes the contract the principal offers and establishes that a pure-strategy equilibrium exists and is unique. I first validate the first-order approach in Section 3.1, meaning that it is without loss for the principal to replace the agent's incentive-compatibility constraint with his first-order condition. I then characterize the cheapest contract implementing a given effort in Section 3.2. Section 3.3 establishes existence and uniqueness.

3.1 Regularity and the first-order approach

I establish the validity of the first-order approach, meaning that it is without loss for the principal to replace the agent's incentive-compatibility constraint with the agent's first-order condition when designing the contract.⁵ To establish validity, I show that the agent's objective in each period is strictly concave in effort, at every contract and retention rule he may face. I first show that conditional on the strict concavity of the agent's objective in the second period, the agent's continuation incentive in the first period is twice continuously differentiable in first-period effort: the agent's first-period objective involves $U_2^*(y_1, e_1, \hat{e}_1)$, whose dependence on the true effort goes through the agent's own posterior and his resulting second-period effort choice.

Fix any admissible second-period contract w_2 and define

$$u_2(w_2; \theta, y_1, e_1) := \int_{\mathbb{R}} w_2(y_2; y_1) g_{\varepsilon}(y_2 - e_2^*(w_2; y_1, e_1) \theta) dy_2 - c(e_2^*(w_2; y_1, e_1)),$$

the state-conditional continuation rent at that contract, evaluated at the agent's second-period response, so that $U_2 = \mathbb{E}_p[u_2]$. The second-period objective is twice continuously differentiable in e_2 for every admissible contract, with second derivative $\int_{\mathbb{R}} w_2 \partial_{e_2}^2 f_2 dy_2 - c''(e_2)$, so the concavity condition of the next lemma is a statement about a well-defined object.

Lemma 1 (Own-effort regularity and the continuation incentive). *Suppose the second-period objective is strictly concave in e_2 at every admissible contract. Then for every admissible contract and retention rule, $U_2(w_2; y_1, \cdot) \in C^2(\mathbb{R}_+)$ and*

$$\frac{d}{de_1} \mathbb{E}[\rho U_2] = - \int_{\Theta} \int_{\mathbb{R}} \rho(w_1, y_1) u_2(w_2; \theta, y_1, e_1) \theta g'_{\varepsilon}(y_1 - e_1 \theta) dy_1 g_{\theta}(\theta) d\theta.$$

By the envelope theorem the agent's second-period re-optimization drops out, since he is already

⁵The alternative is to impose incentive compatibility globally, a continuum of constraints comparing the target to every other effort level, which is generally intractable. The validity of the first-order approach is therefore a question commonly faced in contracting models with a continuum of action choices.

choosing it optimally. What remains is the effect of his effort on the distribution of output given the state: the rent he collects at each realization is unchanged, because the second-period contract and the retention rule are built on the principal's conjecture. As discussed in the introduction, this differentiability is a key feature that distinguishes the model from [Bhaskar and Roketskiy \(2023\)](#), where full extraction in the second period leaves the agent's continuation with a kink exactly at the conjectured effort. Here limited liability keeps the agent's second-period payoff nonnegative after any deviation, so the quit option never binds and the continuation is smooth.⁶

The analysis characterizes the agent's optimal effort at a given contract by his first-order condition, replacing [\(IC₂\)](#) and [\(IC₁\)](#) in the principal's programs. In general, whether the first-order condition identifies a global maximum may depend on the contract proposed. The classical sufficient conditions (cf. [Rogerson, 1985](#); [Jewitt, 1988](#); [Mirrlees, 1999](#)) do not apply here: strongly log-concave noise fails the convexity-of-the-distribution-function condition even at a known complementarity,⁷ and when effort can be counterproductive the monotone likelihood ratio property of the output distribution with respect to effort fails as well, since more effort raises the likelihood of very low outputs. The solution to static problems with bounded payments has been characterized without the approach ([Chi and Olsen, 2025](#)), and [Azevedo and Wolff \(2025\)](#) restore it through a high reservation utility when the agent is risk averse. Neither route is directly available in this model with dynamic incentives, unknown complementarity, and risk neutrality. I instead directly require the agent's objective to be strictly concave in effort in each period, uniformly over the contract space, and show that this is satisfied when the cost has sufficient curvature.

Assumption 2 (First-order approach). *For $t \in \{1, 2\}$ and all admissible contracts and retention rules, the agent's period- t objective is strictly concave in e_t :*

$$\text{(stage 2)} \quad c''(e_2) > \int_{\mathbb{R}} w_2 \frac{\partial^2 f_2}{\partial e_2^2} dy_2 \quad \forall e_1, e_2 \geq 0, \quad (8)$$

$$\text{(stage 1)} \quad c''(e_1) > \int_{\mathbb{R}} \left[(w_1 + \rho U_2) \frac{\partial^2 f_1}{\partial e_1^2} + 2\rho \frac{dU_2}{de_1} \frac{\partial f_1}{\partial e_1} + \rho \frac{d^2 U_2}{de_1^2} f_1 \right] dy_1 \quad \forall e_1 \geq 0. \quad (9)$$

The following proposition shows [Assumption 2](#) is nonvacuous.

Proposition 1 (Cost curvature validates the first-order approach). *There exists $\alpha^{\text{FOA}} \in (0, +\infty)$ such that [Assumption 2](#) holds for all costs with $c'' > \alpha^{\text{FOA}}$. In particular it holds for the quadratic cost $\frac{\alpha}{2}e^2$ whenever $\alpha > \alpha^{\text{FOA}}$.*

⁶The agent's true continuation at a retained history is $\max\{0, U_2^*\}$, since he may decline the second-period contract. Under limited liability $U_2^* \geq 0$ pointwise, so the maximum coincides with its smooth branch.

⁷At a known complementarity θ , $\partial_e^2 F(y | e) = \theta^2 g'_e(y - e\theta)$, which is positive below the mode of the noise and negative above it, so the distribution function is convex in effort on one side of the mode and concave on the other.

The proposition rests on two observations. First, since payments are bounded, no contract can introduce more curvature into the agent's problem through the wage than the wage cap times the curvature of the output density in effort, which is finite because the complementarity is bounded and the noise is regular. Second, cost curvature dampens the agent's second-period response to what he has learned: his second-period effort moves with the posterior at a rate inversely proportional to the slack in his second-period problem, so the continuation's sensitivity to first-period effort through his own response falls as the cost becomes more convex. The only channel that survives at high curvature is the direct one, through the output density, and the wage cap bounds it. The left-hand sides grow without bound in the curvature while the right-hand sides are bounded or falling, so sufficient curvature dominates.

3.2 The optimal contract

I characterize the optimal contract *before* proving existence. The statement is about the cheapest way to implement a target effort level at a given conjecture, supposing that such effort is implementable.⁸ This ordering allows us to reduce the principal's problem in each period to a single scalar, which greatly simplifies the proof of existence and uniqueness that follows.

As in the canonical contracting model of [Innes \(1990\)](#) with limited liability and risk-neutral parties, the optimal contract is a bonus contract: the principal pays either the full wage cap or zero, depending on the agent's output realization. When effort is always productive, the principal pays the agent when output surpasses an optimally defined threshold. She pays on both tails, above one threshold and below another, when it is possible for effort to be counterproductive. The second tail is the novel feature here, and it arises from a specific failure of the monotone likelihood ratio property of the output distribution with respect to effort: when the support extends below $\theta = 0$, low output is indicative of high effort applied to a low complementarity, and rewarding the lower tail becomes part of the cheapest incentive scheme. Note that zero effort is implemented most cheaply by the null contract, which pays zero everywhere, and the agent's continuation incentive may be enough to induce strictly positive effort under the null contract in the first period. So the threshold description concerns effort that requires strictly positive marginal wage incentive to be undertaken by the agent.

Theorem 1 (Shape of the optimal contract). *Fix a period $t \in \{1, 2\}$ and a strictly positive target effort at which the wage delivers strictly positive marginal incentive.⁹ The cheapest contract*

⁸Even in the first period, this problem is well defined whether or not the agent follows the principal's conjecture at the target.

⁹In period 2 this is the only way to implement positive effort, since nothing follows. In period 1 the agent's first-order

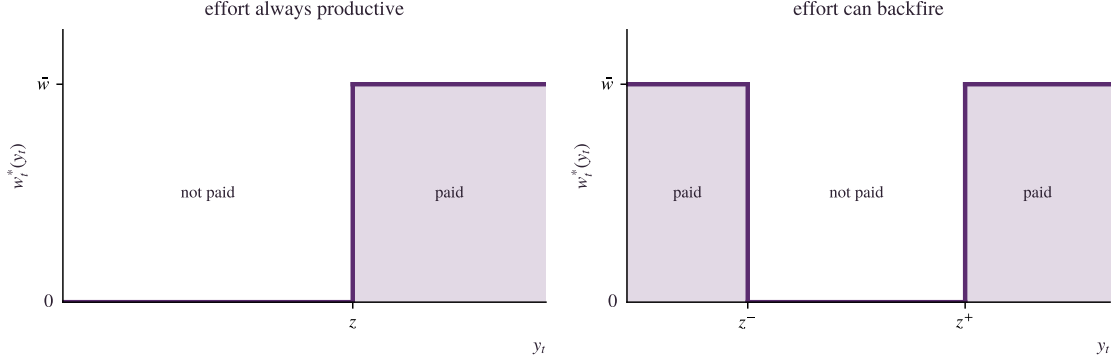


Figure 1: The optimal contract of Theorem 1. Left: when effort is productive at every complementarity, the principal pays the cap above a single threshold z , where high output is the evidence of effort. Right: when effort can be counterproductive, she pays on both tails, $(-\infty, z^-] \cup [z^+, +\infty)$, and nothing on the middle interval. A very low output is then evidence that effort was exerted and backfired.

implementing it pays either $\bar{w}$ or nothing at almost every output, and:

- (i) if $\underline{\theta} \geq 0$, so that effort's marginal product is always positive, it has exactly one threshold z_t , paying $\bar{w}$ on $[z_t, +\infty)$ and nothing below;
- (ii) if $\underline{\theta} < 0$, so that effort can be counterproductive with strictly positive probability, it has exactly two thresholds $z_t^- < z_t^+$, paying $\bar{w}$ on both tails, $(-\infty, z_t^-] \cup [z_t^+, +\infty)$, and nothing on the middle interval.

The shape comes from the cheapest implementation of effort under bounded payments. The wage instrument is bounded on both sides: the agent cannot be made to pay, so incentives must come from rewards, and rewards cannot exceed the cap. Pay therefore spreads over the regions of outputs that are most indicative of effort. Effort shifts the location of output at each complementarity and, because complementarities differ, widens the spread of the output distribution. Both movements push probability mass into the tails: when the complementarity makes effort productive, very high output is far more likely under hard work than under shirking, and the right tail is paid. Very low output is ordinarily evidence of shirking and is never rewarded. It becomes evidence of effort only when effort can be counterproductive, that is, when the support extends below $\theta = 0$. Then hard work applied to a low complementarity pushes output far down, a very low realization signals high effort rather than noise, and the cheapest scheme pays the left tail as well.

condition reads $c'(\tilde{e}_1) = M_w + T$, with M_w the marginal incentive delivered by the wage and T by the continuation. The hypothesis is $M_w > 0$, which holds at every target above the continuation-only effort e_0 .

Note that the presence of the second threshold is not a learning phenomenon. It also arises in the static problem, where learning has no value because what output reveals about the complementarity is never used: it is unknown, and when the support extends below $\theta = 0$ the output distribution fails the monotone likelihood ratio property. The cheapest incentive scheme is then non-monotonic in output. Although that value of information may affect how much effort the principal chooses to implement in the first period, the shape of the contract is the same, so long as the target effort requires strictly positive marginal wage incentive. The proof relies on a counting argument. The noise is variation diminishing at the order Assumption 1 provides, so smoothing cannot create sign changes. The number of crossings of the wage coefficient in the agent's Lagrangian, which is a smoothing by the noise density of an explicit function of the belief, one or two according to the support, is exactly the number of thresholds. At least one crossing is guaranteed, and intuitively so: in the far right tail the likelihood score favors effort without bound, so the weight ends positive and the right tail of the distribution is always paid. It ends positive in the far left tail exactly when effort can be counterproductive, which forces the second crossing.¹⁰

3.3 Existence and uniqueness of equilibrium

This section establishes that the model is well posed: an equilibrium exists, and it is unique once the cost of effort is sufficiently convex. I first establish that the optimal second-period retention rule is a threshold rule in first-period output which moves smoothly with the conjecture.

Lemma 2 (Retention is a threshold rule). *There exists $\alpha^M \geq \alpha^{\text{FOA}}$ such that for all costs with $c'' > \alpha^M$ and all conjectures $\hat{e}_1 > 0$, the optimal retention rule is the cutoff rule*

$$\rho^*(y_1, \hat{e}_1) = \mathbf{1}\{y_1 \geq \underline{y}(\hat{e}_1)\},$$

where $\underline{y}(\hat{e}_1) := \inf\{y_1 : \Pi_2^*(y_1, \hat{e}_1) \geq \pi_A\}$ and $\Pi_2^*(\cdot, \hat{e}_1)$ is strictly increasing on a neighborhood of $\underline{y}(\hat{e}_1)$.

To see why, start from the optimal second-period contract at output y_1 and move to $y'_1 > y_1$. The posterior improves in the likelihood ratio order, so at the implemented effort under y_1 , expected output is higher. The old contract is no longer exactly incentive compatible at the better belief, but incentives can be restored by perturbing the payment regions, at a price: at the optimum a

¹⁰The exact two-threshold count uses that the likelihood score is unbounded in both tails when $\underline{\theta} < 0$, which strong log-concavity of the noise guarantees: $(\log g_\varepsilon)'(u) \leq -m(u - m_\varepsilon)$ to the right of the mode and symmetrically to the left. Under mere log-concavity the score can have finite tail limits, and a cutoff above the smaller of them leaves only one tail paying. That the two tails are the only paid regions uses the order-three total positivity of Assumption 1, as the footnote there explains.

unit of wage bill buys at most $\kappa > 0$ units of incentive, so a unit of incentive costs at least $1/\kappa$, and no admissible change to the contract does better.¹¹ The gain in expected output is first order in the improvement, while restoring incentives costs the incentive deficit divided by κ , which is second order. The old bonus is also paid more often, but sufficient curvature makes the output gain dominate, since it implies that the bonus is not paid too often to begin with. So the principal is better off at the better belief induced by y'_1 , implementing the effort under y_1 , and her optimum at y'_1 cannot do worse. The monotonicity just below the threshold ensures the single crossing.

In Section 2, I note that the principal's first-period program involves solving a fixed point problem for the agent: solving for the second period and imposing Bayes' rule, the optimal effort choice at the first-period contract must be consistent with the principal's conjecture going into the second period. Showing that a fixed point exists is the main obstacle to existence of pure-strategy equilibrium, which fails in similar environments without limited liability (Bhaskar and Roketskiy, 2023).

Theorem 2 (Existence and uniqueness). *There exists $\alpha^E \geq \max\{\alpha^M, \alpha^{\text{FOA}}\}$ such that for all costs with $c'' > \alpha^E$, a pure-strategy perfect Bayesian equilibrium exists and is unique.*

By Assumption 2, the first-order approach is valid, and hence (IC₁) at wage schedule w_1 and conjecture $\hat{e}_1$ is completely characterized by the agent's first-order condition:

$$c'(e_1) = \int_{\mathbb{R}} w_1(y_1) \partial_{e_1} f_1(y_1 | e_1) dy_1 + T(e_1, \hat{e}_1), \quad (10)$$

where $T(e_1, \hat{e}_1)$ denotes the total derivative $\frac{d}{de_1} \mathbb{E}[\rho^* U_2^*]$ of Lemma 1, evaluated at the contract and retention rule selected under the conjecture $\hat{e}_1$ and at the effort e_1 the agent actually takes. By Assumption 2 the best response of the agent is a singleton, and thus, at a given contract, a function of the conjecture defined implicitly by (10). Sufficient curvature implies that this function is a contraction on a compact interval of conjectures, giving a unique fixed point at each admissible contract. By Theorem 1, any optimal contract is indexed by a single scalar κ , so the principal's problem reduces to maximizing her objective over $\kappa \in [0, +\infty]$, where $\kappa = +\infty$ is the null contract.¹² The domain is therefore compact, profit is continuous on it, and a maximizer exists. Uniqueness follows for sufficient effort cost curvature, which implies that the principal's problem is strictly concave in the target effort.

¹¹The old contract pays exactly where the likelihood score is high, so every admissible rearrangement moves pay from high-score to low-score outputs. Moving the threshold is the cheapest of them, and it attains the rate κ up to an error that vanishes as the two beliefs approach.

¹²Cutoffs below zero are dominated: lowering the cutoff below zero enlarges the payment region to outputs at which the score is negative, which raises the wage bill and lowers the incentive the wage delivers.

The uniqueness of equilibrium licenses the comparison of Section 4, where I compare the unique equilibrium effort in the first period to that of a manipulation-free benchmark.

4 The belief-manipulation channel

In this section, I compare the equilibrium characterized in Section 3 with that of a manipulation-free benchmark. In particular, I compare the equilibrium first-period effort under the two regimes across a family of priors. I say that there is *over-experimentation* at a prior when the equilibrium first-period effort exceeds the benchmark's, and *under-experimentation* when it falls short: with multiplicative technology the informativeness of first-period output about the complementarity is increasing in effort, so more first-period effort is a more informative experiment about the complementarity going into the second period.¹³ I first present the benchmark in Section 4.1. I then decompose the manipulation term in Section 4.2. Section 4.3 states the comparative static in the prior and the intuition behind it, and Section 4.4 shows that the curvatures for which the two forces oppose form a nonempty set when the noise is large enough.

4.1 The manipulation-free benchmark

The benchmark model is identical to that presented in Section 2, except that the agent does not account for the effect of his first-period effort choice on his continuation value. This particular benchmark allows us to isolate the effect of the dynamic incentive of the agent to manipulate the principal's belief, while preserving the three key frictions of the model: moral hazard, limited liability, and limited commitment.¹⁴

In the first period, given the contract $w_1(\cdot)$, the agent's problem is equivalent to¹⁵

$$\max_{e_1 \geq 0} \int w_1(y_1) f_1(y_1|e_1) dy_1 - c(e_1). \quad (11)$$

¹³At effort $e > 0$ the statistic $y_1/e = \theta + \varepsilon_1/e$ is sufficient for θ , a Gaussian location experiment with noise variance σ^2/e^2 ; for $e' > e$ the experiment at e is obtained from the one at e' by adding Gaussian noise of variance $\sigma^2/e^2 - \sigma^2/e'^2$, drawn independently of θ , so the experiment at e' is more informative in the Blackwell order.

¹⁴Other natural benchmarks include the following. Full commitment maintains the agent's dynamic incentive, which is accounted for by the principal ex ante: there is no reoptimization or interim belief to manipulate. Observable effort removes the agent's ability to manipulate beliefs and removes the rents that generate the retention incentive. A planner removes all frictions at once.

¹⁵There are several interpretations of this formulation. First, the agent could simply be myopic. Second, he could be forward-looking but naive in that he does not internalize the effect of his actions today on his future contract terms. And lastly, the agent could be both forward-looking and sophisticated but simply one of many similar agents who share the same underlying complementarity. In this last case, he is informationally small and does not affect the principal's future belief, whereas he is informationally large in the single-agent model.

Similar to Theorem 2, the existence of a pure-strategy equilibrium and its uniqueness rely on sufficient convexity of the agent's effort cost. This ensures a unique best response of the agent at each contract in the optimal class and that, given this, a unique contract maximizes the principal's objective.¹⁶

Proposition 2 (Benchmark equilibrium). *There exists $\alpha^B \in (0, +\infty)$ such that for all costs with $c'' > \alpha^B$, a pure-strategy perfect Bayesian equilibrium of the benchmark game exists and is unique.*

The comparison between the equilibrium effort of the model and that of the benchmark is then well defined.

4.2 Decomposition of the manipulation term

Section 3.3 defined $T(e_1, \hat{e}_1)$, the marginal value to the agent of his continuation with respect to first-period effort, the manipulation term. Write $u_2^*(\theta', y_1; \hat{e}_1)$ for the second-period rent of an agent whose complementarity is θ' , after first-period output y_1 , under the contract and the retention rule the principal sets at conjecture $\hat{e}_1$.

Lemma 3 (Continuation incentive decomposition). *Let $c'' > \alpha^M$.*

$$T(e_1, \hat{e}_1) = \int_{\Theta} \theta' \left[u_2^*(\theta', \underline{y}(\hat{e}_1); \hat{e}_1) g_{\varepsilon}(\underline{y}(\hat{e}_1) - e_1 \theta') + \int_{\underline{y}(\hat{e}_1)}^{+\infty} \frac{\partial u_2^*(\theta', y_1; \hat{e}_1)}{\partial y_1} g_{\varepsilon}(y_1 - e_1 \theta') dy_1 \right] g_{\theta}(\theta') d\theta'. \quad (12)$$

The decomposition follows from Lemma 1 with the threshold retention rule substituted, and integrating by parts in y_1 . The first term of the decomposition is an extensive, *retention* effect. It captures the value of being pushed just into the retention region, a discrete jump in value for the agent. The second term is an intensive, *wage* effect. It evaluates, given retention, how the agent's second-period rent evolves when the realized output changes the contract through the principal's posterior belief. While the first term is strictly positive, the sign of the second term is ambiguous in general. This continuation incentive is the main object we study below to characterize the effect of the belief manipulation channel on equilibrium effort.

¹⁶The principal's problem is simpler here, since removing the dynamic incentive of the agent eliminates the fixed point problem.

4.3 The comparative static in the prior

To conduct the comparative static, I first define a family of priors indexed by a single parameter γ and ordered by the likelihood ratio.¹⁷ This parameter allows us to measure how optimism in the initial belief influences the agent's incentive to manipulate the principal's belief upwards or downwards as compared to the benchmark.

Definition 1 (MLR-ordered family on fixed support). *Fix a log-concave base density g_0 on $\Theta = [\underline{\theta}, \bar{\theta}]$. For $\gamma \in \mathbb{R}$ define the exponentially tilted prior*

$$g_\theta(\theta \mid \gamma) = \frac{e^{\gamma\theta} g_0(\theta)}{\int_{\Theta} e^{\gamma\theta'} g_0(\theta') d\theta'}.$$

For $\gamma_2 > \gamma_1$ the ratio $g_\theta(\theta \mid \gamma_2)/g_\theta(\theta \mid \gamma_1)$ is proportional to $e^{(\gamma_2 - \gamma_1)\theta}$ and hence strictly increasing in θ , so the family is strictly ordered by the likelihood ratio. Intuitively, as γ moves from $-\infty$ to $+\infty$, the prior is initially concentrated near $\underline{\theta}$, then spreads out, and eventually concentrates near $\bar{\theta}$.¹⁸ The priors drawn in Figure 2 below are members of this family.

Assumption 3 (Gaussian noise, quadratic cost, informative prior). *The following are maintained for the remainder of the paper and are not restated in the results below.*

- (i) Gaussian noise: $\varepsilon_t \sim N(0, \sigma^2)$.
- (ii) Quadratic cost: $c(e) = \alpha e^2/2$, where $\alpha > \alpha^E(\gamma)$ for every $\gamma \in \mathbb{R}$, with $\alpha^E(\gamma)$ the threshold of Theorem 2 at prior γ .
- (iii) Strongly log-concave base: $g_0 = e^{-\psi}$ on Θ with $\psi'' \geq \varsigma > 0$.

Conditions (i) and (ii) are for tractability. Under a general noise density, two prior-output pairs induce posteriors that need not be comparable, since the MLR order is partial. Fixing the effort conjecture, I show in Lemma 4 that Gaussian noise allows us to express the posterior as an exponential tilt of a modified base prior, where the tilt parameter is an affine function of the prior index and the output realization.¹⁹ Similarly, the restriction to quadratic cost allows us to index the family of cost functions by a single parameter. The restriction on the parameter is taken uniformly

¹⁷This is the order in which one belief is more favorable than another in the sense of Milgrom (1981). It implies first-order stochastic dominance but is not implied by it.

¹⁸I adopt an exponential-tilt family rather than a location-shift family because the complementarity parameter space is compact: a location shift would move the support, whereas the tilt preserves it.

¹⁹Among additive noise structures, the Gaussian family is the only one with this property: a single index requires $\partial^2 \log g_\varepsilon(y_1 - \hat{e}_1\theta)/\partial y_1 \partial \theta$ to factor, which forces the log-curvature to be of the form $-Ce^{ku}$, and $k \neq 0$ is excluded by the floor m of Assumption 1, leaving the Gaussian. Comparative statics on a partially ordered set of beliefs is a different problem from the one solved here, and I do not attempt it.

over the family of priors, so that the equilibrium exists and is unique at every prior in the family I compare. Condition (iii) says that the principal does not enter the relationship without information, in that the prior is sufficiently concentrated. This allows us to get a handle on how the wage effect moves with posterior index, and sufficient concentration ensures that it tracks that of the known complementarity at the posterior mean, in particular in the number of sign changes and whether a ratchet effect appears.

Lemma 4 (The posterior depends on a single index). *Fix the principal's conjecture $\hat{e}_1$, and define the modified base and the index*

$$\tilde{g}_0(\theta) := g_0(\theta) e^{-\hat{e}_1^2 \theta^2 / 2\sigma^2}, \quad b := \gamma + \frac{\hat{e}_1}{\sigma^2} y_1.$$

Under Assumption 3(i), the posterior at prior γ after first-period output y_1 is the tilt of $\tilde{g}_0$ at b :

$$p(\theta \mid y_1; \gamma) \propto e^{b\theta} \tilde{g}_0(\theta). \quad (13)$$

Fixing the conjecture, the retention rule is a threshold in the index: the agent is retained if and only if $b \geq \bar{b}$, with $\bar{b}$ fixed by π_A , and the second-period contract at a retained history depends on the history only through b . This index is the object the appendix works with throughout.

Retention regimes. Let $\pi^- := \Pi_2(\delta_{\underline{\theta}})$ and $\pi^+ := \Pi_2(\delta_{\bar{\theta}})$ denote the principal's second-period profit when the complementarity is known with certainty to be the lowest and the highest possible. Where the outside option sits relative to $[\pi^-, \pi^+]$ determines which of three retention regimes obtains.

- (P1) $\pi_A \leq \pi^-$ (*always retain*): replacement is never worthwhile at any posterior, and the retention margin is absent.
- (P2) $\pi^- < \pi_A < \pi^+$ (*credible replacement*): replacement is worthwhile at poor posteriors and not at good ones, so the retention decision turns on what the principal has learned.
- (P3) $\pi_A \geq \pi^+$ (*never retain*): the outside option exceeds the retention value at every posterior and the relationship does not form.

Write γ_L for the prior at which the relationship forms: the smallest γ at which the principal's value from initiating reaches $2\pi_A$. The theorem is stated for (P2), where both effects are present. (P1) is discussed afterward, and in (P3) the relationship does not form, so there is no equilibrium effort to compare.

Theorem 3 (Over- and under-experimentation). *There exists $\sigma^* \in (0, +\infty)$ such that, for any fixed $\sigma \geq \sigma^*$ and $\pi_A \in (\pi^-, \pi^+)$, there exists $\varsigma^* \in (0, +\infty)$ such that for all $\varsigma \geq \varsigma^*$:*

- (i) *if $\alpha < \bar{\alpha}$, there is a unique γ^* such that there is over-experimentation for all $\gamma \in [\gamma_L, \gamma^*)$ and under-experimentation for all $\gamma \in (\gamma^*, +\infty)$;*
- (ii) *if $\alpha \geq \bar{\alpha}$, there is over-experimentation for all $\gamma \in [\gamma_L, +\infty)$.*

When the prior is sufficiently pessimistic, so that the agent is just retained, the retention effect dominates, leading to *more* experimentation than in the benchmark: the agent will likely not be retained in the second period, so he places more weight on getting something at all rather than how his rent will change conditional on retention. When the cost curvature is not too high and the prior sufficiently optimistic, the principal can get away with *tightening* the contract upon receiving good news while still providing sufficient incentives for the agent to increase effort. In that case, the wage force constitutes a *ratchet effect*, and both forces balance. As the prior becomes increasingly optimistic, concentrating on the upper endpoint of the support, the agent is fired with vanishingly small probability, and the ratchet effect leads to *less* experimentation than in the benchmark. When the cost curvature is very high, incentives are costlier, and the principal optimally *loosens* the contract with good news at every prior in the family. There, the wage force constitutes a *reverse ratchet effect*, and both wage and retention forces push in the same direction, leading to *more* experimentation at *every* prior. The result is depicted in Figure 2.

In regime (P1) the retention force is absent and the manipulation term is the wage effect alone, so whether there is over- or under-experimentation is driven by the shape of the rent alone. The single peak of that rent, established in the proof of Theorem 3, then gives at most one crossing, from over- to under-experimentation.

4.4 When both forces are present

Note that case (i) of Theorem 3, where the retention force opposes the ratchet effect, relies on the cost curvature being sufficiently low, whereas existence and uniqueness require that the curvature be sufficiently high. I establish in the following result that when the prior is sufficiently concentrated and the output sufficiently noisy, there exists a nonempty interval of curvatures such that case (i) obtains, and hence it is nonvacuous.

Proposition 3 (The two-sided case is nonempty under enough noise). *There exist σ^{**} and ς^{**} such that for all $\sigma \geq \sigma^{**}$ and $\varsigma \geq \varsigma^{**}$, $\sup_{\gamma \in \mathbb{R}} \alpha^E(\gamma) < \bar{\alpha}$.*

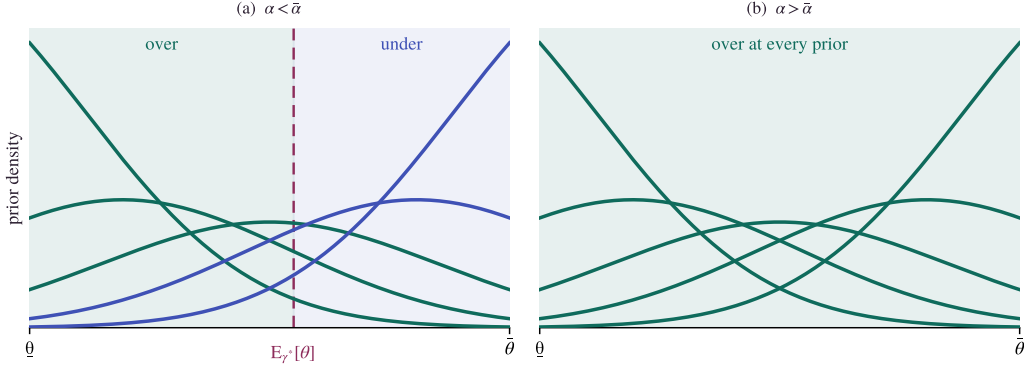


Figure 2: The two cases of Theorem 3. Each curve is a prior from the tilted family, ordered from pessimistic to optimistic. A prior drawn in teal induces over-experimentation and one drawn in indigo induces under-experimentation. In (a), the two-sided case $\alpha < \bar{\alpha}$, the sign changes once, at the threshold γ^* whose mean is marked. In (b), performance pay is scarce, $\alpha > \bar{\alpha}$, and the agent over-experiments at every prior.

At a known complementarity both thresholds fall as the noise grows, but the existence threshold falls faster, so for σ large enough the curvature needed for existence lies below the curvature at which the ratchet appears. A concentrated enough base then ensures the same relationship holds across the family it generates, since what the assumption requires is the supremum of the existence thresholds over its members.

5 Application: technology adoption

One application is to the adoption of a new technology. Suppose that before the game starts, the principal must decide whether to adopt the technology. The investment is irreversible and costs the principal $k > 0$. If the principal adopts, then the game is as described in Section 2. If she does not adopt, the game is identical except that the principal's outside option is zero and the complementarity is known and equal to one:

$$y_t = e_t + \varepsilon_t.$$

So the investment provides the principal with both the option to fully automate, valued at π_A in each period, as well as the possibility to use the technology alongside the agent, where the complementarity is unknown following installation. Without the technology there is nothing to learn, so the agent has no belief incentive and the model and the benchmark coincide.

Write $\Pi(\gamma)$ for the principal's equilibrium value in the adopted game at prior γ , and Π^0 for her

value in the game without the technology. Adoption occurs if and only if $k \leq \Delta(\gamma) := \Pi(\gamma) - \Pi^0$, so a higher Δ means weakly more adoption. Let Π^m and Δ^m denote the corresponding objects in the manipulation-free benchmark. The belief-manipulation channel then determines whether the technology is worth more or less to the principal than in the benchmark, and hence who adopts.

Proposition 4 (Technology adoption). *With σ^* and ς^* as in Theorem 3, for all $\sigma \geq \sigma^*$ and $\varsigma \geq \varsigma^*$:*

- (i) *if $\alpha < \bar{\alpha}$, there is a unique γ^{**} such that there is weakly more adoption than in the benchmark at every prior below γ^{**} and weakly less at every prior above it;*
- (ii) *if $\alpha \geq \bar{\alpha}$, there is weakly more adoption than in the benchmark at every prior.*

The threshold γ^{**} is the prior at which the principal's value under manipulation crosses her benchmark value, and the proof bounds its distance from γ^* . Below it, the continuation delivers part of the effort incentive for free, so the principal implements the effort she wants with a lower wage bill, and the technology is worth more to her than in the benchmark. Above it, the continuation works against her, every target costs more, and the technology is worth less. Replacement risk thus raises the principal's willingness to invest in workers near the replacement margin and lowers it for workers with a record to protect.

6 Conclusion

This paper studies dynamic contracting when the complementarity between the agent and a task is commonly unknown, under moral hazard, limited commitment, and limited liability. There are three main findings. First, optimal spot contracts are one- or two-tailed: the principal pays a bonus on a single tail of output when effort is always productive, and on two tails when effort can be counterproductive, with no payment in between. Second, for sufficient curvature of the effort cost, a pure-strategy equilibrium exists and is unique, which the limited-liability rent ensures. Third, the agent has a dynamic incentive to manipulate the principal's belief about the complementarity through his private action choice, and this pushes towards more learning about the underlying state at pessimistic priors and less learning once the prior becomes sufficiently optimistic. Whether the channel raises or lowers learning, and with it the value of a new technology to the principal, therefore depends on how secure the agent using it is.

A natural extension would let the relationship run over an infinite horizon, with the principal choosing when to end it and draw a new agent. Replacement is then a stopping problem, and because it is observed, continuation play can condition on the principal's past replacements, which

may sustain behavior that the two-period game cannot. The questions are which contracts and replacement policies can be sustained, and how much is learned about each agent compared to the single-agent case.

References

- Eduardo M. Azevedo and Ilan Wolff. Broad validity of the first-order approach in moral hazard. Working paper, Wharton; arXiv:2506.18873, 2025.
- Mark Bagnoli and Ted Bergstrom. Log-concave probability and its applications. *Economic Theory*, 26(2):445–469, 2005.
- Dirk Bergemann and Ulrich Hege. The financing of innovation: Learning and stopping. *RAND Journal of Economics*, 36(4):719–752, 2005.
- V. Bhaskar and George J. Mailath. The curse of long horizons. *Journal of Mathematical Economics*, 82:74–89, 2019.
- V. Bhaskar and Nikita Roketskiy. The ratchet effect: A learning perspective. *Journal of Economic Theory*, 214:105762, 2023.
- Christer Borell. Complements of Lyapunov’s inequality. *Mathematische Annalen*, 205:323–331, 1973.
- Herm Jan Brascamp and Elliott H. Lieb. On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation. *Journal of Functional Analysis*, 22(4):366–389, 1976.
- Chang Koo Chi and Trond E. Olsen. Optimal incentive contracts with bonus caps. *RAND Journal of Economics*, 56(1):55–73, 2025.
- Gonzalo Cisternas. Two-sided learning and the ratchet principle. *Review of Economic Studies*, 85(1):307–351, 2018.
- Peter DeMarzo and Yuliy Sannikov. Learning, termination, and payout policy in dynamic incentive contracts. *Review of Economic Studies*, 84(1):182–236, 2017.
- Mathias Dewatripont, Ian Jewitt, and Jean Tirole. The economics of career concerns, part I: Comparing information structures. *Review of Economic Studies*, 66(1):183–198, 1999.

- Mehmet Ekmekci, Leandro Gorno, Lucas Maestri, Jian Sun, and Dong Wei. Learning from manipulable signals. *American Economic Review*, 112(12):3995–4040, 2022.
- Xavier Freixas, Roger Guesnerie, and Jean Tirole. Planning under incomplete information and the ratchet effect. *Review of Economic Studies*, 52(2):173–191, 1985.
- Drew Fudenberg and Jean Tirole. A “signal-jamming” theory of predation. *RAND Journal of Economics*, 17(3):366–376, 1986.
- Drew Fudenberg and Jean Tirole. Moral hazard and renegotiation in agency contracts. *Econometrica*, 58(6):1279–1319, 1990.
- Hong Gao and Haibo Xu. Learning, belief manipulation and optimal relationship termination. *Economics Letters*, 190:109108, 2020.
- Dino Gerardi and Lucas Maestri. Dynamic contracting with limited commitment and the ratchet effect. *Theoretical Economics*, 15(2):583–623, 2020.
- Faruk Gul. Unobservable investment and the hold-up problem. *Econometrica*, 69(2):343–376, 2001.
- Marina Halac, Navin Kartik, and Qingmin Liu. Optimal contracts for experimentation. *Review of Economic Studies*, 83(3):1040–1091, 2016.
- Bengt Holmström. Managerial incentive problems: A dynamic perspective. *Review of Economic Studies*, 66(1):169–182, 1999.
- Johannes Hörner and Larry Samuelson. Incentives for experimenting agents. *RAND Journal of Economics*, 44(4):632–663, 2013.
- Robert D. Innes. Limited liability and incentive contracting with ex-ante action choices. *Journal of Economic Theory*, 52(1):45–67, 1990.
- Ian Jewitt. Justifying the first-order approach to principal-agent problems. *Econometrica*, 56(5): 1177–1190, 1988.
- Samuel Karlin. *Total Positivity, Volume I*. Stanford University Press, Stanford, CA, 1968.
- Byunghoon Kim. Signal jamming by shirking under replacement risk. Working paper, University of Pennsylvania, 2026.

- Chris A. J. Klaassen, Philip J. Mokveld, and Bert van Es. Squared skewness minus kurtosis bounded by 186/125 for unimodal distributions. *Statistics & Probability Letters*, 50(2):131–135, 2000.
- Jean-Jacques Laffont and Jean Tirole. Comparative statics of the optimal dynamic incentive contract. *European Economic Review*, 31(4):901–926, 1987.
- Jean-Jacques Laffont and Jean Tirole. The dynamics of incentive contracts. *Econometrica*, 56(5):1153–1175, 1988.
- Gustavo Manso. Motivating innovation. *Journal of Finance*, 66(5):1823–1860, 2011.
- Paul R. Milgrom. Good news and bad news: Representation theorems and applications. *Bell Journal of Economics*, 12(2):380–391, 1981.
- James A. Mirrlees. The theory of moral hazard and unobservable behaviour: Part I. *Review of Economic Studies*, 66(1):3–21, 1999.
- Julien Prat and Boyan Jovanovic. Dynamic contracts when the agent’s quality is unknown. *Theoretical Economics*, 9(3):865–914, 2014.
- William P. Rogerson. The first-order approach to principal-agent problems. *Econometrica*, 53(6):1357–1367, 1985.
- H. F. Weinberger. A characterization of the Pólya frequency functions of order 3. *Applicable Analysis*, 15(1-4):53–69, 1983.

A Proofs

Notation. Throughout the appendix, $\theta_{\max} := \max\{|\underline{\theta}|, \bar{\theta}\}$ and $\tau_{\max}^2 := (\bar{\theta} - \underline{\theta})^2/4$ bound the mean and variance of any distribution on Θ ; $\nu_1 := \|g'_\varepsilon\|_{L^1}$ and $\nu_2 := \|g''_\varepsilon\|_{L^1}$, finite by Assumption 1; m and M are the curvature bounds of Assumption 1; $\bar{e} := \bar{w}\theta_{\max}\nu_1/\inf c''$ bounds the effort implemented in either period, since no wage delivers marginal incentive above $\bar{w}\theta_{\max}\nu_1$; and $g_\varepsilon^{(k)}$ denotes the k -th derivative of g_ε . For a period- t output density $f_t(\cdot | e)$, $\ell_t(y) := \partial_e f_t(y | e)/f_t(y | e)$ is its likelihood score in effort. φ and Φ denote the standard normal density and distribution function, and $\bar{\Phi} := 1 - \Phi$.

Proof of Lemma 1 (Own-effort regularity and the continuation incentive)

Throughout, $C^2(\mathbb{R}_+)$ is understood with right-derivatives at $e = 0$.

Part (i): smoothness. Since Θ is compact and $g_\varepsilon \in C^2(\mathbb{R})$, the density f_2 and its first two e_1 -derivatives are obtained by differentiating $g_\varepsilon(y_1 - e_1\theta)$ under the θ -integral in (1), the derivatives $-\theta g'_\varepsilon(y_1 - e_1\theta)$ and $\theta^2 g''_\varepsilon(y_1 - e_1\theta)$ being dominated on bounded sets of e_1 by integrable functions of θ .²⁰ Hence the second-period objective $\int w_2 f_2 dy_2 - c(e_2)$ is C^2 in (e_1, e_2) . By the strict concavity assumed in the lemma, $e_2^*(w_2; y_1, e_1)$ is the unique solution of the stage-2 first-order condition

$$c'(e_2) = \int w_2(y_2; y_1) \partial_{e_2} f_2 dy_2, \quad (14)$$

whose e_2 -derivative $\int w_2 \partial_{e_2}^2 f_2 dy_2 - c''(e_2)$ is strictly negative, so by the implicit function theorem e_2^* is C^1 in e_1 . The agent's maximized continuation $U_2(w_2; y_1, e_1)$, the objective evaluated at $e_2^*(w_2; y_1, e_1)$, is then C^2 , as the statement claims: by the envelope condition its first derivative is the e_1 -derivative of the objective evaluated at e_2^* , which is C^1 because the objective is C^2 and e_2^* is C^1 .

Part (ii): the incentive identity. Write the continuation state-conditionally, using $U_2 = \mathbb{E}_p[u_2]$, the posterior (1), and Fubini's theorem, which applies since u_2 is bounded by $\bar{w}$:

$$\mathbb{E}[\rho U_2] = \int_{\Theta} \int_{\mathbb{R}} \rho(y_1; \hat{e}_1) u_2(w_2; \theta, y_1, e_1) g_\varepsilon(y_1 - e_1\theta) dy_1 g_\theta(\theta) d\theta,$$

Differentiating in e_1 under the integral is justified by a dominating function of the same form as $\bar{h}_1$. Since the integral is now over $y_1 \in \mathbb{R}$ rather than the compact Θ , that function must be integrable in y_1 , and it is, because $|g'_\varepsilon|$ is integrable and, by log-concavity, eventually monotone in each tail, so its supremum over a bounded window is attained at an endpoint. Effort e_1 enters twice, through $g_\varepsilon(y_1 - e_1\theta)$ and, inside u_2 , through $e_2^*(w_2; y_1, e_1)$. The e_2^* -channel contributes

$$\int \frac{\partial e_2^*}{\partial e_1} \rho f_1(y_1) \mathbb{E}_{p(\cdot|y_1; e_1)} \left[\frac{\partial u_2}{\partial e_2} \right] dy_1 = 0,$$

by the envelope theorem: the inner posterior expectation is the agent's stage-2 first-order condition (14). The remaining term, from differentiating $g_\varepsilon(y_1 - e_1\theta)$, is the displayed expression. $\square$

²⁰Explicitly, by $\bar{h}_k(\theta) := |\theta|^k \sup_{|u| \leq \delta|\theta|} |g_\varepsilon^{(k)}(y_1 - e^0\theta - u)| g_\theta(\theta)$ for $k = 1, 2$ on $|e_1 - e^0| < \delta$. The same domination justifies every differentiation under a θ -integral in the appendix, and is not repeated.

Proof of Proposition 1 (Cost curvature validates the first-order approach)

Claim 1. For any probability density q on Θ , $|\mathbb{E}_q \theta| \leq \theta_{\max}$ and $\text{Var}_q \theta \leq \tau_{\max}^2$. In particular this holds for the prior and for every posterior, which share the support Θ because the noise has full support.

Proof. A mean lies in the convex hull of the support. The variance bound is Popoviciu's inequality. $\square$

Claim 2. Let q be a probability density on Θ and let $f(y | e) = \int_{\Theta} g_{\varepsilon}(y - e\tilde{\theta}) q(\tilde{\theta}) d\tilde{\theta}$. For every $e \geq 0$,

$$1. \int_{\mathbb{R}} \left| \frac{\partial f}{\partial e} \right| dy \leq \nu_1 \theta_{\max},$$

$$2. \int_{\mathbb{R}} \left| \frac{\partial^2 f}{\partial e^2} \right| dy \leq \nu_2 [\theta_{\max}^2 + \tau_{\max}^2].$$

Proof. Differentiating f with respect to e ,

$$\frac{\partial f}{\partial e} = - \int_{\Theta} \tilde{\theta} g'_{\varepsilon}(y - e\tilde{\theta}) q(\tilde{\theta}) d\tilde{\theta}, \quad (15)$$

$$\frac{\partial^2 f}{\partial e^2} = \int_{\Theta} \tilde{\theta}^2 g''_{\varepsilon}(y - e\tilde{\theta}) q(\tilde{\theta}) d\tilde{\theta}, \quad (16)$$

differentiating under the integral as in the proof of Lemma 1. Taking absolute values, integrating in y , and switching the order of integration by Tonelli's theorem, translation invariance gives $\int_{\mathbb{R}} |g_{\varepsilon}^{(k)}(y - e\tilde{\theta})| dy = \nu_k$ for every $\tilde{\theta}$, so

$$\begin{aligned} \int_{\mathbb{R}} \left| \frac{\partial f}{\partial e} \right| dy &\leq \nu_1 \int_{\Theta} |\tilde{\theta}| q(\tilde{\theta}) d\tilde{\theta}, \\ \int_{\mathbb{R}} \left| \frac{\partial^2 f}{\partial e^2} \right| dy &\leq \nu_2 \int_{\Theta} \tilde{\theta}^2 q(\tilde{\theta}) d\tilde{\theta}. \end{aligned}$$

Finally $\mathbb{E}_q[\tilde{\theta}^2] = (\mathbb{E}_q \tilde{\theta})^2 + \text{Var}_q \tilde{\theta} \leq \theta_{\max}^2 + \tau_{\max}^2$ and $\mathbb{E}_q |\tilde{\theta}| \leq \theta_{\max}$ by Claim 1. $\square$

Claim 3. For every $e_1, e_2 \geq 0$, $y_1 \in \mathbb{R}$, and all admissible w_2 ,

$$\int w_2 \frac{\partial^2 f_2}{\partial e_2^2} dy_2 \leq \bar{w} \nu_2 [\theta_{\max}^2 + \tau_{\max}^2] =: \bar{M}'_2 < \infty.$$

Proof. Since $0 \leq w_2 \leq \bar{w}$, the left side is at most $\bar{w} \int |\partial_{e_2}^2 f_2| dy_2$, and Claim 2 with $q = p(\cdot | y_1, e_1)$ bounds the integral by $\nu_2 [\theta_{\max}^2 + \tau_{\max}^2]$. $\square$

Claim 4. Suppose $c''(e_2) - \int w_2 \partial_{e_2}^2 f_2 dy_2 \geq \underline{D} > 0$, uniformly in e_1, e_2, w_2 . Then:

1. for all y_1 , $0 \leq U_2(w_2; y_1, e_1) \leq \bar{w}$;
2. for all e_1 , $\int \left| \frac{dU_2}{de_1} \right| |\partial_{e_1} f_1| dy_1 \leq \bar{w} J_1 < \infty$, where $J_1 := \sup_{e_1} \int (\int |\partial_{e_1} f_2| dy_2) |\partial_{e_1} f_1| dy_1$;
3. for all e_1 , $\int \left| \frac{d^2 U_2}{de_1^2} \right| f_1 dy_1 \leq \bar{w} J_2 + \frac{\bar{w}^2 \Lambda_{\text{FOA}}}{\underline{D}} < \infty$, where $J_2 := \sup_{e_1} \int (\int |\partial_{e_1}^2 f_2| dy_2) f_1 dy_1$ and $\Lambda_{\text{FOA}} := \sup_{e_1} \int (\int |\partial_{e_1} \partial_{e_2} f_2| dy_2)^2 f_1 dy_1$.

Proof. The first statement is immediate from limited liability and $c(0) = 0$: retention yields $U_2 \geq 0$, and $U_2 \leq \bar{w}$ since $w_2 \leq \bar{w}$. At a bonus contract the tighter $U_2 \leq \bar{w} P_A$ holds, with P_A the payment probability.

For the derivative bounds, the claim's hypothesis

$$c''(e_2) - \int w_2 \partial_{e_2}^2 f_2 dy_2 \geq \underline{D} > 0$$

is the stage-2 problem's strict second-order condition, which Part (i) below establishes with $\underline{D} = \alpha - \bar{M}'_2$. Under it the best response $e_2^*(w_2; y_1, e_1)$ is unique. If the marginal incentive at zero effort is nonpositive, $e_2^* = 0$. Output at zero effort is then pure noise, f_2 does not depend on the posterior, and U_2 does not depend on e_1 , so the bounds below hold trivially. Otherwise e_2^* is interior, the envelope theorem applies, and by the implicit function theorem e_2^* is differentiable in e_1 with the denominator below bounded away from zero. By the envelope theorem,

$$\partial_{e_1} U_2 = \int w_2 \partial_{e_1} f_2 dy_2.$$

Differentiating again along $e_2^*(w_2; y_1, e_1)$,

$$\partial_{e_1}^2 U_2 = \int w_2 \partial_{e_1}^2 f_2 dy_2 + \left(\int w_2 \partial_{e_1} \partial_{e_2} f_2 dy_2 \right) \frac{\partial e_2^*}{\partial e_1},$$

where

$$\frac{\partial e_2^*}{\partial e_1} = \frac{\int w_2 \partial_{e_1} \partial_{e_2} f_2 dy_2}{c''(e_2^*) - \int w_2 \partial_{e_2}^2 f_2 dy_2}.$$

Bounding $w_2 \leq \bar{w}$ and $|\partial e_2^* / \partial e_1| \leq (\int |\partial_{e_1} \partial_{e_2} f_2| dy_2) \bar{w} / \underline{D}$ (the last by the stage-2 SOC), so that the second term of $\partial_{e_1}^2 U_2$ is at most $\bar{w}^2 (\int |\partial_{e_1} \partial_{e_2} f_2| dy_2)^2 / \underline{D}$, then integrating against $|\partial_{e_1} f_1|$ (resp. f_1) over y_1 yields the stated forms with $J_1, J_2, \Lambda_{\text{FOA}}$.

It remains to show these are finite. Effort e_1 enters f_2 only through the posterior $p(\cdot | y_1; e_1)$, and differentiating the posterior gives

$$\partial_{e_1} p(\tilde{\theta} | y_1) = p(\tilde{\theta} | y_1) [\mathbf{s}(\tilde{\theta}) - \bar{\mathbf{s}}],$$

where

$$\mathbf{s}(\tilde{\theta}) := -\tilde{\theta} (\log g_\varepsilon)'(y_1 - e_1 \tilde{\theta})$$

is the score of the stage-1 likelihood in e_1 and $\bar{\mathbf{s}} := \mathbb{E}_p[\mathbf{s}]$ is its posterior mean, arising from differentiating the normalizing constant. Since $|\tilde{\theta}| \leq \theta_{\max}$ on the compact support, each inner y_2 -integral is bounded by a constant multiple of

$$\int_{\Theta} |(\log g_\varepsilon)'(y_1 - e_1 \tilde{\theta})| p(\tilde{\theta} | y_1) d\tilde{\theta}$$

for the first derivative, and of its square together with $\int_{\Theta} |(\log g_\varepsilon)''(y_1 - e_1 \tilde{\theta})| p(\tilde{\theta} | y_1) d\tilde{\theta}$ for the second, since differentiating the score once more produces $\tilde{\theta}^2 (\log g_\varepsilon)''$. Integrating against f_1 (resp. $|\partial_{e_1} f_1|$) over y_1 therefore yields finite expected-score and expected-squared-score quantities, the square in Λ_{FOA} being handled by Jensen's inequality: the Fisher information is finite, $\int (\log g_\varepsilon)'^2 g_\varepsilon = \int (-(\log g_\varepsilon)'') g_\varepsilon \leq M$, where the equality is an integration by parts whose boundary term g'_ε vanishes at $\pm\infty$ because $g''_\varepsilon \in L^1$, and the inequality is the curvature bound of Assumption 1. The same bound gives $\int |(\log g_\varepsilon)''| g_\varepsilon \leq M$; while the marginal product $\tilde{\theta}$ is bounded on the compact Θ . Hence $J_1, J_2, \Lambda_{\text{FOA}} < \infty$. $\square$

Claim 5. *For every $e_1 \geq 0$, every retention rule ρ with values in $[0, 1]$, and every admissible first-period contract w_1 , the right-hand side of (9) is at most*

$$\bar{M}_1 := 2\bar{w} \nu_2 [\theta_{\max}^2 + \tau_{\max}^2] + 2\bar{w} J_1 + \bar{w} J_2 + \frac{\bar{w}^2 \Lambda_{\text{FOA}}}{\underline{D}} < \infty,$$

with $J_1, J_2, \Lambda_{\text{FOA}}, \underline{D}$ the finite constants of Claim 4, none of which depends on w_1 or ρ .

Proof. Bound the three terms of (9).

- Term I: $w_1 + \rho U_2 \leq 2\bar{w}$ by limited liability and Claim 4, and $\int |\partial_{e_1}^2 f_1| dy_1 \leq \nu_2 [\theta_{\max}^2 + \tau_{\max}^2]$ by Claim 2 with $q = g_\theta$.
- Term II: by Claim 4, with $\rho \leq 1$, $\int 2\rho |dU_2/de_1| |\partial_{e_1} f_1| dy_1 \leq 2\bar{w} J_1$.
- Term III: by Claim 4, with $\rho \leq 1$, $\int \rho |d^2 U_2/de_1^2| f_1 dy_1 \leq \bar{w} J_2 + \frac{\bar{w}^2 \Lambda_{\text{FOA}}}{\underline{D}}$.

The second-period objects entering $J_1, J_2, \Lambda_{\text{FOA}}$ are those of arbitrary admissible second-period contracts, for which part (i) supplies the slack $\underline{D} \geq \alpha - \bar{M}'_2$. $\square$

Remark 1. The bounds in Claims 3 and 5 use only the wage cap. On the class of contracts the analysis produces, a bonus $w_t = \bar{w} \mathbf{1}_A$ with A a union of at most two tails, they can be tightened:

integrating by parts on each tail, $\int_A \partial_{e_t}^2 f_t dy_t = - \int_{\Theta} \theta^2 g'_\varepsilon(z - e_t \theta) p(\theta) d\theta$ at a cutoff z , so the wage terms in both periods are at most $2\bar{w}\theta_{\max}^2 \sup|g'_\varepsilon|$, which is small when the cutoffs lie in the tails of the noise. Substituting these constants for $\bar{M}'_2$ and for the first term of $\bar{M}_1$ lowers both thresholds on that class.

Part (i). By Claim 3 the right-hand side of (8) is at most $\bar{M}'_2$ for every admissible w_2 , independent of e_2 and of the history, so $\alpha_2^{\text{FOA}} := \bar{M}'_2$ works. At this curvature the stage-2 problem at every history has a unique agent best response.

Part (ii). By Claim 5 the right-hand side of (9) is at most $\bar{M}_1$, independent of e_1 . The threshold is not $\max\{\bar{M}_1, \bar{M}'_2\}$, however, because $\bar{M}_1$ is not free of the curvature: Term III contains $\bar{w}^2 \Lambda_{\text{FOA}} / \underline{D}$, and $\underline{D}$ is the stage-2 slack, at least $\alpha - \bar{M}'_2$ once (8) holds. Writing A for the part of $\bar{M}_1$ that does not involve $\underline{D}$, the stage-1 requirement $\alpha > \bar{M}_1$ reads

$$\alpha > A + \frac{\bar{w}^2 \Lambda_{\text{FOA}}}{\alpha - \bar{M}'_2} \iff (\alpha - A)(\alpha - \bar{M}'_2) > \bar{w}^2 \Lambda_{\text{FOA}},$$

a quadratic in α whose larger root is

$$\alpha^{\text{FOA}} := \frac{(A + \bar{M}'_2) + \sqrt{(A - \bar{M}'_2)^2 + 4\bar{w}^2 \Lambda_{\text{FOA}}}}{2},$$

finite and exceeding $\bar{M}'_2 = \alpha_2^{\text{FOA}}$, so that $\alpha_1^{\text{FOA}} := \alpha^{\text{FOA}}$ delivers part (ii) and, being larger, part (i) as well. The quadratic cost $c(e) = \alpha e^2 / 2$ has $c'' = \alpha$, so it qualifies if and only if $\alpha > \alpha^{\text{FOA}}$. This proves Proposition 1. $\square$

Proof of Theorem 1 (Shape of the optimal contract)

The proof has six steps. Step 1 derives the bang-bang form of the wage from the pointwise Lagrangian. Step 2 defines the function D_t whose zeros are the thresholds. Steps 3–4 write D_t as a smoothing of an explicit weight, so that its crossings are bounded by the sign changes of that weight, first for beliefs that equal zero at the endpoints of the support. Step 5 removes that restriction by approximation. Step 6 makes the counts exact.

Fix a period $t \in \{1, 2\}$, a strictly positive target effort, and the conjecture at which the continuation objects are built, conditional on retention when $t = 2$. All period-2 objects depend on the realized first-period output, which is suppressed in the notation. Write p_t for the period- t belief over Θ , the prior at $t = 1$ and the posterior at $t = 2$, which is log-concave and C^1 on the interior of

Θ ;²¹ write $e_t > 0$ for the target effort and $f_t, \partial_e f_t$ for the output density of Section 3.2 and its effort derivative, both evaluated at e_t and at the belief p_t .

Step 1: the wage is bang-bang on the score.

Consider period 1. Period 2 is identical with the continuation terms absent. The cheapest contract implementing the target e_1 minimizes the expected wage subject to the agent's first-order condition and the wage bounds, with $\rho(y_1) \in \{0, 1\}$ the retention rule and $U_2(y_1)$ the agent's continuation rent when retained, both with their remaining arguments suppressed. Attaching multiplier λ_1 to the first-order condition and $\underline{\eta}(\cdot) \geq 0, \bar{\eta}(\cdot) \geq 0$ to the liability floor and the cap,²²

$$\begin{aligned} \mathcal{L}(w_1; \lambda_1, \underline{\eta}, \bar{\eta}) = & - \int w_1(y_1) f_1(y_1) dy_1 \\ & + \lambda_1 \left[\int (w_1(y_1) + \rho(y_1) U_2(y_1)) \partial_e f_1(y_1) dy_1 - c'(e_1) \right] \\ & + \int \underline{\eta}(y_1) w_1(y_1) dy_1 + \int \bar{\eta}(y_1) [\bar{w} - w_1(y_1)] dy_1. \end{aligned}$$

Because w_1 enters $\mathcal{L}$ only through the integrals above, the first-order condition may be taken pointwise in y_1 : differentiating the integrand with respect to the value $w_1(y_1)$ and setting the result to zero gives

$$-f_1(y_1) + \lambda_1 \partial_e f_1(y_1) + \underline{\eta}(y_1) - \bar{\eta}(y_1) = 0,$$

with complementary slackness $\underline{\eta}(y_1) w_1(y_1) = 0$ and $\bar{\eta}(y_1) [\bar{w} - w_1(y_1)] = 0$. The multiplier is strictly positive: by hypothesis the wage must deliver strictly positive marginal incentive, so lowering the wage schedule is not free, and $\lambda_1 > 0$. Set $\kappa_1 := 1/\lambda_1 > 0$ and

$$D_1(y_1) := \partial_e f_1(y_1) - \kappa_1 f_1(y_1),$$

so that stationarity reads $\lambda_1 D_1(y_1) = \bar{\eta}(y_1) - \underline{\eta}(y_1)$.

Since $\bar{w} > 0$, the floor and the cap cannot bind at the same y_1 . Where $D_1(y_1) > 0$, necessarily $\bar{\eta}(y_1) > \underline{\eta}(y_1) \geq 0$, so the cap binds and $w_1(y_1) = \bar{w}$. Where $D_1(y_1) < 0$, the floor binds and $w_1(y_1) = 0$. At an interior wage both multipliers are slack, forcing $D_1(y_1) = 0$. In each case the constraint that binds is the one that helps: the principal pays the most it may where the coefficient on the wage is positive and the least where it is negative.

On the level set $\{D_1 = 0\} = \{\ell_1 = \kappa_1\}$ the principal is indifferent pointwise, so the wage there

²¹The prior by Assumption 1. The posterior because it is proportional to $g_\varepsilon(y_1 - e_1\theta) g_\theta(\theta)$, a product of functions log-concave in θ , the first by composition of a log-concave function with an affine map.

²²Participation does not appear: the constraint never binds, so its multiplier is zero by complementary slackness.

is undetermined by stationarity alone. Step 6 shows this set is null, so that the cheapest contract is unique up to a null set and pays $\bar{w}$ on $\{D_1 \geq 0\}$ and nothing elsewhere.

Hence the payment set is $\{D_1 \geq 0\}$ and the number of thresholds is $S^-(D_1)$, where for a function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ the number of sign changes is

$$S^-(\varphi) := \sup \left\{ n \geq 0 : \exists y_0 < y_1 < \dots < y_n \text{ with } \varphi(y_{i-1})\varphi(y_i) < 0 \text{ for } i = 1, \dots, n \right\}.$$

This count ignores zeros and is the functional the variation-diminishing theorem bounds.

The same argument at $t = 2$ gives the payment set $\{D_2 \geq 0\}$ with $D_2 := \partial_e f_2 - \kappa_2 f_2$, $\kappa_2 = 1/\lambda_2 > 0$. From here on the periods are treated uniformly, writing D_t .

In terms of the score ℓ_t , the payment set is $\{\ell_t \geq \kappa_t\}$, so the cutoff is a level of the score.

Write $\lambda_-^t := \inf_{y \in \mathbb{R}} \ell_t(y)$ and $\lambda_+^t := \sup_{y \in \mathbb{R}} \ell_t(y)$ for the range of the score.

Claim 6 (Interior cutoff). *The payment set $\{\ell_t \geq \kappa_t\}$ is a nonempty proper subset of $\mathbb{R}$ if and only if $\lambda_-^t < \kappa_t \leq \lambda_+^t$, in which case ℓ_t takes values on both sides of κ_t . Outside that range the wage is constant and delivers zero marginal incentive.*

In period 2 the condition is implied by positive effort, since the wage is then the only source of incentive. In period 1 it is the theorem's hypothesis: the target lies above the continuation-only effort e_0 .

Proof. The set is nonempty exactly when some y has $\ell_t(y) \geq \kappa_t$, that is when $\kappa_t \leq \lambda_+^t$, and proper exactly when some y has $\ell_t(y) < \kappa_t$, that is when $\kappa_t > \lambda_-^t$. If it is empty the wage is identically 0. If it is all of $\mathbb{R}$ the wage is identically $\bar{w}$, and then

$$M_w = \bar{w} \int \partial_e f_t dy = \bar{w} \partial_e \int f_t dy = 0.$$

Either way the wage delivers no marginal incentive. In period 2 there is no continuation, so the agent's first-order condition reads $c'(e_2) = 0$ and $c'(0) = 0$ gives $e_2 = 0$, contradicting the hypothesis. Since ℓ_t is continuous its range is an interval, so on the displayed range it takes values on both sides of κ_t . $\square$

No condition on the tail behavior of the score is used here: the claim is about where κ_t sits relative to the range of ℓ_t , whether that range is bounded or not.

Step 2: the sign function.

Explicitly,

$$D_t(y) = \int_{\Theta} \left[-\theta g'_\varepsilon(y - e_t \theta) - \kappa_t g_\varepsilon(y - e_t \theta) \right] p_t(\theta) d\theta.$$

Step 3: reduction to a single convolution.

Reparametrize by the location $x = e_t \theta$, strictly increasing in θ since $e_t > 0$, so sign counts are preserved. The reparametrized belief is the density of x ,

$$n_t(x) = \frac{1}{e_t} p_t\left(\frac{x}{e_t}\right)$$

on $[\underline{x}_t, \bar{x}_t] := [e_t \underline{\theta}, e_t \bar{\theta}]$, which is log-concave because $\log n_t(x) = \log p_t(x/e_t) - \log e_t$ is a concave function composed with an affine map, plus a constant, and it is C^1 on the interior because p_t is. In these variables $\theta = x/e_t$ and

$$\begin{aligned} f_t(y) &= \int_{\underline{x}_t}^{\bar{x}_t} g_\varepsilon(y-x) n_t(x) dx = (g_\varepsilon * n_t)(y), \\ \partial_e f_t(y) &= -\frac{1}{e_t} \int_{\underline{x}_t}^{\bar{x}_t} x g'_\varepsilon(y-x) n_t(x) dx, \end{aligned}$$

so the output density is the convolution of the noise with the reparametrized belief. Suppose first that the belief equals zero at the endpoints of its support, $n_t(\underline{x}_t) = n_t(\bar{x}_t) = 0$. Step 5 removes this restriction. Using $g'_\varepsilon(y-x) = -\partial_x [g_\varepsilon(y-x)]$ and integrating by parts in x , the boundary terms vanishing by the endpoint condition,

$$D_t(y) = (g_\varepsilon * h_t)(y), \tag{17}$$

where

$$h_t(x) := -\frac{1}{e_t} (x n_t(x))' - \kappa_t n_t(x),$$

so D_t itself is a single convolution: the noise density against the explicit weight h_t .

Step 4: the crossing count.

The variation-diminishing theorem ([Karlin, 1968](#), Ch. 5, Thm. 3.1(i)) states that if a kernel is totally positive of order r and a weight has at most $r - 1$ sign changes, then their transform has no more sign changes than the weight. Under Assumption 1 the location kernel $(y, x) \mapsto g_\varepsilon(y-x)$ is totally positive of order three, so the theorem applies to weights with at most two sign changes, which covers both cases. Case (i), with one sign change, needs only the order-two property that

log-concavity provides.²³ Applied to (17),

$$S^-(D_t) \leq S^-(h_t). \quad (18)$$

Writing $\psi_t := (\log n_t)'$ for the score of the reparametrized belief,

$$h_t(x) = -\frac{n_t(x)}{e_t} \underbrace{\left[(1 + e_t \kappa_t) + x \psi_t(x) \right]}_{=: \chi_t(x)}.$$

Since $n_t > 0$ on the interior, h_t has the same number of sign changes as χ_t , with the opposite sign pattern: a pattern $-, +, -$ for χ_t is $+, -, +$ for h_t . Display (18) transfers only the count. The sign pattern of D_t is determined in Step 6. Log-concavity of n_t enters from here on through the single fact that ψ_t is nonincreasing. Because n_t is positive on the interior and equals zero at both endpoints, its mode m_t is interior, $\psi_t > 0$ on $(\underline{x}_t, m_t)$, $\psi_t < 0$ on $(m_t, \bar{x}_t)$, and $\psi_t \rightarrow +\infty$ at $\underline{x}_t$, $\psi_t \rightarrow -\infty$ at $\bar{x}_t$, since the slope of a concave function diverges where the function falls to $-\infty$.

Claim 7. *Let n_t be as above. (i) If $\underline{x}_t \geq 0$, then χ_t changes sign exactly once, from $+$ to $-$. (ii) If $\underline{x}_t < 0 < \bar{x}_t$, then χ_t changes sign exactly twice, in the pattern $-, +, -$.*

Proof. Since $1 + e_t \kappa_t > 0$,

$$x \psi_t(x) \geq 0 \implies \chi_t(x) \geq 1 + e_t \kappa_t > 0, \quad (19)$$

so χ_t can equal zero only where x and $\psi_t(x)$ have strictly opposite signs: on one of

$$P := \{x \in (\underline{x}_t, \bar{x}_t) : x > 0, \psi_t(x) < 0\},$$

$$N := \{x \in (\underline{x}_t, \bar{x}_t) : x < 0, \psi_t(x) > 0\}.$$

Both are intervals: P is the part of the support to the right of both 0 and m_t , and N the part to the left of both.

Exactly one crossing on P . The product $x \psi_t(x)$ is strictly decreasing on P : for $x_1 < x_2$ in P ,

$$x_2 \psi_t(x_2) \leq x_2 \psi_t(x_1) < x_1 \psi_t(x_1),$$

the first inequality because ψ_t is nonincreasing and $x_2 > 0$, the second because $\psi_t(x_1) < 0$ and

²³Order three cannot be dropped for case (ii): for the noise $g_\varepsilon \propto \exp(-u^2/2 - 0.03 \cos 5u)$, which is strongly log-concave with curvature in $[0.25, 1.75]$ but not of order three, the belief $p \propto (1 - \theta^2)^2$ on $[-1, 1]$ and target 0.3, the score has four local minima, and for every cutoff in $(0.046, 0.054)$ the payment set has four components.

$x_2 > x_1$. Hence χ_t is strictly decreasing on P . At the left endpoint of P , $\psi_t(x) \geq 0$ by continuity, so $\chi_t \geq 1 + e_t \kappa_t > 0$ there by (19); and

$$\chi_t(x) \longrightarrow -\infty \quad \text{as } x \uparrow \bar{x}_t,$$

because $\bar{x}_t > 0$ and $\psi_t \rightarrow -\infty$. A continuous, strictly decreasing function that starts strictly positive and falls to $-\infty$ equals zero exactly once, crossing from $+$ to $-$.

Exactly one crossing on N , when N is nonempty. Symmetrically, $\psi_t(x)$ is strictly increasing on N : for $x_1 < x_2$ in N ,

$$x_1 \psi_t(x_1) \leq x_1 \psi_t(x_2) < x_2 \psi_t(x_2),$$

the first inequality because ψ_t is nonincreasing and $x_1 < 0$, the second because $\psi_t(x_2) > 0$ and $x_1 < x_2$. Hence χ_t is strictly increasing on N , with

$$\chi_t(x) \longrightarrow -\infty \quad \text{as } x \downarrow \underline{x}_t$$

when $\underline{x}_t < 0$ (there $\psi_t \rightarrow +\infty$ and $x < 0$), while $\chi_t \geq 1 + e_t \kappa_t > 0$ at the right endpoint of N by (19). So χ_t equals zero exactly once on N , crossing from $-$ to $+$.

If $\underline{x}_t \geq 0$ the support contains no $x < 0$, so N is empty and χ_t has exactly one sign change, $+$ to $-$. If $\underline{x}_t < 0 < \bar{x}_t$, each of N and P contributes exactly one crossing, and $\chi_t > 0$ between them by (19): the pattern is $-$, $+$, $-$. $\square$

Since $e_t > 0$, the case split depends only on primitives: $\underline{x}_t \geq 0 \iff \underline{\theta} \geq 0$, and $\underline{x}_t < 0 < \bar{x}_t \iff \underline{\theta} < 0 < \bar{\theta}$, the second inequality automatic since $\bar{\theta} > 0$. Combining (18) with Claim 7: for beliefs that equal zero at the endpoints, $S^-(D_t) \leq 1$ when $\underline{\theta} \geq 0$ and $S^-(D_t) \leq 2$ when $\underline{\theta} < 0$.

Step 5: general log-concave beliefs.

Let p_t be an arbitrary log-concave belief on Θ , possibly positive at the endpoints (e.g. uniform, or a truncated Gaussian). For $m \in \mathbb{N}$ define

$$p_{t,m}(\theta) := \frac{1}{C_m} p_t(\theta) w(\theta)^{1/m},$$

with $w(\theta) := (\theta - \underline{\theta})(\bar{\theta} - \theta)$, where $C_m := \int_{\Theta} p_t w^{1/m} d\theta$ is the normalizing constant. Each $p_{t,m}$ is log-concave, since its log adds the concave terms $\frac{1}{m} \log(\theta - \underline{\theta}) + \frac{1}{m} \log(\bar{\theta} - \theta)$ to $\log p_t$. It is C^1 on the interior, and equals zero at both endpoints, so Steps 3–4 apply to it: writing $D_{t,m}$ for the associated function in (17), $S^-(D_{t,m}) \leq 1$ when $\underline{\theta} \geq 0$ and $S^-(D_{t,m}) \leq 2$ when $\underline{\theta} < 0$, for every m .

Two elementary bounds control the approximation: with $\bar{W} := \max\{1, (\bar{\theta} - \underline{\theta})^2/4\}$ and $c_0 :=$

$\int_{\Theta} p_t \min\{w, 1\} d\theta > 0$, one has $p_{t,m} \leq (\bar{W}/c_0) p_t$ uniformly in m , and $p_{t,m} \rightarrow p_t$ pointwise on the interior.²⁴ The same argument gives $D_{t,m}(y) \rightarrow D_t(y)$ for every y .²⁵

Crossings can disappear in the limit but cannot appear. If $S^-(D_t) \geq k+1$, choose points $y_0 < \dots < y_{k+1}$ at which D_t takes strictly alternating signs. For m large, $D_{t,m}$ takes the same strict signs at those points, so $S^-(D_{t,m}) \geq k+1$. Applying this with $k=1$ and $k=2$ transfers the ceilings of Step 4 to D_t with no restriction on the log-concave belief.

Step 6: exact counts and the shape of the payment set.

First, by Claim 6 the payment set is neither empty nor all of $\mathbb{R}$, so $S^-(D_t) \geq 1$ in both cases.

Case (i), $\underline{\theta} \geq 0$. Steps 4–5 give $S^-(D_t) \leq 1$, so the count is exactly one. For the direction, let m_ε denote a mode of g_ε . Log-concave densities are unimodal, so $g'_\varepsilon \geq 0$ on $(-\infty, m_\varepsilon]$. For any $y \leq m_\varepsilon$ and any $\theta \in \Theta$, the argument satisfies $y - e_t \theta \leq y \leq m_\varepsilon$, because $\theta \geq 0$ on the support and $e_t > 0$. Hence $g'_\varepsilon(y - e_t \theta) \geq 0$ and

$$\partial_e f_t(y) = - \int_{\Theta} \theta g'_\varepsilon(y - e_t \theta) p_t(\theta) d\theta \leq 0 < \kappa_t f_t(y),$$

so $D_t < 0$ on $(-\infty, m_\varepsilon]$. The single sign change therefore runs from $-$ to $+$, and the payment set is a single upper interval $[z_t, +\infty)$. This argument needs no tail asymptotics, and it covers the boundary case $\underline{\theta} = 0$: the state $\theta = 0$ has zero marginal product and drops out of the integrand, so only productive complementarities with positive probability still yield the one-sided bonus.

Case (ii), $\underline{\theta} < 0$.

Claim 8. *Let $\underline{\theta} < 0$. Then $D_t(y) > 0$ for all $|y|$ sufficiently large.*

Proof. Since f_t is the mixture over states of $g_\varepsilon(y - e_t \theta)$, the score of the mixture is the posterior-weighted average of the state-conditional scores,

$$\ell(y) = \frac{\partial_e f_t(y)}{f_t(y)} = \mathbb{E}_{p_t(\cdot|y)} \left[-\theta (\log g_\varepsilon)'(y - e_t \theta) \right],$$

so the sign of the score in a tail is governed by which states the posterior puts weight on there. It is *not* enough to bound the average below by the smallest state-conditional score: as $y \rightarrow +\infty$ the argument $y - e_t \theta$ diverges to $+\infty$ for every state, so $(\log g_\varepsilon)' \rightarrow -\lambda_\varepsilon^+$, where $\lambda_\varepsilon^+ := -\lim_{u \rightarrow +\infty} (\log g_\varepsilon)'(u) \in$

²⁴For every $m \geq 1$, $\min\{w(\theta), 1\} \leq w(\theta)^{1/m} \leq \bar{W}$, so $c_0 \leq C_m \leq \bar{W}$, which gives the uniform bound. For each interior θ , $w(\theta)^{1/m} \rightarrow 1$, so the integrands $p_t w^{1/m}$ converge pointwise to p_t while dominated by $\bar{W} p_t$. Dominated convergence gives $C_m \rightarrow 1$.

²⁵Write $K_t(y, \theta) := -\theta g'_\varepsilon(y - e_t \theta) - \kappa_t g_\varepsilon(y - e_t \theta)$ for the integrand of D_t over states, continuous in θ on the compact Θ , so $B_t(y) := \max_{\Theta} |K_t(y, \theta)| < \infty$. Then $|K_t(y, \theta)| p_{t,m}(\theta) \leq B_t(y) (\bar{W}/c_0) p_t(\theta)$, an integrable dominating function independent of m , and $K_t(y, \cdot) p_{t,m} \rightarrow K_t(y, \cdot) p_t$ pointwise.

$(0, +\infty]$ and $\lambda_\varepsilon^- := \lim_{u \rightarrow -\infty} (\log g_\varepsilon)'(u) \in (0, +\infty]$ are the tail slopes of the noise, and the state- θ term tends to $\theta \lambda_\varepsilon^+$, which is negative for the counterproductive states $\theta < 0$. The average is nonetheless positive because the posterior concentrates on the productive extreme.

Fix $\theta < \bar{\theta}$. By log-concavity, $\log g_\varepsilon$ is concave, and since $e_t \bar{\theta} > e_t \theta$ the argument at $\bar{\theta}$ is smaller, so

$$\log g_\varepsilon(y - e_t \bar{\theta}) - \log g_\varepsilon(y - e_t \theta) \rightarrow +\infty \quad \text{as } y \rightarrow +\infty,$$

the increment of a concave function over a fixed gap growing without bound once its slope diverges. Hence $p_t(\theta | y) \rightarrow 0$ for every $\theta < \bar{\theta}$: the posterior concentrates on $\bar{\theta}$, and

$$\ell(y) \rightarrow -\bar{\theta} (\log g_\varepsilon)'(y - e_t \bar{\theta}) \rightarrow \bar{\theta} \lambda_\varepsilon^+,$$

strictly positive since $\bar{\theta} > 0$. Symmetrically, as $y \rightarrow -\infty$ the posterior concentrates on $\underline{\theta}$, whose marginal product is negative when $\underline{\theta} < 0$, and $\ell(y) \rightarrow |\underline{\theta}| \lambda_\varepsilon^- > 0$.

With the noise score unbounded, $\lambda_\varepsilon^\pm = \infty$, so $\ell > \kappa_t$ and hence $D_t = f_t(\ell - \kappa_t) > 0$ for $|y|$ large, whatever the cutoff. This is the only place unboundedness is used. $\square$

Steps 4–5 give $S^-(D_t) \leq 2$, and Claim 6 gives $S^-(D_t) \geq 1$. Under Assumption 1 the score is unbounded: strong log-concavity gives $(\log g_\varepsilon)'(u) \leq -m(u - m_\varepsilon)$ for $u \geq m_\varepsilon$ and $(\log g_\varepsilon)'(u) \geq m(m_\varepsilon - u)$ for $u \leq m_\varepsilon$, so $\lambda_\varepsilon^\pm = \infty$. Claim 8 then gives $D_t > 0$ in both tails. The payment set is not all of $\mathbb{R}$, so a middle region is unpaid and $S^-(D_t) \geq 2$, hence exactly two, with pattern $+, -, +$. The payment set is $(-\infty, z_t^-] \cup [z_t^+, +\infty)$ with $z_t^- < z_t^+$, the ordering strict because the unpaid middle interval is nonempty.²⁶ Together with Steps 1–2 this establishes both cases of the theorem. *The level set.* In both cases h_t has finitely many sign changes, one or two, and by Assumption 1 the kernel $(y, x) \mapsto g_\varepsilon(y - x)$ is strictly totally positive of order three. The strict form of the variation-diminishing theorem (Karlin, 1968, Ch. 5, Thm. 3.1(ii)) then bounds the sign changes of D_t by those of h_t even when each zero of D_t is counted as a sign change in whichever direction produces more. A function vanishing on an interval would have infinitely many such changes, so D_t vanishes at finitely many points and the level set $\{D_t = 0\}$ of Step 1 is null. $\square$

²⁶Under log-concavity without the floor of Assumption 1(i), the two tail limits $\bar{\theta} \lambda_\varepsilon^+$ and $|\underline{\theta}| \lambda_\varepsilon^-$ can be finite. A cutoff below both still gives two thresholds, while a cutoff between them gives D_t opposite signs at the two ends, an odd count, hence one threshold on the side with the higher limit.

Proof of Lemma 2 (Retention is a threshold rule)

The proof uses two properties of the second-period contract at high curvature, established first: a bound on its cutoff and on the incentive it delivers, and the recession of a lower payment tail as the posterior improves.

The next claim records two properties of the second-period contract at curvature α . It concerns the stage-2 problem alone and is used both here and in the proof of Theorem 2.

Fix a retained history with posterior p and write $\mu := \mathbb{E}_p[\theta]$. Let κ_2^* be the cutoff of the optimal second-period contract, e_2 the effort it implements, P its payment probability, and $\mathcal{I}$ the Fisher information of the output density in effort. Write $J := \theta_{\max}^2 \int |\partial_{e_2}^2 f_2|$ and $I(e_2) := \int_{\{\ell_2 \geq \kappa_2^*\}} \partial_{e_2} f_2$ for the incentive the contract delivers.

Claim 9 (The second-period contract at curvature α). *If $c'' > \alpha$, then*

$$\kappa_2^* = \frac{c''(e_2) - \bar{w} \partial_{e_2} I}{\mu - c'(e_2)} \geq \frac{\alpha - \bar{w} J}{\bar{\theta}}, \quad \text{and, when } \alpha > \bar{w} J, \quad P \leq \frac{\mathcal{I}}{\kappa_2^{*2}}, \quad U_2^* \leq \bar{w} P. \quad (20)$$

Proof. The principal's first-order condition in the target, $\mu - c'(e_2) = (c''(e_2) - \bar{w} \partial_{e_2} I)/\kappa_2^*$, gives the equality. The bound follows from $c'' \geq \alpha$, $|\partial_{e_2} I| \leq J$ and $\mu - c'(e_2) \leq \bar{\theta}$. For the second display, the score is at least κ_2^* on the payment set and, that bound being positive, may be squared there, so $\kappa_2^{*2} P \leq \int \ell_2^2 f_2 = \mathcal{I}$; and $U_2^* \leq \bar{w} P$ since the wage is at most $\bar{w}$ and paid with probability P . $\square$

The following estimate is used in Step 4 of the proof below and again in the proof of Theorem 3: it confines the lower payment tail, and the exposure to it at every productive state, by the counterproductive mass of the posterior.

Fix a posterior p with counterproductive mass $\mu_- := p(\theta < 0) > 0$, and let the second-period contract pay on $A = \{\ell_2 \geq \kappa\}$ with $\kappa \geq \underline{\kappa} > 0$, implementing effort $e_2 \geq 0$.

Claim 10 (Recession of the lower payment tail). *There are functions Δ_g, Δ_G of μ_- , explicit in primitives, nondecreasing, and vanishing as $\mu_- \downarrow 0$, such that for μ_- below an explicit level every point z^- of the lower payment component and every productive state $\theta \in [0, \bar{\theta}]$ satisfy*

$$g_\varepsilon(z^- - e_2\theta) \leq \Delta_g(\mu_-), \quad G_\varepsilon(z^- - e_2\theta) \leq \Delta_G(\mu_-).$$

Proof. Payment in the lower tail requires the score to reach the cutoff there. Left of the noise's mode the productive states push the score down, since raising the state moves output away from the lower region, so only the counterproductive mass pushes it up, by at most $\mu_- |\theta| \|g'_\varepsilon\|_\infty$. The

score can therefore reach the cutoff only where the output density is correspondingly small, which, the density being increasing left of the mode, places the lower payment region below a level that recedes as μ_- vanishes. The functions Δ_g and Δ_G are the density and the distribution function at that level.

Formally, $\|g'_\varepsilon\|_\infty < \infty$ since $g''_\varepsilon \in L^1$. Let m_ε be the mode of g_ε and define

$$t(\mu_-) := \frac{\mu_- |\underline{\theta}| \|g'_\varepsilon\|_\infty}{(1 - \mu_-) \underline{\kappa}}, \quad u^*(t) := \text{the unique } u < m_\varepsilon \text{ with } g_\varepsilon(u) = t \quad (t < g_\varepsilon(m_\varepsilon)),$$

the left level crossing of the noise density. The explicit level is that at which $e_2 \bar{\theta} + u^*(t(\mu_-)) < m_\varepsilon$, and the functions are $\Delta_g(\mu_-) := g_\varepsilon(u^*(t(\mu_-)) + e_2 \bar{\theta})$ and $\Delta_G(\mu_-) := G_\varepsilon(u^*(t(\mu_-)) + e_2 \bar{\theta})$. The bounds follow once the part of the payment set left of the mode is shown to lie below $e_2 \bar{\theta} + u^*(t(\mu_-))$, that is, $A \cap (-\infty, m_\varepsilon) \subseteq (-\infty, e_2 \bar{\theta} + u^*(t(\mu_-))]$.

Fix $y < m_\varepsilon$. At a productive state the score argument satisfies $y - e_2 \theta \leq y < m_\varepsilon$, so it lies left of the mode, where $g'_\varepsilon \geq 0$. Hence $\theta(-g'_\varepsilon)(y - e_2 \theta) \leq 0$ and the productive part of the score's numerator is nonpositive. The counterproductive part is bounded by $\mu_- |\underline{\theta}| \|g'_\varepsilon\|_\infty$, since $|\theta| \leq |\underline{\theta}|$ below 0. The denominator satisfies $f_2(y) \geq (1 - \mu_-) g_\varepsilon(y - e_2 \bar{\theta})$, because for productive states the arguments $y - e_2 \theta$ all lie in $[y - e_2 \bar{\theta}, y]$, left of the mode, where g_ε is increasing, so the smallest density value is at the most negative argument. Combining,

$$\ell_2(y) \leq \frac{m |\underline{\theta}| \|g'_\varepsilon\|_\infty}{(1 - \mu_-) g_\varepsilon(y - e_2 \bar{\theta})} \quad \text{for all } y < m_\varepsilon.$$

Payment at y requires $\ell_2(y) \geq \kappa \geq \underline{\kappa}$, hence $g_\varepsilon(y - e_2 \bar{\theta}) \leq t(\mu_-)$. Left of the mode g_ε is strictly increasing, so this forces $y - e_2 \bar{\theta} \leq u^*(t(\mu_-))$, which is the statement above. For the bounds on Δ_g and Δ_G , any z^- in the lower component satisfies $z^- \leq e_2 \bar{\theta} + u^*(t(\mu_-)) < m_\varepsilon$, and for productive θ the argument $z^- - e_2 \theta$ is smaller still. Since g_ε is increasing on $(-\infty, m_\varepsilon)$ and G_ε is increasing everywhere, both are bounded by their values at $u^*(t(\mu_-)) + e_2 \bar{\theta}$. Monotonicity in μ_- holds because $t(\cdot)$ and $u^*(\cdot)$ are increasing. Vanishing as $\mu_- \downarrow 0$ holds because $u^*(t) \rightarrow -\infty$ as $t \downarrow 0$ by full support, and $g_\varepsilon, G_\varepsilon$ vanish at $-\infty$. $\square$

The main argument. Fix the conjecture $\hat{e}_1$ and write $p_t := p(\cdot \mid y_1 = t; \hat{e}_1)$ for the posterior after first-period output t . By the monotone likelihood ratio property of the location family, $t \mapsto p_t$ is increasing in the likelihood ratio order, and it is C^1 since $g_\varepsilon \in C^2$. Its path score is

$$\varrho_t(\theta) := \partial_t \log p_t(\theta) = (\log g_\varepsilon)'(t - \hat{e}_1 \theta) - \mathbb{E}_{p_t}[(\log g_\varepsilon)'(t - \hat{e}_1 \theta)],$$

strictly increasing in θ by strict log-concavity of g_ε , and for any bounded ϕ on Θ , $\partial_t \mathbb{E}_{p_t}[\phi] = \text{Cov}_{p_t}(\varrho_t, \phi)$. Fix t at a retained history, write $p = p_t$, and let the optimal second-period contract at p implement $e > 0$ and pay $\bar{w}$ on $A = \{\ell_2 \geq \kappa\}$, a union of at most two tails by Theorem 1, with upper cutoff z^+ and, when the support extends below zero, lower cutoff z^- . $\kappa > 0$ is the value of the score at either cutoff. For a payment set A and a state θ , write

$$P_A(\theta) := \int_A g_\varepsilon(y - e\theta) dy, \quad D_A(\theta) := \int_A \partial_e g_\varepsilon(y - e\theta) dy = -\theta \int_A g'_\varepsilon(y - e\theta) dy,$$

the state-conditional payment probability and incentive, so that $\mathbb{E}_p[P_A]$ is the payment probability and the agent's condition reads $c'(e) = \bar{w} \mathbb{E}_p[D_A]$. Two properties of the contract at a retained history are used below, and both hold for $c'' > \alpha^M$: the cutoff lies to the right of the modal output of every complementarity, $z^+ \geq e\theta + m_\varepsilon$ for all $\theta \in \Theta$, and the incentive-price condition

$$\kappa e \leq 1. \quad (21)$$

Retention supplies a lower bound on the posterior mean: since the wage bill is nonnegative, $\pi_A \leq \Pi_2(p) \leq e \mathbb{E}_p[\theta]$, so $\mathbb{E}_p[\theta] \geq \pi_A/e > 0$. Claim 9 gives $\kappa \geq (\alpha - \bar{w}J)/\bar{\theta}$ and $P \leq I/\kappa^2$, so the paid region moves into the tail as curvature rises. An upper tail of probability below the mass g_ε places above its mode lies beyond $e\bar{\theta} + m_\varepsilon$, which is the first property for α above an explicit level. For the second, on the quadratic class $\kappa e \leq c'(e)/(\mathbb{E}_p[\theta] - c'(e))$, and the delivered incentive $c'(e) = \bar{w} \mathbb{E}_p[\theta g_\varepsilon(z^+ - e\theta)]$ falls with the cutoff while $\mathbb{E}_p[\theta]$ is bounded below by retention, so $\kappa e \leq 1$ once $2c'(e) \leq \mathbb{E}_p[\theta]$, again for α above an explicit level. For general costs in the admissible class the same follows from $e(c''(e) + \bar{w}J) \leq \mathbb{E}_p[\theta] - c'(e)$, using $\kappa \leq (c''(e) + \bar{w}J)/(\mathbb{E}_p[\theta] - c'(e))$ from Claim 9 and $\partial_e I \geq 0$ in the configuration. Let α^M be the largest of these two levels and, when $\underline{\theta} < 0$, the level α^{mass} of Step 4 below.

Step 1: keep the effort, move the upper cutoff. Fix e and, when present, z^- . For t' near t let $z^+(t')$ solve

$$c'(e) = \bar{w} \mathbb{E}_{p_{t'}}[D_{A(z^+)}], \quad (22)$$

so that the contract paying $\bar{w}$ on $A(z^+(t'))$ implements e at $p_{t'}$: the agent's condition holds, and his objective is strictly concave at every admissible contract by Assumption 2. Since $\partial_{z^+} \mathbb{E}_p[D_A] = -\partial_e f_2(z^+) = -\ell_2(z^+) f_2(z^+) = -\kappa f_2(z^+) < 0$, the implicit function theorem gives $z^+(\cdot) \in C^1$ with

$$\frac{dz^+}{dt} = \frac{\text{Cov}_p(\varrho, D_A)}{\kappa f_2(z^+)}.$$

The wage bill $\bar{w} \mathbb{E}_{p_{t'}}[P_A(z^+(t'))]$ therefore changes at the rate

$$\bar{w} \left[\text{Cov}_p(\varrho, P_A) + \partial_{z^+} \mathbb{E}_p[P_A] \frac{dz^+}{dt} \right] = \bar{w} \left[\text{Cov}_p(\varrho, P_A) - \frac{\text{Cov}_p(\varrho, D_A)}{\kappa} \right],$$

using $\partial_{z^+} \mathbb{E}_p[P_A] = -f_2(z^+)$, while expected output $e \mathbb{E}_{p_{t'}}[\theta]$ changes at the rate $e \text{Cov}_p(\varrho, \theta)$.

Step 2: the lower slope. The principal's optimum at $p_{t'}$ does at least as well as the contract of Step 1, and the two coincide at $t' = t$. Hence the lower right Dini derivative $\underline{D} \Pi_2(t) := \liminf_{t' \downarrow t} [\Pi_2(p_{t'}) - \Pi_2(p_t)] / (t' - t)$ satisfies²⁷

$$\underline{D} \Pi_2(t) \geq e \text{Cov}_p(\varrho, \theta) + \frac{\bar{w}}{\kappa} \text{Cov}_p(\varrho, \xi), \quad \xi := D_A - \kappa P_A. \quad (23)$$

The function $\xi(\theta)$ is what state θ contributes to the principal's Lagrangian: the incentive it delivers, net of what it costs in wage bill, priced at κ .

Step 3: the productive regime. Suppose $\theta \geq 0$, so the contract is a single upper tail. Then $\xi(\theta) = \theta g_\varepsilon(z^+ - e\theta) - \kappa[1 - G_\varepsilon(z^+ - e\theta)]$ and

$$\xi'(\theta) = (1 - \kappa e) g_\varepsilon(z^+ - e\theta) - e \theta g'_\varepsilon(z^+ - e\theta). \quad (24)$$

In the configuration $g'_\varepsilon(z^+ - e\theta) \leq 0$ and $\kappa e \leq 1$, so ξ is nondecreasing. A better state both delivers more incentive, since it pushes more output mass past the cutoff, and costs more, since it is paid more often. With the cutoff beyond every state's modal output and the incentive price low relative to the target, both conditions supplied by curvature of the cost, the first effect dominates state by state. Since ϱ is increasing, $\text{Cov}_p(\varrho, \xi) \geq 0$ by Chebyshev's association inequality, and (23) gives $\underline{D} \Pi_2 \geq e \text{Cov}_p(\varrho, \theta) > 0$, strictly, since ϱ is strictly increasing and p is nondegenerate.

Step 4: supports extending below $\theta = 0$. Write $\mu_- := p(\theta < 0)$ for the counterproductive mass. Retention bounds it. The agent's payoff is nonnegative, so the wage bill is at least $c(e)$ and $\Pi_2(p) \leq \mathcal{S}(\mathbb{E}_p[\theta])$, where $\mathcal{S}(\mu) := \max_{e \geq 0} \{e\mu - c(e)\}$ is the full surplus at mean μ , zero for $\mu \leq 0$ and strictly increasing for $\mu > 0$. At a retained history $\mathcal{S}(\mathbb{E}_p[\theta]) \geq \pi_A > 0$, so $\mathbb{E}_p[\theta] \geq \underline{\mu} := \mathcal{S}^{-1}(\pi_A) > 0$. Every posterior with nonpositive mean is dismissed on this bound alone. Since the counterproductive states contribute at most zero to the mean and the productive

²⁷It is used because Π_2 is a value function and is not shown to be differentiable in t . The lower right Dini derivative is defined regardless.

ones at most $\bar{\theta}$, $\mathbb{E}_p[\theta] \leq (1 - \mu_-)\bar{\theta}$, hence

$$\mu_- \leq \bar{\mu}_-(\pi_A) := \frac{\bar{\theta} - \mu}{\bar{\theta}}, \quad (25)$$

which tightens as the cost becomes more convex: $\mathcal{S}$ falls with curvature, so $\underline{\mu} = \mathcal{S}^{-1}(\pi_A)$ rises. Every admissible cost has $c'' > \alpha$ with c'' nondecreasing, so $\mathcal{S}(\mu) \leq \mu^2/(2\alpha)$ and

$$\mu_- \leq 1 - \frac{\sqrt{2\alpha\pi_A}}{\bar{\theta}}$$

at every retained history. Two cases now exhaust. If no history is retained, the retention rule is the cutoff rule with $\underline{y}(\hat{e}_1) = +\infty$ and the lemma holds. Otherwise the display above applies at every retained history, and its right side falls below the mass $\bar{\mu}$ required in (26) below once $\alpha \geq \alpha^{\text{mass}} := \bar{\theta}^2(1 - \bar{\mu})^2/(2\pi_A)$.

The lower tail is what prevents ξ from being monotone, and the reason is economic: raising the state pushes output up, so it makes the lower payment region less likely to be reached. The lower tail therefore buys incentive from the counterproductive states, whose output falls with effort, and wastes wage on the productive ones, who are paid there only after bad luck. That waste is the term to be controlled, and Claim 10 controls it by showing that when the counterproductive mass is small the lower payment region sits so far into the left tail of the noise that no productive state reaches it. Formally, decompose $\xi = \xi^+ + \xi^-$, where $\xi^+(\theta) := \theta g_\varepsilon(z^+ - e\theta) - \kappa[1 - G_\varepsilon(z^+ - e\theta)]$ is the upper-tail term, defined for all θ , and $\xi^-(\theta) := -\theta g_\varepsilon(z^- - e\theta) - \kappa G_\varepsilon(z^- - e\theta)$ is the lower-tail term, absent when the contract is single-threshold. Let $\tilde{\xi}(\theta) := \xi^+(\max\{\theta, 0\})$: it equals ξ^+ on the productive states, where (24) makes it nondecreasing, and is constant on the counterproductive ones, so it is nondecreasing on Θ and $\text{Cov}_p(\varrho, \tilde{\xi}) \geq 0$. The difference $\xi - \tilde{\xi}$ is $[\xi^+ - \xi^+(0)]\mathbf{1}\{\theta < 0\} + \xi^-$. With $\bar{\xi} := \sup_\Theta(|\xi^+| + |\xi^-|)$, $\Lambda^- := \sup_{\theta \geq 0} |\xi^-(\theta)|$ the lower-tail exposure of the productive states, and $\bar{\varrho} := \sup_\Theta |\varrho - \mathbb{E}_p \varrho|$, the bound $|\text{Cov}_p(\varrho, h)| \leq \bar{\varrho} \mathbb{E}_p |h|$ gives

$$\text{Cov}_p(\varrho, \xi) \geq \text{Cov}_p(\varrho, \tilde{\xi}) - \bar{\varrho} \mathbb{E}_p |\xi - \tilde{\xi}| \geq -\bar{\varrho} [\Lambda^- + 3\mu_- \bar{\xi}].$$

If

$$\frac{\bar{w}}{\kappa} \bar{\varrho} [\Lambda^- + 3\mu_- \bar{\xi}] \leq \frac{1}{2} e \text{Cov}_p(\varrho, \theta), \quad (26)$$

then (23) gives $\underline{D} \Pi_2 \geq \frac{1}{2} e \text{Cov}_p(\varrho, \theta) > 0$. Both terms on the left shrink with μ_- : the second directly, and the first by Claim 10, which gives $\Lambda^- \leq \bar{\theta} \Delta_g(\mu_-) + \kappa \Delta_G(\mu_-)$, vanishing as $\mu_- \downarrow 0$. So (26) is a smallness requirement on μ_- . Near the retention margin it is met by (25) once π_A exceeds

an explicit level. Far above the margin it holds as well: as $y_1 \rightarrow +\infty$ the posterior concentrates at $\bar{\theta}$, so μ_- and Λ^- vanish exponentially in y_1 , while $e \geq \pi_A/\bar{\theta}$ is bounded below, $\bar{\varrho} \leq M\hat{e}_1(\bar{\theta} - \underline{\theta})$ is bounded above by the curvature ceiling, and $\text{Cov}_p(\varrho, \theta)$, of the order of the posterior variance, vanishes only polynomially. Hence (26) holds on the whole retained set.

Step 5: from the slope to the cutoff. The properties used above hold at every retained history, hence, by continuity, on a neighborhood of the retained set in t . On that neighborhood $\underline{D}\Pi_2 > 0$. Π_2 is continuous in t by the maximum theorem, and a continuous function whose lower right Dini derivative is strictly positive at every point of an interval is strictly increasing there. So Π_2 is strictly increasing on a neighborhood of the retained set and crosses π_A at most once, from below: if it crossed upward and later fell back, it would have to decrease while still inside the neighborhood. The retention rule is therefore the single cutoff $\underline{y}(\hat{e}_1)$. $\square$

Proof of Theorem 2 (Existence and uniqueness)

The argument runs in five steps. Contracts are indexed by a single cutoff. At each cutoff the agent's problem has a unique fixed point, so the implemented effort is a well-defined function of the cutoff. That function is continuous and strictly decreasing. The principal's problem is therefore one-dimensional and attains its maximum, and strict concavity makes the maximizer unique. The supporting claims are stated and proved in order of use.

Throughout, $\underline{D} = \alpha - \bar{M}'_2 > 0$ is the stage-2 concavity slack of Proposition 1 and $T(e, \hat{e})$ the manipulation term.

Claim 11. *Under Assumption 1 and Proposition 1, the posterior $p(\cdot \mid y_1; \hat{e})$, the agent's continuation $U_2^*(y_1; \hat{e})$, and the retention cutoff $\underline{y}(\hat{e})$ are continuously differentiable in $\hat{e}$, with derivative bounds integrable against the output density.*

This is smoothness in the conjecture, distinct from the smoothness in the agent's true effort supplied by Lemma 1. There a deviation moves the output distribution and his own posterior and stage-2 response while the contract and retention rule stay fixed. Here the contract and retention rule are what move.

Proof. The posterior map. Differentiating (1) in $\hat{e}$ under the integral gives

$$\partial_{\hat{e}} p(\theta \mid y_1; \hat{e}) = p(\theta \mid y_1; \hat{e}) \left[-\theta \frac{g'_\varepsilon(y_1 - \hat{e}\theta)}{g_\varepsilon(y_1 - \hat{e}\theta)} + \int_{\Theta} \theta' \frac{g'_\varepsilon(y_1 - \hat{e}\theta')}{g_\varepsilon(y_1 - \hat{e}\theta')} p(\theta' \mid y_1; \hat{e}) d\theta' \right],$$

the posterior times the centered conjecture-score, the differentiation being justified as in the proof of Lemma 1. Its L^1 norm is $\int_{\Theta} |\partial_{\hat{e}} p| d\theta \leq 2\theta_{\max} \mathbb{E}_p |(\log g_\varepsilon)'(y_1 - \hat{e}\theta)|$, finite at every history.

What the estimates below use is its integral against f_1 or $|\partial_e f_1|$ over y_1 , which is finite by the Fisher-information argument in the proof of Claim 4.

The agent's stage-2 response. Any feasible wage delivers marginal incentive at most $\bar{w} \nu_1 \theta_{\max}$, so second-period effort lies in $[0, \bar{e}_2]$ with $\bar{e}_2 = (c')^{-1}(\bar{w} \nu_1 \theta_{\max})$. On this compact set the first-order approach makes the agent's objective strictly concave with slack $\underline{D}$, so his best response to any contract is unique and, by the implicit function theorem, C^1 in the contract's cutoff and in the posterior.

The principal's stage-2 optimum. With the contract indexed by its cutoff (Theorem 1, second period), the principal's reduced objective in the target is strictly concave for $c'' > \alpha^M$ by the argument of Step 5 below applied to the second period, where no continuation term appears. Her optimal cutoff $\kappa_2^*(y_1, \hat{e})$ is therefore unique and, since her first-order condition is C^1 in $(y_1, \hat{e})$ by the first paragraph, C^1 in $(y_1, \hat{e})$ by the implicit function theorem.

The continuations. The agent's continuation $U_2^*(y_1; \hat{e})$ in (3) is his value at the contract with cutoff $\kappa_2^*(y_1, \hat{e})$, so $\hat{e}$ enters it through that cutoff and through the posterior, both C^1 . By the envelope theorem on the agent's problem the derivative contains no term in his own response, and U_2^* is C^1 in $\hat{e}$. The principal's value $\Pi_2(p(\cdot | y_1; \hat{e}))$ is C^1 in $(y_1, \hat{e})$ by the envelope theorem on her problem, whose maximizer is unique by the previous paragraph.

The retention cutoff. $\underline{y}(\hat{e})$ solves $\Pi_2(p(\cdot | y_1; \hat{e})) = \pi_A$. The left side is C^1 in $(y_1, \hat{e})$, and its y_1 -derivative is nonzero at the crossing: a C^1 function's derivative coincides with its lower right Dini derivative, and the proof of Lemma 2 bounds that derivative in y_1 below by a strictly positive constant on a neighborhood of the retained set (its display (23) and Step 5). The implicit function theorem therefore gives $\underline{y}(\hat{e})$ continuously differentiable. $\square$

The arguments below use a bound on the slope of the score where the cutoffs lie. Write $q(z, \theta) := (\log g_\varepsilon)'(z - e_t \theta)$; the following depends only on Assumption 1.

Claim 12 (Slope of the score in the tails).

$$\ell'_t(z) = \mathbb{E}_{p(\cdot|z)} \left[\theta \left(-(\log g_\varepsilon)''(z - e_t \theta) \right) \right] + \text{Cov}_{p(\cdot|z)}(-\theta q(z, \theta), q(z, \theta)). \quad (27)$$

There are $\bar{z}^+$, $\bar{z}^-$ and $\underline{s} > 0$, depending only on primitives and on a compact range of e_t , such that $\ell'_t(z) \geq \underline{s}$ for $z \geq \bar{z}^+$ and, when $\underline{\theta} < 0$, $\ell'_t(z) \leq -\underline{s}$ for $z \leq \bar{z}^-$. On any compact set of outputs $|\ell'_t| \leq \bar{S} < \infty$.

Proof. Since $\partial_e \log g_\varepsilon(z - e\theta) = -\theta q(z, \theta)$, the score is $\ell_t(z) = \mathbb{E}_{p(\cdot|z)}[-\theta q]$, and since $\partial_z \log p(\theta | z) = q(z, \theta) - \mathbb{E}_{p(\cdot|z)}[q]$, differentiating gives (27). As $z \rightarrow +\infty$ the posterior is the prior tilted by the likelihood, which under the monotone likelihood ratio property concentrates on $\bar{\theta}$: writing

$\bar{q} := q(z, \bar{\theta})$, the posterior is within a tilt of parameter $e_t|\bar{q}|$ of a point mass at $\bar{\theta}$, with variance of order $1/(e_t\bar{q})^2$, and $|\bar{q}| \geq m|z - e_t\bar{\theta} - m_\varepsilon| \rightarrow +\infty$ by the floor m of Assumption 1. The first term of (27) then tends to $\bar{\theta}(-(\log g_\varepsilon)''(z - e_t\bar{\theta})) \geq \bar{\theta}m > 0$. In the second, q is increasing in θ with $\partial_\theta q = e_t(-(\log g_\varepsilon)'')$, so to first order in the posterior's dispersion the covariance is $-\bar{q} e_t c \text{Var}(\theta)$ with $c := -(\log g_\varepsilon)''(z - e_t\bar{\theta})$, of order $c/(e_t|\bar{q}|)$, which vanishes relative to the first term since $c \leq M$ by the curvature ceiling of Assumption 1. $\bar{z}^+$ is the point beyond which it is below half of it. The lower tail is the mirror image when $\bar{\theta} < 0$, with the posterior concentrating on $\bar{\theta}$ and the first term tending to $\bar{\theta}(-(\log g_\varepsilon)'') \leq \bar{\theta}m < 0$. The upper bound $\bar{S}$ on compact sets follows from C^2 regularity. $\square$

Where the slope bound is invoked below, the cutoffs in question lie in the tails: by Claim 9 the score at the cutoff is at least $(\alpha - \bar{w}J)/\bar{\theta}$, which exceeds the score at $\bar{z}^\pm$ once c'' exceeds an explicit level, included in α^C below, and the cutoff then lies beyond $\bar{z}^\pm$.

Fix a target effort $\tilde{e}$ and a conjecture $\hat{e}$, and write $r_w := c'(\tilde{e}) - T(\tilde{e}, \hat{e}) > 0$ for the incentive the wage must deliver. By Theorem 1 the least-cost contract pays $\bar{w}$ on

$$A(\kappa, \hat{e}) := \{y : \ell_{\hat{e}}(y) \geq \kappa\},$$

$$\ell_{\hat{e}} := \frac{\partial_e f_1(\cdot | \hat{e})}{f_1(\cdot | \hat{e})},$$

so the contract is indexed by the single scalar κ .²⁸ Write

$$W(\kappa, \hat{e}; e) = \bar{w} \int_{A(\kappa, \hat{e})} f_1(y | e) dy,$$

$$M_w(\kappa, \hat{e}; e) = \bar{w} \int_{A(\kappa, \hat{e})} \partial_e f_1(y | e) dy$$

for the expected wage bill and the incentive it delivers.

Claim 13. (i) As κ falls the lower threshold rises and the upper threshold falls, at rates $1/|\ell'(z^\pm)|$.

(ii) $\partial_\kappa M_w = \kappa \partial_\kappa W < 0$ and $dW/dM_w = 1/\kappa$, so the cutoff solving $M_w(\kappa) = r_w$ is unique.

(iii) Holding stage-2 effort fixed, the continuation rent at every state is strictly decreasing in κ .

Proof. (i) As κ falls the set $\{\ell \geq \kappa\}$ grows, so its endpoints, the roots of $\ell(z) = \kappa$, move outward. Implicit differentiation gives the rates, finite and bounded under Claim 12 since the cutoffs lie in the tails.

²⁸Two efforts enter and must be kept apart: the region is fixed by the principal's belief, since that is what she designs with, while the density integrated over it is the true one.

(ii) Differentiating W and M_w in κ , using the Leibniz rule, produces boundary terms at $z^\pm$ only. Substituting the boundary identity $\partial_e f_1(z^\pm) = \kappa f_1(z^\pm)$ converts each boundary term of M'_w into κ times the corresponding term of W' , and $W' < 0$ strictly by full support, giving $dW/dM_w = 1/\kappa$. Strict monotonicity makes the solution unique wherever one exists. At or beyond the ends of the score's range the payment set is degenerate and $M_w = 0$.

(iii) With two tails the state- θ payment probability is $G_{2,\theta}(z^-) + 1 - G_{2,\theta}(z^+)$, and since $dz^-/d\kappa < 0 < dz^+/d\kappa$ by (i), both terms fall in κ . With a single tail one term remains, $1 - G_{2,\theta}(z)$ if the upper tail pays and $G_{2,\theta}(z)$ if the lower one does, and it falls in κ for the same reason: raising the cutoff shrinks the payment region from whichever side it lies on. $\square$

Step 1: contracts are indexed by one scalar.

By Claim 13 the least-cost contract implementing a target is the bonus on $A(\kappa, \hat{e})$, so the contract is indexed by its cutoff, and $\partial_\kappa M_w = \kappa \partial_\kappa W < 0$: a higher cutoff delivers strictly less incentive, and the cutoff meeting a given requirement is unique.

Cutoffs outside the range of the score give degenerate contracts: every $\kappa > \sup \ell_{\hat{e}}$ leaves the payment region empty, so the wage is identically zero, and every $\kappa \leq \inf \ell_{\hat{e}}$ makes it all of $\mathbb{R}$, so the wage is the constant $\bar{w}$. Neither delivers any incentive, the second because $\int \partial_e f_1 dy = \partial_e \int f_1 dy = 0$. Cutoffs below zero are dominated: the payment region $\{\ell_{\hat{e}} \geq \kappa\}$ then includes outputs at which $\partial_e f_1 < 0$, so relative to $\kappa = 0$ the wage bill is higher and the delivered incentive lower. Nothing is lost by letting κ range over $[0, +\infty]$, with the empty payment region identified with $\kappa = +\infty$. The domain is compact whether the score's range is bounded or not.

Step 2: a unique implemented effort at each cutoff.

Fix κ . The agent faces continuation objects built at a conjecture $\hat{e}$, which he takes as given since the principal cannot detect a deviation. His first-order condition is

$$c'(e) = M_w(\kappa, \hat{e}; e) + T(e, \hat{e}), \quad (28)$$

and Proposition 1 makes his objective strictly concave, so the solution $\phi_\kappa(\hat{e})$ is unique. The contract implements e exactly when $\phi_\kappa(e) = e$.

The conjecture enters (28) through the continuation-delivered incentive alone: the payment region is a set of outputs fixed before the agent acts, and varying the conjecture inside his best-response map does not rewrite it. Fix κ and write $E = [\underline{e}, \bar{e}]$ for the range of targets the principal can optimally implement, with $\underline{e} > 0$ the level below which no target is consistent with the relationship forming. There are two cases. If $\Pi_2(g_\theta) < 2\pi_A$, then as the target $\bar{e} \rightarrow 0$ the posterior converges to the prior at every history, so the principal's expected continuation converges

to $\max\{\Pi_2(g_\theta), \pi_A\} < 2\pi_A$, while her first-period profit is at most $\tilde{e} \mathbb{E}_{g_\theta}[\theta]^+ \rightarrow 0$. Hence her value from initiating falls below $2\pi_A$ at small targets, and on the initiated set the target is bounded away from zero. If instead $\Pi_2(g_\theta) \geq 2\pi_A > 0$, then $\mathcal{S}(\mathbb{E}_{g_\theta}[\theta]) \geq \Pi_2(g_\theta) > 0$ forces $\mathbb{E}_{g_\theta}[\theta] > 0$, and at a small target the marginal product of effort is this positive mean, the marginal wage cost of raising the target, $(c'' - \partial_{\tilde{e}}T)/\kappa$ by Step 5, vanishes as the cutoff moves into the tail, and the derivative of the expected continuation in the target vanishes at $\tilde{e} = 0$: at zero effort output is pure noise, and the first-order change in the posterior, $p(\theta)(\log g_\varepsilon)'(y_1)(\mathbb{E}_{g_\theta}[\theta] - \theta)$, integrates to zero against g_ε since $\int g'_\varepsilon = 0$. So the principal always prefers a higher target. In neither case is a positive prior mean assumed. A conjecture below $\underline{e}$ is therefore never part of an equilibrium, and the zero conjecture in particular, although self-confirming at the null contract, is not. The map ϕ_κ is a self-map of E , and the bounds below are suprema over pairs $(e, \hat{e}) \in E \times E$, since F depends on the effort and the conjecture separately. The sensitivities of the retention margin that enter them are bounded there by Claim 11, and the lower bound $\underline{e}$ is what keeps them so.

(i) *The slope.* Writing the first-order condition as $F(\hat{e}, e) = 0$, which is C^1 by Claim 11 with $\partial_e F \neq 0$ by Proposition 1, the implicit function theorem gives

$$\phi'_\kappa(\hat{e}) = \frac{\partial_{\hat{e}} F}{D}, \quad D := -\partial_e F > 0, \quad (29)$$

the numerator a *partial* derivative at fixed e . Composing with the best response before differentiating would reintroduce ϕ'_κ itself.

(ii) *The denominator.* D is $c''(e)$ less the curvature the wage contributes and less the curvature the continuation contributes. The first is bounded by $C_1 := \bar{w} \theta_{\max}^2 \int |g''_\varepsilon|$, free of the cost. The second is *not* free of the cost, since every term contains U_2^* or one of its derivatives, but it falls as the curvature rises, by (iv) below, so that

$$\inf_{E \times E} D \geq \alpha - C_1 - \frac{C_2}{\alpha^2}, \quad \alpha := \inf c'', \quad (30)$$

with C_2 explicit in $\bar{w}$, $\theta_{\max}$ and the noise. Taking $\alpha \geq 2\alpha_0$ with $\alpha_0 := 2C_1$ and evaluating C_2 there closes the estimate without circularity.

(iii)–(iv) *The bar rises with the curvature and levels fall.* By Claim 9, at each retained history $\kappa_2^* \geq (\alpha - \bar{w}J)/\bar{\theta}$, and for $\alpha > \bar{w}J$ the payment probability and the continuation rent satisfy $P \leq I/\kappa_2^{*2}$ and $U_2^* \leq \bar{w}P$. The positivity of the bar is implied by the threshold assembled in (vii). The continuation's contribution to (30) is therefore of size $1/\alpha^2$.

(v) *The numerator.* The wage term of F does not depend on the conjecture, so $\partial_{\hat{e}} F = \partial_{\hat{e}} T$, and

T depends on $\hat{e}$ through two objects only: the retention cutoff $\underline{y}(\hat{e})$ and the state-conditional rent $u_2^*(\theta', y_1; \hat{e})$ at each retained history, which moves with the posterior $p(\cdot \mid y_1; \hat{e})$ and with the second-period contract built at it. In the form of Lemma 3, differentiating in $\hat{e}$ therefore produces a boundary term, $\partial_{\hat{e}} \underline{y}$ times the y_1 -derivative of the atom's integrand at the bar, and an interior term, $\partial_{\hat{e}} u_2^*$ and $\partial_{\hat{e}} \partial_{y_1} u_2^*$ integrated against the output density.

The first factor is bounded by Claim 11: $|\partial_{\hat{e}} \underline{y}| \leq C_y$, the ratio of the bound on $\partial_{\hat{e}} \Pi_2$ there to the positive lower bound on $\partial_{y_1} \Pi_2$ near the retained set.

For the interior term, write the second-period contract by its cutoff κ_2^* and differentiate the principal's second-period first-order condition, not the identity (20), in the belief: the condition is an equation in (κ_2^*, e_2) whose coefficients involve $c'(e_2)$ and $c''(e_2)$ but no higher derivative, and the agent's condition $c'(e_2) = \bar{w}I$ gives $|\partial_{\hat{e}} e_2| \leq \bar{w} \sup |\partial_{\hat{e}} I| / \underline{D}$, so that $|\partial_{\hat{e}} \kappa_2^*| \leq C_3$ with C_3 assembled from $\bar{w}$, $\theta_{\max}$, the noise and $\underline{M}$, the lower bound on the extraction margin $\mu - c'(e_2)$ on the retained set.

Then $\partial_{\hat{e}} u_2^* = \bar{w} \partial_{\hat{e}} P_2 - c'(e_2) \partial_{\hat{e}} e_2$, and $\partial_{\hat{e}} P_2$ collects the movement of the posterior, bounded by the L^1 bound of Claim 11, and the movement of the cutoff, $\mathcal{B}(\kappa_2^*) \partial_{\hat{e}} \kappa_2^*$, where $\mathcal{B}(\kappa)$ is the density of the second-period score at level κ under f_2 , the density of the score at the cutoff, written $\mathcal{B}$. The same bound holds for $\partial_{\hat{e}} \partial_{y_1} u_2^*$ with one more y_1 -derivative of the posterior, controlled by the same dominations.

(vi) *The density of the score at the cutoff.* By the change of variables $z \mapsto \ell_2(z)$, $\mathcal{B}(\kappa) = \sum f_2(z) / |\ell_2'(z)|$ over the roots of $\ell_2(z) = \kappa$, and for cutoffs in the tails Claim 12 bounds the slope below by $\underline{s}$, so

$$\Lambda_\sigma := \sup_{\kappa} \mathcal{B}(\kappa) \leq \frac{2 \sup g_\varepsilon}{\underline{s}},$$

the factor two for the two roots of a two-tail contract. The floor m of Assumption 1 enters the argument here, through $\underline{s}$. Collecting,

$$\sup_{E \times E} |\partial_{\hat{e}} T| \leq \bar{N} := C_y C_4 + C_5 \Lambda_\sigma + C_6,$$

with C_4, C_5, C_6 assembled from $\bar{w}$, $\theta_{\max}$, the noise and the constants above, none depending on the cutoff of the first-period contract.

(vii) *Assembly.* Collecting (30) and (vi),

$$\sup_{E \times E} |\phi'_\kappa| \leq \frac{\bar{N}}{\alpha - C_1 - C_2/\alpha^2} < 1 \quad (31)$$

for every α above a threshold assembled from $C_1, C_2, \bar{N}$ and $\bar{w}J$, the last securing positivity of the

bar in (iv) and never binding, since the others require a larger curvature. The bound is uniform in the first-period cutoff κ : no constant in it depends on κ , and at the endpoint $\kappa = +\infty$, the null contract, the wage delivers no incentive, the wage's contribution to D vanishes, and the same bound applies with C_1 replaced by zero. Banach's theorem then gives a unique fixed point $e(\kappa)$ at each bar.

(viii) *The fixed point is continuous in the bar.* Write $q < 1$ for the modulus in (31), uniform in κ by (vii). For two bars, inserting $\phi_\kappa(e(\kappa'))$ and using that each effort is a fixed point of its own map,

$$|e(\kappa) - e(\kappa')| \leq \underbrace{|\phi_\kappa(e(\kappa)) - \phi_\kappa(e(\kappa'))|}_{\leq q |e(\kappa) - e(\kappa')|} + |\phi_\kappa(e(\kappa')) - \phi_{\kappa'}(e(\kappa'))|,$$

so that

$$|e(\kappa) - e(\kappa')| \leq \frac{1}{1-q} |\phi_\kappa(e(\kappa')) - \phi_{\kappa'}(e(\kappa'))|. \quad (32)$$

The right-hand side vanishes as $\kappa' \rightarrow \kappa$ because $\phi_\kappa(e)$ is continuous in the bar at fixed effort, which is the implicit function theorem applied to F in κ rather than in $\hat{e}$, the denominator being the same D and so nonzero. Uniformity of q keeps the factor $1/(1-q)$ from degenerating as the bar moves.

Step 3: the implemented effort is continuous and strictly decreasing.

Differentiating $\phi_\kappa(e) = e$ totally in κ ,

$$\begin{aligned} \frac{de}{d\kappa} &= \frac{\partial_\kappa \phi_\kappa}{1 - \phi'_\kappa}, \\ \partial_\kappa \phi_\kappa &= \frac{\kappa \partial_\kappa W}{D} < 0 \end{aligned}$$

by Step 1, with $D = -\partial_e F > 0$ the denominator of (29). Hence $e'(\kappa) < 0$: raising the cutoff lowers the implemented effort, and the belief feedback dampens the direct effect without overturning it. Since $\partial_e(\phi_\kappa(e) - e) = \phi'_\kappa - 1$ is bounded away from zero, the implicit function theorem gives $e(\cdot) \in C^1$, extending continuously to the endpoints, where it takes the effort the continuation alone motivates.

The slope bound of Claim 12 holds for cutoffs in the tails, where the cheapest contract's cutoffs lie for $c'' > \alpha^M$, and by Claim 6 in the proof of Theorem 1 every cutoff delivering positive incentive is interior to the score's range. The endpoint $\kappa = +\infty$ is the null contract and is treated separately in Step 5.

Step 4: the principal's problem is one-dimensional.

Substituting,

$$\tilde{\Pi}(\kappa) = B(e(\kappa)) - W(\kappa, e(\kappa); e(\kappa)),$$

with B the output-plus-continuation value. Both effort arguments of W take the same value here, since at a fixed point the conjecture equals the effort actually played. By Step 3 this is continuous on $[0, +\infty]$, which is compact, the upper endpoint being the null contract, at which the payment set is empty and the objective extends by continuity, so a maximizer exists by Weierstrass.

Step 5: strict concavity gives a unique optimum.

The maximizer of Step 4 either sits at the endpoint $\kappa = +\infty$ or is interior, and both cases are settled.

At $\kappa = +\infty$ the payment set is empty, so the wage delivers no marginal incentive and the implemented effort is the fixed point e_0 of the continuation channel alone, unique by the contraction of Step 2, which is uniform in κ and covers the endpoint. At the null contract the wage delivers no incentive and the principal implements e_0 at no wage cost. Every interior cutoff implements a target above e_0 with a positive wage bill, and the comparison between them is part of the maximization over the compact domain in Step 4.

By Step 3, $e(\cdot)$ is a strictly decreasing C^1 bijection from the feasible cutoffs onto an interval of targets with lower end e_0 , so the principal's objective may be written in the target. Reparametrize as follows. For a target $\tilde{e}$ the required wage incentive is $r_w(\tilde{e}) := c'(\tilde{e}) - T(\tilde{e}, \tilde{e})$, and the cutoff $\kappa(\tilde{e})$ that delivers it at least cost solves $M_w(\kappa; \tilde{e}) = r_w(\tilde{e})$ on the branch of Claim 13 on which M_w is decreasing in κ : a given incentive can be delivered by a low cutoff or by a high one, and the principal uses the high one, where the payment probability is smaller. Write $\mathcal{K} := \{\kappa \geq 0 : \partial_\kappa M_w(\kappa; \tilde{e}) < 0 \text{ for some admissible } \tilde{e}\}$ for the set of such cutoffs, and take every supremum and infimum below over $\mathcal{K}$. These cutoffs are bounded below by a positive $\underline{\kappa}$: for $c'' > \alpha^M$ the cheapest contract's cutoff lies beyond the modal output of every complementarity, by the configuration in the proof of Lemma 2, where the score is bounded away from zero. Selecting it is cost minimization: Theorem 1 gives the form of the contract, and the two solutions are the two cutoffs delivering the same incentive. Write

$$\tilde{\Pi}(\tilde{e}) = B(\tilde{e}) - W(\tilde{e}), \quad B(\tilde{e}) := \tilde{e} \mathbb{E}_{g_\theta}[\theta] + \Psi(\tilde{e}), \quad W(\tilde{e}) := \bar{w} P(\kappa(\tilde{e}); \tilde{e}),$$

with Ψ the principal's expected continuation at conjecture $\tilde{e}$ and W the least-cost wage bill. Three constants, all free of c'' : $\bar{s}_T := \sup|\partial_{\tilde{e}} T|$, the sensitivity of the continuation incentive to the target, bounded on the admissible parameter set. $\bar{d} := \sup|B''| + \sup|\partial_{\tilde{e}}^2 T|/\underline{\kappa}$, where the second derivatives

are bounded by the dominations of Lemma 1, and

$$\Xi := \inf_{\kappa \in \mathcal{K}} \frac{1}{\kappa^3 \bar{w} \mathcal{B}(\kappa)} > 0,$$

where $\mathcal{B}(\kappa) = |\partial_\kappa W|/\bar{w}$ is the density of the score at the cutoff, so that $\kappa^3 \mathcal{B}(\kappa)$ is bounded on $\mathcal{K}$: by $\kappa^3 \Lambda_\sigma$ on any bounded range of cutoffs, and by the decay of $\mathcal{B}$ at deep cutoffs, which is faster than any power of κ , since the output density has Gaussian-or-faster tails while the score grows at most linearly.

The marginal bill. By the pricing identity of Claim 13, $\partial_\kappa M_w = \kappa \partial_\kappa W$, so along the branch $dW/dr = 1/\kappa$ and

$$W'(\tilde{e}) = \frac{r'_w(\tilde{e})}{\kappa(\tilde{e})} = \frac{c''(\tilde{e}) - \partial_{\tilde{e}} T}{\kappa(\tilde{e})}, \quad \frac{d}{dr} \left(\frac{1}{\kappa} \right) = \frac{1}{\kappa^3 \bar{w} \mathcal{B}(\kappa)} \geq \Xi.$$

The increment. Fix $\tilde{e} < \tilde{e}'$ with $\Delta := \tilde{e}' - \tilde{e}$, and write κ, κ' for the two cutoffs. Then

$$W'(\tilde{e}') - W'(\tilde{e}) = \underbrace{\frac{c''(\tilde{e}') - c''(\tilde{e})}{\kappa'}}_{\geq 0} + \underbrace{(c''(\tilde{e}) - \partial_{\tilde{e}} T) \left(\frac{1}{\kappa'} - \frac{1}{\kappa} \right)}_{\geq (c'' - \bar{s}_T) \Xi (c'' - \bar{s}_T) \Delta} - \underbrace{\frac{\partial_{\tilde{e}} T(\tilde{e}') - \partial_{\tilde{e}} T(\tilde{e})}{\kappa'}}_{|\cdot| \leq \sup |\partial_{\tilde{e}}^2 T| \Delta / \underline{\kappa}}.$$

The first term is nonnegative because c'' is nondecreasing. For the second, $1/\kappa' - 1/\kappa = \int_{r_w}^{r'_w} d(1/\kappa) \geq \Xi (r'_w - r_w)$ and $r'_w - r_w \geq (c''(\tilde{e}) - \bar{s}_T) \Delta$, since $r'_w = c'' - \partial_{\tilde{e}} T \geq c'' - \bar{s}_T$ and c'' is nondecreasing. Meanwhile $B'(\tilde{e}') - B'(\tilde{e}) \leq \sup |B''| \Delta$. Collecting, $\tilde{\Pi}'$ decreases by at least

$$\left[\Xi (c''(\tilde{e}) - \bar{s}_T)^2 - \bar{d} \right] \Delta,$$

so $\tilde{\Pi}$ is strongly concave in the target, with modulus at the cutoff κ at least $\Xi(\kappa)(\inf c'' - \bar{s}_T)^2 - \bar{d}(\kappa)$, where $\bar{d}(\kappa) := \sup |B''| + \sup |\partial_{\tilde{e}}^2 T|/\kappa$. This is positive at every cutoff on the branch for all costs with

$$\inf c'' > \alpha^P := \bar{s}_T + \sup_{\kappa \in \mathcal{K}} \sqrt{\bar{d}(\kappa)/\Xi(\kappa)} = \bar{s}_T + \sup_{\kappa \in \mathcal{K}} \sqrt{\bar{w} \mathcal{B}(\kappa) (\kappa^3 \sup |B''| + \kappa^2 \sup |\partial_{\tilde{e}}^2 T|)},$$

a threshold in $\inf c''$ fixed by primitives: the supremum is finite because $\mathcal{B}$ is bounded and the powers of κ are bounded on any bounded range, while $\mathcal{B}$ decays faster than any power of κ at deep cutoffs, as above.

By Step 3 the map $e(\cdot)$ is a continuous, strictly decreasing bijection from the feasible cutoffs onto an interval of targets, on which $\tilde{\Pi}$ is continuous and strictly concave. A strictly concave function on an interval has at most one maximizer, and since $e(\cdot)$ is a bijection exactly one cutoff

attains it. Call it κ^* ; it may be the null contract, in which case the equilibrium is the threat-only one above. Whether the relationship forms at all is then the comparison between $\tilde{\Pi}(\kappa^*)$ and $2\pi_A$: it forms when the first exceeds the second, the principal takes her outside option in both periods when it falls short, and she is indifferent when they are equal.

Either way the maximizer is unique, and by Step 2 it pins down a unique implemented effort. Collecting the conditions: the first-order approach needs $c'' > \alpha^{\text{FOA}}$, the configuration of Lemma 2 needs $c'' > \alpha^M$, the contraction (31) holds for $\inf c''$ above a level α^C assembled from $C_1, C_2, \bar{N}$, and the principal's strict concavity holds for $\inf c'' > \alpha^P$. Each is a threshold in $\inf c''$, so $\alpha^E := \max\{\alpha^{\text{FOA}}, \alpha^M, \alpha^C, \alpha^P\}$ is the threshold of the theorem, and every cost with $c'' > \alpha^E$, in particular every quadratic of sufficient curvature, is admissible.

Proof of Proposition 2 (Benchmark equilibrium)

The proof follows the two steps of Theorem 2, with the agent's problem simplified, since the benchmark agent's first-period objective has no continuation term.

Step 1: the agent's problem. Fix any first-period contract. By Theorem 1 it is a schedule $w_1 = \bar{w}\mathbf{1}_A$ on a payment set A that is a union of at most two tails. Write $P_1(A, e) = \int_A f_1(y_1 \mid e) dy_1$. The benchmark agent's objective is $V^B(e) = \bar{w} P_1(A, e) - c(e)$, with no term in the conjecture. Effort enters f_1 as a location shift of each state's output, so $\partial_e^2 P_1(A, e)$ is a sum of one boundary term per tail, each carrying a factor $\theta^2 g'_\varepsilon$, and $\|g'_\varepsilon\|_\infty < \infty$ by Assumption 1. Hence

$$|\partial_e^2 [\bar{w} P_1(A, e)]| \leq W^B := n_A \bar{w} \theta_{\max}^2 \|g'_\varepsilon\|_\infty,$$

with $n_A \in \{1, 2\}$ the number of tails. This is the wage-curvature bound of Theorem 2 with the continuation term deleted, and it is uniform in (A, e, γ) . For $c'' > \alpha^B := W^B$ the objective is strictly concave, so the agent's best response is a singleton $e^B(A)$, lying in $[0, \bar{e}]$, and continuous in the contract by the implicit function theorem, the second derivative being bounded above by $W^B - \inf c'' < 0$.²⁹

Step 2: the principal's problem. The principal maximizes

$$\Pi^B(A) = e^B(A) \mathbb{E}_\gamma[\theta] - \bar{w} P_1(A, e^B(A)) + \Psi(e^B(A)),$$

with Ψ her expected continuation, the same function as in the model since the second period is

²⁹The best response is interior when $V^{B'}(0) > 0$. At $e = 0$ every state's output has the same density, so $\partial_e P_1(A, 0) = \mathbb{E}_\gamma[\theta] g_\varepsilon(z)$ for a single upper tail and $\mathbb{E}_\gamma[\theta] [g_\varepsilon(z^+) - g_\varepsilon(z^-)]$ for two. With two tails, or a prior of nonpositive mean, the corner $e = 0$ is possible, and such contracts implement no effort and are never chosen.

identical given the posterior. By Theorem 1 the choice is over a compact set of cutoffs, the three terms are continuous, and a maximizer exists. Uniqueness follows from Steps 3 and 5 of the proof of Theorem 2, which apply with the continuation term absent: for $c'' > \alpha^B$ the implemented effort is continuous and strictly monotone in the cutoff, and the reduced objective is strictly concave in the target. The second period and the retention rule are unchanged from the model, so a unique first-period contract, a unique implemented effort and the unique continuation give a unique equilibrium, with $\alpha^B = W^B$ below the model's α^E . $\square$

Proof of Lemma 3 (Continuation incentive decomposition)

Proof. Fix the conjecture $\hat{e}_1$, which fixes the retention bar $\underline{y} = \underline{y}(\hat{e}_1)$ and, for each state θ' , the second-period rent $u_2^*(\theta', y_1; \hat{e}_1)$ as a function of the realized first-period output through the principal's posterior. The true effort e_1 also enters the agent's second-period response through his own posterior, but by Lemma 1 that channel contributes nothing to the derivative, so for the purpose of differentiating it enters only through the distribution of y_1 . The agent's continuation value at true effort e_1 is

$$\int_{\Theta} \int_{\underline{y}}^{+\infty} u_2^*(\theta', y_1; \hat{e}_1) g_{\varepsilon}(y_1 - e_1 \theta') dy_1 g_{\theta}(\theta') d\theta',$$

the inner integral being over retained outputs. Effort enters only through the location of the output density, $\partial_{e_1} g_{\varepsilon}(y_1 - e_1 \theta') = -\theta' g'_{\varepsilon}(y_1 - e_1 \theta')$, so differentiating under the integral as in Lemma 1,

$$T(e_1, \hat{e}_1) = \int_{\Theta} \theta' \left[- \int_{\underline{y}}^{+\infty} u_2^* g'_{\varepsilon}(y_1 - e_1 \theta') dy_1 \right] g_{\theta}(\theta') d\theta'.$$

Integrating by parts in y_1 on $[\underline{y}, +\infty)$, with $u_2^* g_{\varepsilon} \rightarrow 0$ at $+\infty$, since the rent is bounded by $\bar{w}$ and $g_{\varepsilon} \rightarrow 0$ at infinity under Assumption 1,

$$\begin{aligned} - \int_{\underline{y}}^{+\infty} u_2^* g'_{\varepsilon} dy_1 &= u_2^*(\theta', \underline{y}; \hat{e}_1) g_{\varepsilon}(\underline{y} - e_1 \theta') \\ &\quad + \int_{\underline{y}}^{+\infty} \frac{\partial u_2^*}{\partial y_1} g_{\varepsilon}(y_1 - e_1 \theta') dy_1, \end{aligned}$$

which is (12). The first term is the retention effect, the rent at the bar times the density of reaching it, and the second the wage effect, the change in rent from moving the posterior, integrated over retained outputs. $\square$

Proof of Lemma 4 (The posterior depends on a single index)

Proof. Fix the conjecture $\hat{e}_1$ and the prior index γ . By (1) and Definition 1, the posterior at output y_1 is proportional in θ to $g_\varepsilon(y_1 - \hat{e}_1\theta) e^{\gamma\theta} g_0(\theta)$. With $\varepsilon_t \sim N(0, \sigma^2)$,

$$g_\varepsilon(y_1 - \hat{e}_1\theta) \propto \exp\left(-\frac{(y_1 - \hat{e}_1\theta)^2}{2\sigma^2}\right) \propto \exp\left(\frac{\hat{e}_1 y_1}{\sigma^2} \theta - \frac{\hat{e}_1^2 \theta^2}{2\sigma^2}\right),$$

the factor $\exp(-y_1^2/2\sigma^2)$ being free of θ and absorbed into the normalizing constant. Multiplying by $e^{\gamma\theta} g_0(\theta)$ and collecting the terms linear in θ ,

$$p(\theta \mid y_1; \gamma) \propto \exp\left(\left[\gamma + \frac{\hat{e}_1}{\sigma^2} y_1\right] \theta\right) g_0(\theta) e^{-\hat{e}_1^2 \theta^2 / 2\sigma^2} = e^{b\theta} \tilde{g}_0(\theta),$$

which is (13). □

Proof of Theorem 3 (Over- and under-experimentation)

All constants in this appendix are suprema or infima over a compact set of parameter values fixed by the primitives, the *admissible parameter set*, listed under *The constants* at the end of this proof.

The argument has four parts: the identities for exponential tilts, in which every belief along the learning path is a tilt of one base and derivatives are cumulants. The shape of the agent's second-period rent along that path, single-peaked at a known complementarity and, for a concentrated prior, along the family. The sign of the manipulation term T , positive where the relationship forms, negative at optimistic priors, and crossing zero only downwards, and the assembly, which transfers the sign of T to the comparison of efforts.

The learning path. Fix the conjecture $\hat{e}_1$. The realized index is $b := \gamma + \omega y_1$ with $\omega := \hat{e}_1/\sigma^2$. The posterior after realized index b is $q_b \propto e^{b\theta} \tilde{g}_0$, where $\tilde{g}_0 := g_0 e^{-\hat{e}_1^2 \theta^2 / 2\sigma^2}$ is the prior damped by the Gaussian factor the signal contributes, referred to below as the *modified base*. Its precision is at least ς , and it depends on the conjecture through $\hat{e}_1$. Write $L_\gamma(b)$ for the density of b under prior γ , which depends on γ through the tilt as well as through $b - \gamma$ and $\lambda := \sigma^2/\hat{e}_1^2$. For any density h on Θ and its tilts $h_t \propto e^{t\theta} h$, write $m_t, V_t, \kappa_k(t)$ for the mean, variance and k -th cumulant and $\gamma_k := \kappa_k/V^{k/2}$ for the standardized cumulants (γ_4 excess).

Retention. Retention is $\{b \geq \bar{b}\}$ with $\Pi_2(q_{\bar{b}}) = \pi_A$, where $\Pi_2(q)$ is the principal's second-period profit at belief q . By Lemma 2, Π_2 is nondecreasing on a neighborhood of the retained set, so retention is a cutoff in the index. Monotonicity away from that neighborhood is not used. The

theorem's range is $\pi_A \in (\pi^-, \pi^+) = (\Pi_2(\delta_{\underline{\theta}}), \Pi_2(\delta_{\bar{\theta}}))$. When $\underline{\theta} \leq 0$, $\pi^- = 0$ and the range is $(0, \pi^+)$.

The second-period contract. In the second period at belief q the optimal wage is $\bar{w}\mathbf{1}\{y \geq z^+\} + \bar{w}\mathbf{1}\{y \leq z^-\}$, the lower tail present iff $\mu_-(q) := q(\theta < 0) > 0$. Write $u_\theta^\pm := (\theta e - z^\pm)/\sigma$. Retained posteriors satisfy $m_b \geq \underline{m} := \pi_A/\bar{e}$ with $\bar{e}$ the effort bound of the appendix's notation (the configuration of Lemma 2), so $\mu_-(q_b) \leq C_0 e^{-s\underline{m}^2/2}$. Every quantity below that comes from the lower tail is a bounded function times μ_- , written $O(\mu_-)$, and is included in the constants at the end of the proof of Theorem 3.

The reweighted posterior. The reweighted posterior is $\tilde{q} \propto \varphi(u_\theta^+)q$, with expectation $\widetilde{\mathbb{E}}[\cdot]$, covariance $\widetilde{\text{Cov}}$, and $M_{00} := \mathbb{E}_q[\varphi(u_\theta^+)]$. Completing the square, $\tilde{q}$ is the tilt at $t + ez^+$ of q damped by $e^{-\theta^2 e^2/2\sigma^2}$. Choosing z^+ is choosing the agent's rent a . $\hat{e}(a, q)$ is the effort supplied at rent a , $\hat{\pi}(a, q) := m\hat{e} - \alpha\hat{e}^2/2 - a$, $U(q) := \arg \max_a \hat{\pi}$, and $\Gamma(a, q) := (m - \alpha\hat{e})\hat{e}_a$. The principal's condition is $\Gamma = 1$.

The objects to be signed. $T(\gamma)$ denotes the manipulation term at prior γ and the conjecture, and $\beta(b) := -\mathbb{E}_{q_b}[\theta \partial_b u_2^*]$ the wage effect, with the sign convention that $\beta > 0$ when better news lowers the agent's rent.

Derivatives along the tilt family

Claim 14 (Tilt identities). *For bounded g , $\frac{d}{dt}\mathbb{E}_{h_t}[g] = \text{Cov}_{h_t}(\theta, g)$ and $\frac{d}{dt}\kappa_k(t) = \kappa_{k+1}(t)$. For $g \in C^1$, $\text{Cov}_{h_t}(\theta, g) = \int g'K_t$ with $K_t \geq 0$, $\int K_t = V_t$. If $h = e^{-\psi}$ with $\psi'' \geq \varsigma$, so is h_t , and $V_t \leq 1/\varsigma$.*

Proof. Since $h_t \propto e^{t\theta}h$, differentiating the log-density gives $\frac{d}{dt}\log h_t(\theta) = \theta - m_t$, and differentiating $\mathbb{E}_{h_t}[g] = \int g h_t$ under the integral gives the first identity. For the second, if $K(s) := \log \int e^{s\theta}h$ is the cumulant generating function of h , that of h_t is $s \mapsto K(t+s) - K(t)$, so the k -th cumulant of h_t is $K^{(k)}(t)$ and its t -derivative is $K^{(k+1)}(t) = \kappa_{k+1}(t)$. The third is Hoeffding's covariance identity, $\text{Cov}(\theta, g(\theta)) = \int g'(x)[H(x)(1-H(x))]dx$ for a distribution function H , applied to H_t with $K_t := H_t(1-H_t) \geq 0$, whose integral is the variance V_t . For the last, $h_t = e^{-(\psi-t\theta)+K(t)}$, and $\psi-t\theta$ has the same second derivative as ψ , so h_t is ς -strongly log-concave and the Brascamp–Lieb inequality (Brascamp and Lieb, 1976) gives $V_t \leq 1/\varsigma$. $\square$

Claim 15 (Score of the realized index). $\partial_\gamma \log L_\gamma(b) = \lambda(b - \mathbb{E}_\gamma[b])$.

Proof. $L_\gamma(b) \propto \int e^{\gamma\theta} g_0(\theta) \varphi((b - \gamma - \omega\theta\hat{e}_1)/(\omega\sigma)) d\theta \propto e^{-\lambda(b-\gamma)^2/2} \tilde{Z}(b)/Z(\gamma)$ after completing the square in θ . The derivative of the logarithm in γ is affine in b and integrates to zero against L . $\square$

The rent along the learning path

Claim 16 (Rent coordinate). $\text{sgn } U'(t) = \text{sgn } \partial_t \Gamma(a, t)|_{a=U(t)}$. If $t \mapsto \Gamma(a, t)$ is strictly quasi-concave on V for every a in the range of U , then U is single-peaked along the family.

Proof. $\hat{\pi}_a(U(t), t) = \Gamma - 1 = 0$ and $\hat{\pi}_{aa} < 0$ give $\text{sgn } U' = \text{sgn } \partial_t \Gamma$. With $\tau(a)$ the maximizer of $\Gamma(a, \cdot)$, $U'(t) > 0 \iff t < \tau(U(t))$. $h(t) := \tau(U(t)) - t$ has $h' = -1$ at any zero, so it has at most one, and U' has the sign of h . $\square$

Take the complementarity known and equal to $m > 0$, so $q = \delta_m$, and fix a rent $a \in (0, \bar{w}/2)$. The agent's first-order condition $\alpha e = (\bar{w}/\sigma)m\varphi(v)$ has two solutions in the cutoff, since φ rises and then falls. The principal uses the one with the cutoff above expected output, where the same incentive is delivered with a lower payment probability. Selecting it is cost minimization, not a restriction: Theorem 1 gives the form of the contract, and the two solutions are the two cutoffs delivering the same incentive. Write V for the interval of standardized cutoffs on which this solution exists, and

$$v := \frac{z^+ - em}{\sigma}, \quad \varpi := \frac{em}{\sigma}, \quad \eta := pv, \quad \nu := \frac{\varpi}{v}, \quad \kappa := \frac{\bar{w}}{\sigma}\varphi(v),$$

where κ , the incentive price $\alpha e/m$, is not the score cutoff κ of the earlier proofs. All statements below are for $v \in V$.

Claim 17 (Known complementarity). Let $Q := (1 - \kappa)(1 - \nu) + (1 - \eta)(1 - \eta - \kappa)$, so that $m(1 - \eta)^2(1 - \kappa) \partial_m \log \Gamma = Q$. For $\sigma \geq \sigma^*$, Q decreases strictly along the path from $2(1 - \kappa) > 0$ to $-\infty$. Hence $\partial_m \log \Gamma$ changes sign once, and $\Gamma(a, \cdot)$ is single-peaked.

Proof. Along fixed a ,

$$\begin{aligned} \varpi &= \frac{2(\bar{\Phi}(v) - a/\bar{w})}{\varphi(v)}, \quad \nu_m = -\frac{\varpi}{m(1 - \eta)}, \\ \nu_m &= \frac{v(v + 2 - \eta)}{m(1 - \eta)}, \quad \eta_m = \frac{\eta(2 - \eta - v)}{m(1 - \eta)}, \quad \kappa_m = \frac{\kappa\eta}{m(1 - \eta)}, \end{aligned}$$

obtained by differentiating the first-order condition and the rent identity, using $\bar{\Phi}' = -\varphi$ and

$\varphi' = -v\varphi$. Hence

$$m \partial_m \log \hat{e}_a = \frac{1-v}{(1-\eta)^2}, \quad m \partial_m \log(m - \alpha \hat{e}) = 1 - \frac{\kappa \eta}{(1-\eta)(1-\kappa)},$$

and adding the two gives $m(1-\eta)^2(1-\kappa) \partial_m \log \Gamma = Q$. Differentiating Q along the path,

$$m(1-\eta)Q_m = -[(1-\kappa)v(v+2-\eta) + 2\eta(1-\eta)(2-\eta-v)].$$

For $\sigma \geq \sigma^*$, $\bar{w} \leq 2\sigma$, so $\kappa \leq (\bar{w}/\sigma)\varphi(0) \leq 2\varphi(0) < 5/6$. The bracket is then positive for every $v > 0$ and $\eta \in (0, 1)$: as a quadratic in v it has positive leading coefficient $1-\kappa$ and constant term $2\eta(1-\eta)(2-\eta)$, and for $\kappa \leq 5/6$ either its linear coefficient $(1-\kappa)(2-\eta) - 2\eta(1-\eta)$ is nonnegative or its discriminant is negative. Q is therefore strictly decreasing. At the pessimistic end $v, \eta \rightarrow 0$ and $Q \rightarrow 2(1-\kappa) > 0$. At the other end $v \rightarrow 0$, $v \rightarrow +\infty$ and $Q \rightarrow -\infty$. $\square$

Fix a rent a and a point t on the path. Let $\tilde{m}$, V , γ_3 , γ_4 be the mean, variance, skewness and excess kurtosis of the posterior q_t , and write Q and Q_m for the function of Claim 17 and its m -derivative, evaluated at $\tilde{m}$. The counterpart of Q at the actual posterior is

$$\Psi(t) := \frac{\tilde{m}(1-\tilde{\eta})^2(1-\tilde{\kappa})}{V} \partial_t \log \Gamma(a, t),$$

where $\tilde{\eta}, \tilde{\kappa}$ are the quantities η, κ of Claim 17 evaluated at q_t , defined in the proof below. The normalization is chosen so that $\Psi = Q$ at a point mass.

Claim 18 (Expansion around the known complementarity). *For $\varsigma \geq \varsigma_{\text{exp}} := \max\{8\bar{e}^2/\sigma^2, 16/\underline{m}^2\}$,*

$$\begin{aligned} \Psi &= Q + c_1 \gamma_3 \sqrt{V} + R_1, & |R_1| &\leq K_1 V + O(\mu_-), \\ \partial_t \Psi &= V[Q_m + c_1(\gamma_4 - \gamma_3^2)] + R_2, & |R_2| &\leq K_2 V^{3/2} + O(\mu_-), \end{aligned}$$

where $c_1 = c_1(\tilde{m}, e, z^+)$ is given explicitly in the proof and K_1, K_2 are constants of the admissible parameter set.

Proof. Here $\tilde{\kappa} := \bar{w}M_{00}/\sigma$ and $\tilde{\eta} := (\bar{w}/\alpha\sigma^2)M_{00}\tilde{\mathbb{E}}[-\theta^2 u^+]$ are the κ, η of Claim 17 evaluated at the posterior, D denotes ∂_t along fixed a , and the hypothesis on ς ensures $V(e/\sigma)^2 \leq \frac{1}{8}$, $V \leq \tilde{m}^2/16$ and $\tilde{\mathbb{E}}[-\theta u^+]$ bounded away from 0, so that every denominator stays within a factor two of its point-mass value. Every object in Γ and $D\Gamma$ is a rational function of expectations $\mathbb{E}_q[F(\theta)]$ for explicit smooth F , namely the payment density, its moments against θ and u^+ , and the payment probability, of the reweighted posterior's means, and of their D -derivatives, which are covariances

by Claim 14.

The expansion. Expand each expectation around $\tilde{m}$ in central moments,

$$\mathbb{E}_q[F] = \sum_{j \leq 5} F^{(j)}(\tilde{m}) \frac{\mu_j}{j!} + O(V^3),$$

where $\mu_3 = \gamma_3 V^{3/2}$, $\mu_4 = (\gamma_4 + 3)V^2$ and $\mu_5 = (\gamma_5 + 10\gamma_3)V^{5/2}$. The reweighted posterior's cumulants are the tilt series $\tilde{\kappa}_k = \sum_j \beta^j \kappa_{k+j}/j!$, where $\beta := ev/\sigma$, corrected by the damping factor. Collecting powers of $\sqrt{V}$ gives the first display. The $\sqrt{V}$ coefficient contains γ_3 because every odd contribution does.

Differentiating along the path. For $D\Psi$ apply the family rule on the state $(\tilde{m}, e, z^+, \sqrt{V}, \gamma_3, \gamma_4, \gamma_5)$ with $D\sqrt{V} = \gamma_3 V/2$, $D\gamma_3 = \sqrt{V}(\gamma_4 - \frac{3}{2}\gamma_3^2)$, $D\gamma_4 = \sqrt{V}(\gamma_5 - 2\gamma_3\gamma_4)$ (from $\dot{\kappa}_k = \kappa_{k+1}$) and the rent-neutral $\dot{e}, \dot{z}^+$. In particular $D(\gamma_3\sqrt{V}) = V(\gamma_4 - \gamma_3^2)$, which produces the leading bracket.

The coefficients. $c_1 = N_1/(2\tilde{m})$, where, in units $\sigma = \bar{w} = 1$ and with η, ν, κ the quantities of Claim 17 at $(\tilde{m}, e, z^+)$,

$$N_1 = (1 - \eta)(1 - \kappa)[2\eta - 2\nu - \eta\nu - 2\eta^2 + \eta^2\nu + \eta\nu^2] + \eta(1 - \kappa)[2 + 2\eta - 4\nu + \eta\nu - 2\eta^2 + \eta\nu^2] - \kappa\eta(1 - \eta)(2 - \eta\nu) + \eta\nu(1 - \eta)(\kappa - \eta);$$

N_1 collects the third-derivative terms of every expectation entering Ψ , including the shift in the reweighted posterior's third cumulant from the damping factor. The constants K_1, K_2 collect the higher-order coefficients multiplied by the cumulant bounds of Claim 20, and the remainder is Taylor's theorem with $|F^{(6)}|$ bounded on the stated set, where every denominator stays within a factor two of its point-mass value. The remaining algebra is a direct computation and is omitted. $\square$

Claim 19 (Signs). *For $\eta \in (0, 1)$, $\nu > 0$, $\kappa \in [0, 5/6]$:*

- (i) *on $\{Q = 0\}$, $N_1 < 0$;*
- (ii) *$G := \text{bracket}/(1 - \eta) + N_1 > 0$ everywhere;*
- (iii) *with $\zeta := 1 - \eta$, on $\{Q = 0\}$, $|N_1|\zeta/\text{bracket} \leq \frac{1}{2}$.*

Proof. On the curve $\nu = 1 + \zeta(\zeta - \kappa)/(1 - \kappa)$ and $N_1 = \zeta A(\zeta, \kappa)/(1 - \kappa)$ with $A = -(\zeta^4 - 3\zeta^3 + 3\zeta^2 + \zeta + 2) + \kappa(4 + 5\zeta - \zeta^3) - 2\kappa^2(1 + 2\zeta - \zeta^2)$. A is concave in κ with vertex at $\kappa^*(\zeta) \geq 5/6$, since the equivalent condition $4 - 10\zeta + 20\zeta^2 - 6\zeta^3 \geq 0$ has no root on $[0, 1]$, so $\max_{[0, 5/6]} A = A(5/6)$, and $36A(5/6) = -2 + 14\zeta - 58\zeta^2 + 78\zeta^3 - 36\zeta^4$ has no root on $[0, 1]$ and equals -2 at 0: (i). For (ii), $\text{bracket} + (1 - \eta)N_1 = av^2 + bv + c$ with $a = (1 - \kappa)(1 + \eta - \eta^2) > 0$,

$c = 2\eta(1 - \eta)[(4 - 2\eta) - \kappa(3 - 2\eta)] > 0$, $b = \eta[-(2\eta^2 - 7\eta + 4) + \kappa(3\eta^2 - 7\eta + 3)]$. For $\eta \geq 0.95$, $b > 0$. For $\eta \leq 0.95$, $4ac - b^2 > 0$ on $\kappa \in [0, 5/6]$: its values at $\kappa = 0$ and $5/6$ are positive without roots, and its κ -derivative at $5/6$ is negative without roots. Root counts are by Sturm sequences on the stated intervals for the displayed polynomials. (iii) is (ii) on the curve. $\square$

Claim 20 (Cumulant bounds for log-concave laws). *Let X have a log-concave density on $\mathbb{R}$. With $p_0 := 1/e$, $q_0 := 1 - 1/e$,*

$$\gamma_3 \leq \frac{6/p_0^2 - 1/q_0^2}{(2/p_0 + 1/q_0)^{3/2}} =: C_3, \quad \gamma_4 \leq \frac{24(1/p_0^3 + 1/(8q_0^3))}{(2/p_0 + 1/q_0)^2} - 3 =: C_4,$$

$$|\gamma_5| \leq C_5, \quad |\gamma_5 - \gamma_3\gamma_4| \leq C_{53},$$

explicit, and $\gamma_4 - \gamma_3^2 \geq -186/125$.

Proof. Let $\bar{x} := \mathbb{E}X$, $Y_{\pm}$ the positive and negative parts of $X - \bar{x}$, $p := P(X > \bar{x}) \in [1/e, 1 - 1/e]$. The conditional laws of $Y_{\pm}$ are nonnegative and log-concave. For such Y , $m_k := \mathbb{E}[Y^k]/k!$ is log-concave in k (Borell, 1973), giving $m_2 \leq m_1^2$, $m_3 \leq m_2^2/m_1$, $m_4 \leq m_2^3/m_1^2$, $m_5 \leq m_2^4/m_1^3$, and Cauchy–Schwarz gives $m_2 \geq \frac{1}{2}m_1^2$, $m_3 \geq \frac{2}{3}m_2^2/m_1$, $m_4 \geq \frac{2}{3}m_2^3/m_1^2$, $m_5 \geq \frac{4}{9}m_2^4/m_1^3$. Mean zero gives $pm_1^+ = (1 - p)m_1^-$, normalized to 1; $V = 2pm_2^+ + 2(1 - p)m_2^-$, $\mu_3 = 6(pm_3^+ - (1 - p)m_3^-)$, $\mu_4 = 24(pm_4^+ + (1 - p)m_4^-)$, $\mu_5 = 120(pm_5^+ - (1 - p)m_5^-)$, $\kappa_5 = \mu_5 - 10\mu_3V$. Substituting them bounds each standardized cumulant by an explicit continuous function of (p, x, y) , $x := m_2^+/(m_1^+)^2$, $y := m_2^-/(m_1^-)^2$, on the compact box $[1/e, 1 - 1/e] \times [\frac{1}{2}, 1]^2$. The constants are the suprema over the box, attained at the corner $(1/e, 1, \frac{1}{2})$. For γ_3 , for instance, the function $[6x^2/p^2 - 4y^2/(1 - p)^2]/[2x/p + 2y/(1 - p)]^{3/2}$ is increasing in x , decreasing in y , and increasing as p decreases. The last inequality is the bound of Klaassen et al. (2000) for unimodal laws. $\square$

Claim 21 (Skewness near the endpoint of the support). *If $X \leq \bar{\theta}$ almost surely and $\bar{\theta} - \mathbb{E}X \leq c \text{sd}(X)$ with $c < 1/\sqrt{2}$, then $\mathbb{E}(X - \mathbb{E}X)^3 < 0$.*

Proof. Let $\mu := \mathbb{E}X$, $s := \text{sd}(X)$, $A := \mathbb{E}[(X - \mu)_+^2] \leq (cs)^2$. Then $\mathbb{E}[(X - \mu)_+^3] \leq csA$, and by Cauchy–Schwarz $\mathbb{E}[(\mu - X)_+^3] \geq \mathbb{E}[(\mu - X)_+^2]^2/\mathbb{E}[(\mu - X)_+] \geq (s^2 - A)^2/(cs)$. Hence $\mathbb{E}(X - \mu)^3 < 0$ whenever $(s^2 - A)^2 > c^2s^2A$, which holds for all $A \leq c^2s^2$ iff $(1 - c^2)^2 > c^4$, i.e. $c < 1/\sqrt{2}$. $\square$

Lemma 5 (Single peak of the rent). *Under Assumption 3 with $\sigma \geq \sigma^*$ and $\varsigma \geq \max\{\varsigma_{\text{exp}}, \varsigma_{\text{bulk}}, \varsigma_{\text{sd}}\}$ (Section A), for every a in the range of U the section $\Gamma(a, \cdot)$ is single-peaked along the family. Hence U is single-peaked, with peak at a belief b° , and $\beta < 0$ on $[\bar{b}, b^\circ)$, $\beta > 0$ on $(b^\circ, +\infty)$ where nonempty. If $\alpha \geq \bar{\alpha}$, U is increasing along the whole family and $\beta \leq 0$ throughout. If $\alpha \leq \underline{\alpha}$ (the boundary at $\underline{\theta}$), decreasing and $\beta \geq 0$ throughout.*

Proof. Fix a and $\delta \in (0, 1)$.

Near the crossing. For $|Q(\tilde{m})| \leq \delta$, by Claims 18, 19(ii) and the last inequality of Claim 20, $D\Psi \leq -(V_t/\tilde{m})[\text{bracket}/(1-\eta) + \frac{93}{125}N_1] + K_2V_t^{3/2} + O(\mu_-)$. The bracket exceeds $G > 0$ since $N_1 < 0$ near the curve, so $D\Psi < 0$ once $K_2\sqrt{V_t} + O(\mu_-)/V_t < G/\tilde{m}$, which is $\varsigma \geq \varsigma_{\text{bulk}}$.

Away from the crossing. For $|Q| \geq \delta$, $|\Psi - Q(\tilde{m})| \leq (|N_1|C_3/(2\tilde{m}))\sqrt{V_t} + K_1V_t + O(\mu_-) < \delta$ for $\varsigma \geq \varsigma_{\text{sd}}$, so Ψ has the sign of Q .

Where the expansion fails. Here $\tilde{\mathbb{E}}[-\theta u^+] \rightarrow 0$, so the coefficient c_1 of Claim 18, which carries that quantity in a denominator, is unbounded and the expansion no longer applies. Its D -derivative is bounded away from zero, because its last term $-\tilde{\mathbb{E}}[\theta(\dot{e}\theta - \dot{z}^+)]/\sigma$ is bounded below by the v_m of Claim 17 up to $O(V)$ and the covariances are $O(V)$, so $\partial_t \log \hat{e}_a \rightarrow -\infty$ and $\Psi \rightarrow -\infty$.

The single peak. Since Q decreases from positive to negative along the section, Ψ single-crosses from $+$ to $-$ and Claim 16 gives the single peak of U . β has the sign of $-U'$ up to $O(\mu_-)$.

The monotone cases. Let $\alpha \geq \bar{\alpha}$. Then $Q(\tilde{m}; \alpha) > 0$ along the family with $Q_m < 0$, so $Q(\tilde{m}) \geq Q(\bar{\theta}; \alpha) + |Q_m|_{\min}(\bar{\theta} - \tilde{m})$, and $Q(\bar{\theta}; \alpha) \geq 0$, with equality only at $\bar{\alpha}$, while $m(\gamma) < \bar{\theta}$ at every prior. Near the top the leading correction is $(N_1/2\tilde{m})\gamma_3\sqrt{V_t}$ with $N_1 < 0$, since the top point at $\alpha = \bar{\alpha}$ lies on the crossing curve. If $\bar{\theta} - \tilde{m} \leq \sqrt{V_t}/\sqrt{2}$, $\gamma_3 < 0$ by Claim 21 and the correction is positive, so $\Psi \geq Q - O(V_t) > 0$. If $\bar{\theta} - \tilde{m} \geq \sqrt{V_t}/\sqrt{2}$, by Claims 20 and 19(iii) (with a continuity margin) the correction is at most $\frac{1}{2}(1+\epsilon)|Q_m|\sqrt{V_t}$ while $Q(\tilde{m}) - Q(\bar{\theta}) \geq |Q_m|_{\min}\sqrt{V_t}/\sqrt{2}$. Since $1/\sqrt{2} > \frac{1}{2}$, $\Psi > 0$ for $\varsigma \geq \varsigma_{\text{sd}}$. In the interior $|\Psi - Q| \leq C/\sqrt{\varsigma}$ is dominated by Q . The case $\alpha \leq \underline{\alpha}$ is the mirror argument at the lower endpoint of the support. $\square$

The manipulation term

Claim 22 (Two forms). *With the realized index of state θ Gaussian around $\hat{b}_\theta := \gamma + \omega\theta\hat{e}_1$,*

$$\begin{aligned} T(\gamma) &= \frac{1}{\omega\sigma^2} \mathbb{E}[\theta u_2^*(\theta, b)(b - \hat{b}_\theta) \mathbf{1}\{b \geq \bar{b}\}] \\ &= \omega \left[r(\bar{b}) L_\gamma(\bar{b}) - \int_{\bar{b}}^{\infty} \beta(b) L_\gamma(b) db \right], \quad r(\bar{b}) := \mathbb{E}_{q_{\bar{b}}}[\theta u_2^*]. \end{aligned}$$

Proof. For the Gaussian density f_θ of the realized index at state θ , $\partial_e f_\theta = \omega\theta(b - \hat{b}_\theta)f_\theta/(\omega\sigma)^2$. The first form follows by differentiating the continuation in e at the conjecture and averaging over θ . The second follows by integrating by parts in b and applying Bayes' rule in the form $\int \psi f_\theta g_\theta(\theta | \gamma) d\theta = L_\gamma(b) \mathbb{E}_{q_b}[\psi]$. It is Lemma 3 in index coordinates. $\square$

Lemma 6 (Retention dominates at initiation). *Let γ_{bar} be the prior whose expected index is the margin, $\mathbb{E}_{\gamma_{\text{bar}}}[b] = \bar{b}$. For $\varsigma \geq \varsigma_{\text{init}}$:*

- (i) $T(\gamma) > 0$ for all $\gamma \leq \gamma_{\text{bar}}$;
- (ii) $\gamma_L \leq \gamma_{\text{bar}}$.

Hence $T(\gamma_L) > 0$.

Proof. Two conditions are used and verified in Step 5: for some $k > 0$,

$$\left(\frac{k\hat{e}_1}{\sigma\sqrt{\varsigma}}\right)^2 < \frac{2u_2^-}{u_2^+} \quad \text{and} \quad OV(\gamma_{\text{bar}}) > D_1(\gamma_{\text{bar}}) + |\Pi^{\text{st}}(\gamma_{\text{bar}}) - \pi_A|, \quad (33)$$

where $u_2^\pm$ are the minimum and maximum of u_2^* over retained index values within $k/\sqrt{\varsigma}$ of the mean, $OV := \mathbb{E}[(\Pi_2 - \pi_A)^+]$ is the option value of learning, D_1 the principal's learning distortion, and Π^{st} her static first-period profit at the prior. The objects in the second condition are defined in Step 3.

Step 1: state by state. Fix $\gamma \leq \gamma_{\text{bar}}$ with prior mean m , and use the first form of Claim 22: T is a sum over states θ of θ times the retained part of the derivative of the state's index density. The state's expected index is $\hat{b}_\theta = \bar{b} + c_\theta$ with $c_\theta := \omega\hat{e}_1(\theta - m)$, so $c_\theta \leq 0$ for $\theta \leq m$ since $\mathbb{E}_\gamma[b] \leq \bar{b}$.

Step 2: states at or below the mean. For $\theta \leq m$ every retained index value satisfies $b - \hat{b}_\theta \geq 0$, so the derivative of the state's density is nonnegative on the retained set. For θ within $k/\sqrt{\varsigma}$ of m , the state-conditional rent $u_2^*(\theta, \cdot)$ is within a bounded multiple of $\sqrt{V}$ of the average rent $U_2 > 0$ (Claim 18), hence positive for ς large. Such states contribute positively.

Step 3: states above the mean. For $\theta > m$, with $x := b - \hat{b}_\theta$ and $\varsigma_b := \omega\sigma$, the retained set contains index values on both sides of $\hat{b}_\theta$, and the state's contribution is at least

$$\frac{\theta}{\omega\sigma^2} \left[\frac{u_2^- \varsigma_b}{\sqrt{2\pi}} - \frac{u_2^+ c_\theta^2}{2\varsigma_b \sqrt{2\pi}} \right],$$

positive whenever $((\theta - m)\hat{e}_1/\sigma)^2 < 2u_2^-/u_2^+$, which the first condition in (33) gives for $\theta - m \leq k/\sqrt{\varsigma}$.

Step 4: the remaining states. Under a ς -strongly log-concave posterior the mass of states with $|\theta - m| > k/\sqrt{\varsigma}$ is at most $e^{-k^2/2}$, and their contribution is bounded by a constant times that mass. Counterproductive states contribute at most a constant times their retained mass, $O(\mu_-)$. For k large both are dominated by Steps 2–3, which proves (i) at γ_{bar} . Lowering γ moves every $\hat{b}_\theta$ down, which enlarges the positive contributions and shrinks the negative ones, so (i) holds at all $\gamma \leq \gamma_{\text{bar}}$.

Step 5: initiation. The principal's value from initiating is $\mathcal{V}(\gamma) = \Pi_1(\gamma) + \pi_A + OV(\gamma)$, with $\Pi_1 = \Pi^{\text{st}} + S_1 - D_1$: $S_1 \geq 0$ is the reduction in the wage bill for the implemented target relative to buying its whole marginal cost, nonnegative because $T > 0$ at every conjecture at γ_{bar} , Step 4 having been proved at a fixed conjecture without regard to which, and $D_1 \geq 0$ is the loss from moving the target away from the static optimum. Hence

$$\mathcal{V}(\gamma_{\text{bar}}) - 2\pi_A \geq OV - D_1 - |\Pi^{\text{st}} - \pi_A| > 0$$

under the second condition in (33), and $\gamma_L \leq \gamma_{\text{bar}}$ by the definition of γ_L as the smallest prior with $\mathcal{V} \geq 2\pi_A$.

Step 6: the conditions. The posterior $q_{\bar{b}}$ has mean $m(\gamma_{\text{bar}}) + O(V^{3/2})$, since the posterior mean is a martingale in b and $m'' = \kappa_3$ along the family, and variance below the prior's by $O(V^2)$, so $|\Pi^{\text{st}} - \pi_A| = |\Pi^{\text{st}} - \Pi_2(q_{\bar{b}})| = O(V^{3/2})$; $D_1 = O(\Pi_2'^2) = O(V^2)$; and $OV = \Pi_2'(\bar{b}) \mathbb{E}[(b - \bar{b})^+] (1 + o(1)) = cV(1 + o(1))$ with $c > 0$, since $\Pi_2' \asymp V$ and $\mathbb{E}[(b - \bar{b})^+] = \varsigma_b / \sqrt{2\pi}$ at the margin. The first condition in (33) holds with $k = \sqrt{2 \log(1/\epsilon)}$ once $\varsigma \geq k^2 \bar{e}^2 u_2^+ / (2\sigma^2 u_2^-)$, and the second once cV exceeds the lower-order terms. ς_{init} is the larger of the two levels. $\square$

Claim 23 (Optimistic end). *If $\alpha < \bar{\alpha}$, $T(\gamma) < 0$ for all γ large.*

Proof. The retention term. $\bar{b}$ is fixed by π_A , so the retention term contains $L_\gamma(\bar{b}) \leq C e^{-\lambda(\gamma - \bar{b} - c)^2/2}$ by Claim 15: Gaussian in γ .

The wage effect. It integrates β against a kernel concentrated on a window of width $O(\lambda^{-1/2})$ around $\gamma + c$. At the top of the family the posterior concentrates at $\bar{\theta}$: the base $g_0 = e^{-\psi}$ with $\psi'' \geq \varsigma$ is bounded and bounded away from zero on its compact support, so its tilts place their mass within $O(1/\gamma)$ of $\bar{\theta}$ and $V(\gamma) \asymp \gamma^{-2}$. By Claim 14, β is a tilt derivative, so $\beta = c(\alpha)V(\gamma)(1 + o(1))$ with $c(\alpha)$ minus the known-complementarity slope of the rent in the mean at $\bar{\theta}$. For $\alpha < \bar{\alpha}$ that slope is negative, since $\bar{\alpha}$ is the curvature at which the two effects on the rent balance at $\bar{\theta}$ and below it the tightening of the contract dominates. So $c(\alpha) > 0$ and the wage effect is positive and of order γ^{-2} .

The retention term therefore vanishes at a Gaussian rate in γ while the wage effect vanishes only polynomially, so the wage effect dominates for γ large and $T < 0$. $\square$

Claim 24 (Down-crossing). *At any zero of T (conjecture fixed), $T'(\gamma) = -M(\gamma)$ with*

$$\begin{aligned} M(\gamma) &:= \lambda\omega \int_{\bar{b}}^{\infty} (b - \bar{b})\beta(b)L_\gamma(b)db \\ &\geq \lambda\omega \max\left\{(b^\circ - \bar{b})^+ r(\bar{b})L_\gamma(\bar{b}), \delta[r(\bar{b})L_\gamma(\bar{b}) - \delta\bar{\beta}\bar{L}]\right\} > 0 \end{aligned}$$

for any $\delta > 0$, with $\bar{\beta} := \sup_{[\bar{b}, \bar{b} + \delta]} \beta$ and $\bar{L} := \sup L$, the second expression positive for δ small.

Proof. Differentiate the second form of Claim 22 under the integral, using Claim 15 for the derivative of the index density. At a zero of T , $rL_\gamma(\bar{b}) = \int \beta L$, so the terms in $\mathbb{E}_\gamma b$ cancel and $T' = -M$. By Lemma 5, $\beta < 0$ on $[\bar{b}, b^\circ)$ and > 0 beyond. Since $b - \bar{b} \leq b^\circ - \bar{b}$ on the first set and $\geq$ on the second, $\int (b - \bar{b})\beta L \geq (b^\circ - \bar{b}) \int \beta L$. When $\bar{b} \geq b^\circ$, $\beta \geq 0$ on the retained range and $\int (b - \bar{b})\beta L \geq \delta \int_{\bar{b}+\delta}^\infty \beta L \geq \delta [\int \beta L - \delta \bar{\beta} \bar{L}]$. Both bounds are positive since $\int \beta L = r(\bar{b})L_\gamma(\bar{b}) > 0$ at a zero. $\square$

Claim 25 (Conjecture correction). *In equilibrium $\hat{e}_1 = \hat{e}_1(\gamma)$ and $dT/d\gamma = T'(\gamma) + \hat{e}'_1(\gamma) \partial_{\hat{e}_1} T$, with $|\hat{e}'_1| \leq C_E V(\gamma)$ and $|\partial_{\hat{e}_1} T| \leq N^*$ for $c'' > \alpha^E$. Hence at every zero of T , $dT/d\gamma < 0$ once $C_E N^* V(\gamma) < M(\gamma)$, which holds for $\varsigma \geq \varsigma_{\text{conj}}$.*

Proof. The equilibrium conjecture is the agent's fixed point at the principal's optimal cutoff, $\hat{e}_1(\gamma) = e(\kappa^*(\gamma); \gamma)$, so

$$\hat{e}'_1(\gamma) = \partial_\gamma e + \partial_\kappa e \cdot \frac{d\kappa^*}{d\gamma}.$$

The agent at a fixed cutoff. $e(\kappa; \gamma)$ solves $F(e, \gamma; \kappa) = 0$, so $\partial_\gamma e = -\partial_\gamma F / \partial_e F$. $\partial_e F$ is bounded below by the concavity slack and $\partial_\gamma F$ is a tilt covariance of the prior, $O(V(\gamma))$ by Claim 14.

The principal's cutoff. $\kappa^*(\gamma)$ solves $\partial_\kappa \tilde{\Pi}(\kappa; \gamma) = 0$, so $d\kappa^*/d\gamma = -\partial_{\kappa\gamma}^2 \tilde{\Pi} / \partial_\kappa^2 \tilde{\Pi}$. The denominator is bounded away from zero by the strict concavity of Theorem 2, Step 5, and the numerator enters through the output distribution and the continuation, again a tilt covariance, $O(V(\gamma))$. $\partial_\kappa e$ is bounded by Step 3 of that proof.

Hence $|\hat{e}'_1| \leq C_E V(\gamma)$, with C_E including the reciprocal of the concavity modulus, and $|\partial_{\hat{e}_1} T| \leq N^*$ is the class bound. Combine with Claim 24. $\square$

Lemma 7 (Pass-through). *Fix γ . Let $e^B(\gamma)$ and $z^B(\gamma)$ be the benchmark's equilibrium target and cutoff at that prior, $T(\gamma; \tilde{e})$ the manipulation term at conjecture $\tilde{e}$, and $C(I; \tilde{e})$ the least-cost wage bill delivering marginal incentive I at target $\tilde{e}$.*

- (i) *Facing $z^B(\gamma)$, the model's agent chooses e_1 with $\text{sgn}(e_1 - e^B(\gamma)) = \text{sgn } T(\gamma; e^B(\gamma))$.*
- (ii) *$\text{sgn}(e^*(\gamma) - e^B(\gamma)) = \text{sgn } \Lambda(\gamma)$, where*

$$\Lambda(\gamma) := T(\gamma; e^B(\gamma)) + \epsilon(\gamma) \partial_{\tilde{e}} T(\gamma; e^B(\gamma)), \quad \epsilon := \frac{C_I}{c'' C_{II} + C_{I\tilde{e}}} > 0,$$

all objects evaluated at $(\gamma, e^B(\gamma), z^B(\gamma))$.

- (iii) *For $\varsigma \geq \varsigma_{\text{conj}}$ every zero of Λ is a strict down-crossing, and any zero γ^* of Λ lies within $\delta^* := \sup |\epsilon \partial_{\tilde{e}} T| / \inf |dT/d\gamma|$ of the zero γ^{**} of $T(\gamma; e^*(\gamma))$, the suprema over a neighborhood of γ^{**} .*

Proof. (i) The two agents' conditions are $c'(e_1) = B + T$ and $c'(e^B) = B$, with B the marginal benefit through the current wage. c' is increasing and both objectives are single-peaked for $\sigma \geq \sigma^*$.

(ii) Both principals maximize $\tilde{e} m_\gamma - C(\cdot; \tilde{e}) + \Psi_1(\tilde{e})$ with the same continuation Ψ_1 . The model's principal needs incentive $c'(\tilde{e}) - T(\tilde{e})$ from the wage where the benchmark's needs $c'(\tilde{e})$, so

$$\Pi^S(\tilde{e}) - \Pi^B(\tilde{e}) = C(c'(\tilde{e}); \tilde{e}) - C(c'(\tilde{e}) - T(\tilde{e}); \tilde{e}).$$

Differentiate at e^B , where $d\Pi^B/d\tilde{e} = 0$, applying the mean value theorem to C_I and $C_{\tilde{e}}$ on an interval of length T :

$$\left. \frac{d\Pi^S}{d\tilde{e}} \right|_{e^B} = T[c''C_{II}(\xi) + C_{I\tilde{e}}(\xi')] + C_I \partial_{\tilde{e}} T.$$

On the threshold family $C_I = 1/\kappa > 0$ by the pricing identity of Claim 13 and $C_{II} = 1/(\kappa^3 \bar{w} \mathcal{B}(\kappa)) > 0$, the score density $\mathcal{B}$ being positive, and for c'' above the level imposed in Step 1 of the assembly below, $c''C_{II} \geq 2|C_{I\tilde{e}}|$, the bracket is positive, so the sign of the derivative is the sign of Λ . Strict concavity of Π^S in the target converts the sign of the derivative at e^B into the sign of $e^* - e^B$.

(iii) By Proposition 2, $e^B(\gamma)$ is C^1 with $|de^B/d\gamma| \leq C_E V(\gamma)$: as in Claim 25, the benchmark effort at a fixed cutoff and the benchmark's optimal cutoff both move with the prior through tilt covariances, over the agent's concavity slack and the principal's concavity modulus. Then

$$\frac{d\Lambda}{d\gamma} = \partial_\gamma T + \frac{de^B}{d\gamma} \partial_{\tilde{e}} T + \frac{d}{d\gamma} [\epsilon \partial_{\tilde{e}} T].$$

At a zero of Λ , $|T(\gamma; e^B)| = |\epsilon \partial_{\tilde{e}} T|$ is small, so the first term is $-M(\gamma)$ up to a term of order $\sup |\epsilon \partial_{\tilde{e}} T|$ by Claim 24. The second is at most $C_E N^* V(\gamma)$. The third is bounded by class constants times $V(\gamma)$. For $\varsigma \geq \varsigma_{\text{conj}}$ the first dominates, and every zero is a strict down-crossing. Since $|T(\gamma; e^B) - T(\gamma; e^*)| \leq N^* |e^B - e^*|$ vanishes at γ^{**} , the distance bound is the mean value theorem applied to $T(\cdot; e^*(\cdot))$ between γ^{**} and γ^* . $\square$

Assembling the proof

Proof of Theorem 3. Step 1: the correction. Let $\Lambda := T + \epsilon \partial_{\tilde{e}} T$ be the corrected comparison of Lemma 7(ii), whose sign is the sign of $e^* - e^B$. In Claim 22 the target enters T in three places: the coefficient $\omega = \tilde{e}/\sigma^2$ on the whole expression, the density L of the realized index, whose mean $\gamma + \iota m$, with $\iota := \tilde{e}^2/\sigma^2$, and scale $s = \tilde{e}/\sigma$ move with $\tilde{e}$; and the modified base, which changes $r(\bar{b})$ and β by relative amounts of order $\tilde{e}\theta_{\max}^2/\sigma^2$. The first two cancel, since $\omega L_\gamma(b) = \varphi(u)/\sigma$ at standardized distance u from the expected index, so each term of T depends on $\tilde{e}$ only through u

and the modification, and

$$\partial_{\bar{e}} \log(\omega L) = \frac{u^2}{\bar{e}} + \frac{2mu}{\sigma} \geq -\frac{m^2 \bar{e}}{\sigma^2}.$$

Hence the $\bar{e}$ -derivative of each term of T is at least $-C_T$ times that term, with $C_T := \theta_{\max}^2 \bar{e} (1 + c_{\text{base}})/\sigma^2$ and c_{base} the constant bounding the modified base's relative effect. The bound is two-sided: the factor $u^2/\bar{e}$ in $\partial_{\bar{e}} \log(\omega L)$ is integrated against $\varphi(u)$, and $\sup u^2 \varphi(u)$ is finite, so after enlarging C_T by a fixed factor the derivative of each term is also at most C_T times the term. For ϵ , on the threshold family $C_I = 1/\kappa$ and $C_{II} = 1/(\kappa^3 \bar{w} \mathcal{B}(\kappa))$ with κ the score at the cutoff (Step 5 of the proof of Theorem 2), so $C_I/C_{II} = \kappa^2 \bar{w} \mathcal{B}(\kappa)$ is bounded on the admissible parameter set by a constant c_ϵ . For $c'' \geq 2 \sup |C_{I\bar{e}}|/\inf C_{II}$, a level included in σ_T below, the bracket in ϵ is at least $c'' C_{II}/2$, so $\epsilon \leq 2c_\epsilon/c''$. Hence

$$\epsilon C_T \leq \frac{2c_\epsilon \theta_{\max}^2 \bar{e} (1 + c_{\text{base}})}{c'' \sigma^2} < \frac{1}{3} \quad \text{for } \sigma \geq \sigma^*,$$

at every admissible curvature. Therefore: in case (ii), where both terms of T are nonnegative, $\Lambda \geq \frac{2}{3}T$ at every prior. In case (i), $\Lambda \geq \frac{2}{3}T$ at γ_L , where the atom dominates, and $\Lambda \leq \frac{2}{3}T$ at optimistic priors, where the atom is negligible against the integral of the wage effect.

Step 2: case (i). By Lemma 6, $T(\gamma_L) > 0$. By Claim 23, $T < 0$ at optimistic priors. T is continuous in γ , so it has a zero, and by Lemma 5 and Claims 24–25 every zero is a strict down-crossing, so the zero γ^{**} is unique. By Step 1, $\Lambda(\gamma_L) > 0$ and $\Lambda < 0$ at optimistic priors. By Lemma 7(iii) every zero of Λ is a down-crossing. Hence Λ has exactly one zero γ^* , within δ^* of γ^{**} , with $e^* > e^B$ on $[\gamma_L, \gamma^*)$ and $e^* < e^B$ beyond.

Step 3: case (ii). For $\alpha \geq \bar{\alpha}$, Lemma 5 gives $\beta \leq 0$ in the family, so both terms of T are nonnegative and $T > 0$ at every prior. By Step 1, $\Lambda \geq \frac{2}{3}T > 0$ at every prior. $\square$

The constants

The admissible parameter set. Every constant is first obtained along the path and then replaced by a bound in primitives, in the direction that keeps the condition sufficient: $e, \hat{e}_1 \leq \bar{e}$; retained means $\tilde{m} \in [\underline{m}, \bar{\theta}]$. The standardized bar, hence η and ν , in the interval spanned by the known-complementarity bar $\alpha\sigma/m^2$, $m \in [\underline{m}, \bar{\theta}]$, with its explicit corrections, so $\eta \leq \bar{\eta} < 1$ and ν in a compact interval away from the end of V ; $\kappa \leq (\bar{w}/\sigma)\varphi(0)$; precision of the reweighted posterior $\geq \varsigma$; $\mu_- \leq C_0 e^{-\varsigma \underline{m}^2/2}$. The denominators of Claim 18 bounded below by explicit functions of the admissible parameter set. Hence $\varsigma^* = \varsigma^*(\alpha, \bar{w}, \sigma, \underline{\theta}, \bar{\theta}, \pi_A)$.

σ^* . $\sigma^*(\alpha, \bar{w}, \bar{\theta}) := \max\{\bar{w}/2, (c_U/((1 - \bar{g})\alpha))^{1/2}, \sigma_T, \sigma_A\}$, with $c_U, \bar{g}$ from Claim 29 in the proof of Proposition 3, σ_T the level at which $\epsilon C_T < \frac{1}{3}$ in Step 1 above, and σ_A the level in the proof

of Proposition 4. The first term gives $\bar{w} \leq 2\sigma$, hence $\kappa \leq 2\varphi(0) < 5/6$ in Claim 17. The second gives $\alpha > \alpha^E$, existence and uniqueness uniformly over the family (Theorem 2 and Claim 30).

ς^* . $\varsigma^* := \max\{\varsigma_{\text{exp}}, \varsigma_{\text{bulk}}, \varsigma_{\text{sd}}, \varsigma_{\text{init}}, \varsigma_{\text{conj}}\}$, where $\varsigma_{\text{exp}} := \max\{8\bar{e}^2/\sigma^2, 16/\underline{m}^2\}$ together with the requirements of Claim 18 that keep $\widetilde{\mathbb{E}}[-\theta u^+]$ bounded away from zero;

$$\varsigma_{\text{bulk}} := \sup_{|Q| \leq \delta} \left(\frac{2\tilde{m}K_2}{G} \right)^2, \quad \varsigma_{\text{sd}} := \sup_{|Q| \geq \delta} \max \left\{ \left(\frac{|N_1|C_3}{\tilde{m}\delta} \right)^2, \frac{2K_1}{\delta} \right\}, \quad \varsigma_{\text{conj}} := \sup \frac{C_E N^*}{M(\gamma)};$$

ς_{init} is from Step 6 of Lemma 6, and the supremum in ς_{conj} is over a neighborhood of the crossing. Each supremum is over the admissible parameter set and is finite.

Proof of Proposition 3 (The two-sided case is nonempty under enough noise)

Take the complementarity known and equal to $\bar{\theta}$, so that first-period output is $y_1 \sim N(\bar{\theta}e, \sigma^2)$ and the retention cutoff of Lemma 2 is $\underline{y}(\hat{e}) = (\bar{b} - \gamma)\sigma^2/\hat{e}$ in index units. Write $W(e; z_1) = \bar{w}\bar{\Phi}((z_1 - \bar{\theta}e)/\sigma)$ for the expected wage and $R(e; \hat{e}) = u_2^\infty \bar{\Phi}(u)$, $u := (\underline{y}(\hat{e}) - \bar{\theta}e)/\sigma$, for the agent's continuation value from retention, with $u_2^\infty = \bar{w}\bar{\Phi}(u^*) - \alpha e_{\text{st}}^2/2$ the second-period rent, $u^* = \alpha\sigma/\bar{\theta}^2$ and $e_{\text{st}} = \bar{w}\bar{\theta}\varphi(u^*)/(\alpha\sigma)$ the known-complementarity bar and effort.

Claim 26 (Three curvature constants).

$$W^* := \sup |W_{ee}| = \frac{\bar{w}\bar{\theta}^2\varphi(1)}{\sigma^2}, \quad K^* := \sup |R_{ee}| = \frac{u_2^\infty\bar{\theta}^2\varphi(1)}{\sigma^2},$$

$$N^* := \sup |R_{e\hat{e}}| \leq \frac{u_2^\infty\bar{\theta}^2\varphi(1)}{\sigma^2} + \frac{2\varphi(\sqrt{2})u_2^\infty\bar{\theta}}{\hat{e}\sigma}.$$

Proof. $W_{ee} = -\bar{w}(\bar{\theta}^2/\sigma^2)x\varphi(x)$ at $x = (z_1 - \bar{\theta}e)/\sigma$, and $\sup |x\varphi(x)| = \varphi(1)$; $R_{ee} = -u_2^\infty(\bar{\theta}^2/\sigma^2)u\varphi(u)$. Since $\partial_{\hat{e}}\underline{y} = -\underline{y}/\hat{e}$ and $\underline{y} = \bar{\theta}\hat{e} + u\sigma$, $R_{e\hat{e}} = u_2^\infty u\varphi(u)\bar{\theta}\underline{y}/(\sigma^2\hat{e}) = u_2^\infty\bar{\theta}^2 u\varphi(u)/\sigma^2 + u_2^\infty\bar{\theta}u^2\varphi(u)/(\sigma\hat{e})$, and $\sup u^2\varphi(u) = 2\varphi(\sqrt{2})$. $\square$

Claim 27 (Ordering of the thresholds). (i) If $\alpha > W^*$ the agent's static problem is strictly concave.

(ii) If $\alpha > W^* + K^*$ his full first-period problem is strictly concave.

(iii) If $\alpha > \alpha_U := W^* + K^* + N^*$ the fixed point $e(\hat{e}) = \hat{e}$ is unique for every contract, so the equilibrium effort is a single-valued function of the contract and the principal's reduced objective is well defined. Its strict concavity is a separate requirement, included in α^E .

The three levels are ordered, $W^* \leq W^* + K^* \leq \alpha_U$, and the gap between the second and the third is N^* , the sensitivity of the agent's incentive to the principal's conjecture. Curvature above the

second level makes the agent's best response unique at a given conjecture. Making the conjecture itself unique requires the third. When $N^* > 0$ the two differ: for $\alpha \in (W^* + K^*, \alpha_U]$ the agent's problem is concave at every conjecture, yet the map from conjecture to best response may have more than one fixed point.

Proof. The agent's second derivative is $-\alpha + W_{ee} + R_{ee} \leq -\alpha + W^* + K^*$: (i)–(ii). Differentiating the first-order condition, $e'(\hat{e}) = R_{e\hat{e}}/(\alpha - W_{ee} - R_{ee})$, so $|e'| \leq N^*/(\alpha - W^* - K^*) < 1$ iff $\alpha > \alpha_U$, and a contraction on a compact interval has one fixed point: (iii). $\square$

Claim 28 (The regime boundary falls like $\sigma^{-3/2}$). $\bar{\alpha} = \bar{\theta}^2 \sqrt{\bar{w}\varphi(0)/2} \sigma^{-3/2} (1 + O(\sigma^{-1}))$.

Proof. Write $u := \alpha\sigma/\bar{\theta}^2$ and $\kappa := \bar{w}\varphi(u)/\sigma$, the incentive price of Claim 17, so that $\bar{\alpha}$ solves $2u^2(1-\kappa) = \kappa$. As $\sigma \rightarrow +\infty$, $\kappa \leq \bar{w}\varphi(0)/\sigma \rightarrow 0$, hence $u \rightarrow 0$ and $u^2 = \bar{w}\varphi(0)/(2\sigma) (1 + O(\sigma^{-1}))$. Substituting back, $\bar{\alpha} = u\bar{\theta}^2/\sigma$ has the stated order. $\square$

Claim 29 (The conjecture-sensitivity term is a fixed fraction of α). *At the equilibrium effort $\hat{e} = e_{st}$, the second term of N^* equals $2\varphi(\sqrt{2})u_2^\infty\alpha/(\bar{w}\varphi(u^*)) \leq \bar{g}\alpha$ with $\bar{g} := 2\varphi(\sqrt{2})\bar{\Phi}(0)/\varphi(0) \approx 0.36$, uniformly in σ and α . Hence $\alpha_U \leq c_U/\sigma^2 + \bar{g}\alpha$ with $c_U := 2\bar{w}\bar{\theta}^2\varphi(1)$, and $\alpha > \alpha_{kn}^E(\sigma) := c_U/((1 - \bar{g})\sigma^2)$ suffices for uniqueness in the known-complementarity model.*

Proof. Substitute $e_{st} = \bar{w}\bar{\theta}\varphi(u^*)/(\alpha\sigma)$ in the second term of Claim 26: $2\varphi(\sqrt{2})u_2^\infty\bar{\theta}/(e_{st}\sigma) = 2\varphi(\sqrt{2})u_2^\infty\alpha/(\bar{w}\varphi(u^*))$. Since $u_2^\infty \leq \bar{w}\bar{\Phi}(u^*)$, this is at most $\alpha \cdot 2\varphi(\sqrt{2})\bar{\Phi}(u^*)/\varphi(u^*)$, and the Mills ratio $\bar{\Phi}/\varphi$ is decreasing on $[0, +\infty)$ with value $\bar{\Phi}(0)/\varphi(0)$ at 0. The remaining terms of α_U , namely W^* , K^* and the first term of N^* , are together at most $2\bar{w}\bar{\theta}^2\varphi(1)/\sigma^2$, since $u_2^\infty \leq \bar{w}\bar{\Phi}(u^*) \leq \bar{w}/2$. Solving $\alpha > c_U/\sigma^2 + \bar{g}\alpha$ gives the threshold. $\square$

Claim 30 (The uniform existence threshold at concentrated priors). *For $\sigma \geq \sigma^{**}$ there is $\varsigma^{**}(\sigma)$ such that for $\varsigma \geq \varsigma^{**}(\sigma)$, the existence threshold $\alpha^E(\gamma)$ of Theorem 2 at every prior in the family, and hence $\sup_\gamma \alpha^E(\gamma)$, is at most $\alpha_{kn}^E(\sigma) + \eta(\sigma, \varsigma)$, with $\eta(\sigma, \varsigma) \rightarrow 0$ as $\varsigma \rightarrow +\infty$ at fixed σ .*

Proof. The threshold of Theorem 2 is the largest of four levels, each bounded below. Throughout, at a cutoff at standardized position v above expected output, the score is $\kappa \approx \bar{\theta}v/\sigma$ and its density is $\mathcal{B}(\kappa) \approx \varphi(v)\sigma/\bar{\theta}$, up to relative corrections of order $\sqrt{V}$ at a concentrated prior (Claim 18).

First-order approach. By Claims 3–5 in the proof of Proposition 1, the threshold α^{FOA} is built from $\bar{w}$, $\theta_{\max}$, $\tau_{\max}^2 \leq (\bar{\theta} - \underline{\theta})^2/4$ and the moments ν_1, ν_2 of the Gaussian noise, which are $O(\sigma^{-1})$ and $O(\sigma^{-2})$, and from the constants $J_1, J_2, \Lambda_{\text{FOA}}$ of Claim 4, each a bound on a derivative of the posterior in e_1 , which under Gaussian noise contributes one factor $1/\sigma$ per derivative. None depends on the shape of the prior beyond its support, so the threshold is $O(\sigma^{-2})$ uniformly over the family.

Configuration. The two conditions defining α^M in the proof of Lemma 2 are satisfied for $\sigma \geq \sigma^{**}$ at every curvature, as recorded below.

Contraction. By (31), uniqueness needs $\bar{N} < \alpha - C_1 - C_2/\alpha^2$, where C_1 and C_2 are wage and continuation curvatures, $O(\sigma^{-2})$ uniformly by the same bounds. $\bar{N}$ bounds $\partial_{\hat{e}}T$, which at a prior with posterior mean m and variance V decomposes as in Claim 26: a term $O(\sigma^{-2})$ from the second-period contract's response to the conjecture, and the retention term $R_{e\hat{e}}$, whose known-complementarity value at m is $u_2^\infty m u \varphi(u) \underline{y}/(\sigma^2 \hat{e})$. By Claim 18 applied to $r(\bar{b})$, β and the density of the realized index, each factor at the actual posterior is its known-complementarity value at m times $1 + O(\sqrt{V})$, uniformly on the admissible parameter set, so at the equilibrium effort $R_{e\hat{e}} \leq (\bar{g} + O(\sqrt{V}))\alpha + O(\sigma^{-2})$ by the computation of Claim 29 with $m \leq \bar{\theta}$. Hence the contraction holds for $\alpha > c'_U/((1 - \bar{g} - O(\sqrt{V}))\sigma^2)$, where c'_U collects $C_1\sigma^2$ and the $O(\sigma^{-2})$ term above. Since the wage and continuation curvatures at the actual posterior are their known-complementarity values, which sum to c_U , up to a relative correction $O(\sqrt{V})$ by Claim 18, $c'_U = c_U + O(\sqrt{V})$, and the threshold is $\alpha_{\text{kn}}^E(\sigma) + O(\sqrt{V}/\sigma^2)$; and $V \leq 1/\varsigma$.

The mass level. When $\underline{\theta} < 0$, α^M includes the level α^{mass} at which retention alone bounds the counterproductive mass by $\bar{\mu}$. Under Assumption 3(iii) that bound holds without curvature: the base is strongly log-concave with curvature ς and the prior mean is positive, so the prior places mass at most $e^{-\varsigma m(\gamma)^2/2}$ on $\theta < 0$, and every retained posterior places less by likelihood ratio dominance. For ς above an explicit level this is below $\bar{\mu}$ at every prior in the family, so α^{mass} is not the binding level and the thresholds below are those of the first-order approach and the configuration.

Principal's concavity. Step 5 of Theorem 2 gives

$$\alpha^P = \bar{s}_T + \sup_{\kappa \in \mathcal{K}} \sqrt{\bar{w} \mathcal{B}(\kappa) (\kappa^3 \sup |B''| + \kappa^2 \sup |\partial_{\hat{e}}^2 T|)}.$$

Here $\bar{s}_T = \sup |\partial_{\hat{e}} T| = O(\sigma^{-2})$, since the target's derivative of each term of T is at most $C_T = O(\sigma^{-2})$ times the term (Step 1 of the proof of Theorem 3) and T is bounded. The target enters the principal's continuation Ψ only through the informativeness $\iota = \tilde{e}^2/\sigma^2$ of the first-period signal, so $B'' = \Psi_u(2\tilde{e}/\sigma^2)^2 + 2\Psi_\iota/\sigma^2 = O(\sigma^{-2})$ with Ψ_ι, Ψ_u bounded on the admissible parameter set, and $\partial_{\hat{e}}^2 T = O(\sigma^{-4})$ by the same bound applied twice. With $\kappa^3 \mathcal{B} \approx \bar{\theta}^2 v^3 \varphi(v)/\sigma^2$ and $\kappa^2 \mathcal{B} \approx \bar{\theta} v^2 \varphi(v)/\sigma$, both bounded in v , the supremum is $O(\sigma^{-2})$. Hence $\alpha^P = O(\sigma^{-2})$ uniformly over the family, up to the same $O(\sqrt{V})$ corrections. Collecting, all four levels are $O(\sigma^{-2})$ uniformly over the family, and $\eta(\sigma, \varsigma) = O(\varsigma^{-1/2} \sigma^{-2})$. $\square$

Proof of Proposition 3. Claims 26–27 give the first-order-approach threshold $W^* = c_W/\sigma^2$, $c_W := \bar{w}\bar{\theta}^2\varphi(1)$, and the ordering. Claim 29 gives $\alpha_{\text{kn}}^E(\sigma) = c_U/((1 - \bar{g})\sigma^2)$, which falls like σ^{-2} ,

and Claim 30 extends it uniformly over the family up to $\eta(\sigma, \varsigma) \rightarrow 0$, and Claim 28 gives $\bar{\alpha} \asymp \bar{\theta}^2 \sqrt{\bar{w}\varphi(0)/2} \sigma^{-3/2}$. Since σ^{-2} decays faster than $\sigma^{-3/2}$, there is σ^{**} beyond which $\alpha_{\text{kn}}^E(\sigma) < \bar{\alpha}(\sigma)$ with a gap $\Delta(\sigma) \geq c\sigma^{-3/2}$; and since $\eta(\sigma, \varsigma) = O(\varsigma^{-1/2}\sigma^{-2})$, the condition $\eta < \Delta/2$ holds for all $\sigma \geq \sigma^{**}$ once $\varsigma \geq \varsigma^{**}$ for a single ς^{**} of order $1/\sigma^{**}$, because the precision required falls as the noise grows. The configuration threshold α^M does not lift the existence threshold above $\bar{\alpha}$: its two conditions, that $2c'(e) \leq \mathbb{E}_p[\theta]$ and that $e(c'' + \bar{w}J) \leq \mathbb{E}_p[\theta] - c'(e)$ on the retained set, are noise requirements in the σ -large regime, since $c'(e) = \alpha e \approx \bar{w}\bar{\theta}\varphi(v)/\sigma \rightarrow 0$ and $e\alpha \rightarrow 0$ at any α in the interval, so they hold for $\sigma \geq \sigma^{**}$. Strict concavity of the principal's reduced objective in the target is covered by the last component of Claim 30. $\square$

Proof of Proposition 4 (Technology adoption)

Proof. Without the technology there is nothing to learn, so the agent's continuation contains no information, $T \equiv 0$, and the equilibrium and benchmark problems coincide: Π^0 is common to the two, and $\Delta(\gamma) - \Delta^m(\gamma) = \Pi(\gamma) - \Pi^m(\gamma)$. With the technology, the continuation is the same function of first-period effort in both problems, so implementing a given target $\tilde{e}$ at prior γ yields the same continuation value, and the two problems differ only in the wage bill. Write $C(I; \tilde{e})$ for the least-cost wage bill delivering marginal incentive I at target $\tilde{e}$ (Lemma 7). The model's principal needs $c'(\tilde{e}) - T(\gamma; \tilde{e})$ from the wage where the benchmark's needs $c'(\tilde{e})$, and C is increasing in I on the branch on which the cutoff is above expected output.

Decomposition. Write $S(\gamma, \tilde{e}) := C(c'(\tilde{e}); \tilde{e}) - C(c'(\tilde{e}) - T(\gamma; \tilde{e}); \tilde{e})$ for the wage saving the continuation incentive provides at target $\tilde{e}$, so that $\Pi^S(\tilde{e}) = \Pi^B(\tilde{e}) + S(\gamma, \tilde{e})$. Then

$$\Pi(\gamma) - \Pi^m(\gamma) = \Pi^S(e^*) - \Pi^B(e^B) = S(\gamma, e^*) - [\Pi^B(e^B) - \Pi^B(e^*)].$$

The bracket is nonnegative, since e^B maximizes Π^B , and second order in the distortion $e^* - e^B$, since the first derivative of Π^B vanishes at e^B . Three constants enter, each a supremum or infimum over the compact ranges of $(\gamma, \tilde{e}, \kappa)$ the two problems visit at the given (σ, ς) , and each finite by the bounds already established: $\bar{\nu} := \sup|\partial_{\tilde{e}}^2 \Pi^B|$, finite by the dominations of Lemma 1. $\underline{C}_I := \inf \partial_I C = 1/\sup \kappa > 0$, since $\partial_I C = 1/\kappa$ by the pricing identity of Claim 13 and the cutoffs on the branch are bounded over the compact range of targets, and $K := \sup|\partial_T e^*|$, the pass-through of Lemma 7(ii), bounded by $C_I/\underline{\nu}$ with $\underline{\nu}$ the concavity modulus of Π^S from Step 5 of Theorem 2. With these, the strong concavity of Π^B gives $\Pi^B(e^B) - \Pi^B(e^*) \leq \frac{1}{2}\bar{\nu}(e^* - e^B)^2$ and Lemma 7(ii) gives $|e^* - e^B| \leq K|T(\gamma; e^B)|$. Since $|T(\gamma; e^B) - T(\gamma; e^*)| \leq N^*|e^B - e^*|$, this yields $|e^* - e^B| \leq K'|T(\gamma; e^*)|$ with $K' := K/(1 - KN^*)$, provided $KN^* < 1$. This holds for $\sigma \geq \sigma^*$, since

$N^* = O(\sigma^{-2})$ while K is bounded on the admissible parameter set, and the level at which it holds is included in σ_A below. The saving is first order: S has the sign of $T(\gamma; e^*)$ and $|S| \geq \underline{C}_I |T(\gamma; e^*)|$.

The sign. Where $T(\gamma; e^*) \leq 0$, $S \leq 0$ and the bracket is nonnegative, so $\Pi \leq \Pi^m$. Where $T(\gamma; e^*) > 0$,

$$\Pi - \Pi^m \geq T(\underline{C}_I - \frac{1}{2} \bar{\nu} K'^2 T) > 0 \quad \text{whenever } T < 2\underline{C}_I / (\bar{\nu} K'^2).$$

The manipulation term is bounded at every prior: by Lemma 3 it is the integral of the retained continuation rent against $\partial_e f_1$, the rent is at most $\bar{w}$, and $\int |\partial_e f_1| \leq \theta_{\max} \nu_1 = 2\varphi(0)\theta_{\max}/\sigma$, so $T \leq \bar{w}\theta_{\max}/\sigma$, so the condition holds at every prior once

$$\sigma > \sigma_A := \frac{\bar{w}\theta_{\max} \bar{\nu} K'^2}{2\underline{C}_I},$$

which is one of the levels in the definition of σ^* under *The constants* at the end of the proof of Theorem 3. Hence $\Pi - \Pi^m$ has the sign of $T(\gamma; e^*(\gamma))$ at every prior.

Locating the threshold. For $\alpha < \bar{\alpha}$, $T(\gamma; e^*(\gamma))$ has a unique zero γ^{**} , positive below and negative above (proof of Theorem 3, Step 2), and $|\gamma^{**} - \gamma^*| \leq \delta^*$ by Lemma 7(iii). So there is weakly more adoption below γ^{**} and weakly less above it. For $\alpha \geq \bar{\alpha}$, $T > 0$ at every prior (Step 3 there), so there is weakly more adoption throughout. $\square$